

The CFS Closure-Quotient Landscape and Cyclotomic Galois Pairs in Discrete Hodge L_1 Spectra on Group-Symmetric 2-Complexes

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Source repository (paper + scripts + findings docs):
<https://github.com/IridiumSoftware/closure-quotient-landscape>

Abstract

We characterise the closure-quotient family of the Closure-Forces-Structure (CFS) programme — the gauge-equivalence quotients Q_n of multiway graphs generated by ternary hypergraph rewriting — using exact Gaussian-integer ray arithmetic and a validity-gated probe corpus. Methodologically, exact arithmetic and validity gating sharpen the published family documentation: the previously-listed Q_{98} is corrected to Q_{102} (a Float64 fidelity-clustering artifact), Q_{180} is shown not to exist ($\mathcal{C}$ -closure is idempotent on $\mathcal{C}$ -closed objects), and the canonical s58 initial-condition (IC) pool used throughout the prior CFS literature is found to pass the $K_6^3 \rightarrow 51$ validity gate but exhibit a specific non-genericity at K_n^3 for $n \geq 9$ — so IC-genericity is a hierarchy (fully-generic / small- n -generic / degenerate), not the binary the literature treats it as. We exhibit the closure laws $\text{cyc}(n) \rightarrow 4n$ and $K_n^3 \rightarrow n(3n-1)/2$ (pentagonal), verify the latter exactly for $n = 6..20$ across two independent strict-generic IC pools, document terminality of Q_{48} , Q_{90} , Q_{102} under both closure-growth and $\mathcal{C}$ -closure, and exhibit a prime closure object Q_{181} via an irregular 12-vertex 132-edge seed. An additive composition rule $Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface})$ with observed interface corrections ($\delta_{\sqcup} = 0$ disjoint, $\delta_{\text{bridge}} = +3$ per bridge triple, $\delta_{\text{glue-}g} = -(4g-3)$ for a g -vertex glue) holds across the tested join modes and forms a commutative monoid under disjoint sum, with the charge-conjugation doubling rule $|Q \cup \mathcal{C}(Q)| = 2|Q|$ recovered as the structurally disjoint case. The resulting landscape is therefore not an iterated-doubling tower but a validity-gated closure family: $\mathcal{C}$ -closed objects are terminal under $\mathcal{C}$, larger quotients require larger or irregular seeds, and IC-genericity is a genuine hierarchy. As a downstream consequence of the exact-arithmetic re-grounding of Q_{51} , we report that its Hodge 1-Laplacian carries the irrational eigenvalue pair $\{3-\sqrt{3}, 3+\sqrt{3}\}$ at multiplicity 30 on each conjugate, decomposing under the natural S_6 action as $[4, 2] \oplus [3, 3] \oplus [3, 2, 1]$ on each eigenspace and tracing to a $\mathbb{Z}_3$ -cyclotomic substructure shared with the triangulated 3×3 flat torus $T_{3,3}^2$.

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The cross-substrate observation motivates the second half of the paper, which tests the cyclotomic-Galois pattern against a 9-substrate corpus of finite group-symmetric 2-complexes: tetrahedron under S_4 , octahedron under O_h , icosahedron and dodecahedron under A_5 , torus $T_{3,3}^2$ under $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$, real projective plane $\mathbb{RP}^2$ (icosahedron antipodal quotient, A_5), Klein bottle K^2 (X-subdivided torus quotient, $\text{Aut} = D_4$), truncated icosahedron (“soccer ball”) under A_5 , and heptagonal bipyramid under D_7 . We find that the Galois *closure field* of $\text{Spec}(L_1)$ is indexed by cyclotomic factors in Aut ’s real character ring, and that this indexing generalises across extension degrees: $\mathbb{Z}_3$ and $\mathbb{Z}_5$ produce quadratic Galois pairs (in $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{5})$); the $\mathbb{Z}_7$ -cyclotomic factor in D_7 produces *cubic* Galois orbits in $\mathbb{Q}(\cos 2\pi/7)$. On four distinct A_5 -symmetric substrates the $\mathbb{Q}(\sqrt{5})$ field is invariant; the specific rational coefficients in the Galois pairs depend on cellular structure. The discrete Hodge kernel-dimension identity $\dim_{\text{ker}} L_1(X) = \dim_{\mathbb{R}} H^1(X)$ is verified at machine zero on the integer L_1 matrices across four topology classes $(S^2, \mathbb{RP}^2, K^2, T^2)$.

The corrections, exact-arithmetic framework, IC-genericity hierarchy, and Galois-pair / cyclotomic-Galois observations all rest on reproducible finite-depth computations; the generative composition algebra, general closure laws, and refined cyclotomic-Galois conjecture are validated on the examined corpora but remain conjectural pending closed-form derivation. Foundational reference: J. von Neumann (1952) [15].

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1 Introduction

1.1 Two-part overview

This paper has two halves, joined by a bridge.

Part 1 (§2–§12) characterises the *CFS closure-quotient family* — the canonical gauge-equivalence quotients Q_n of multiway graphs generated by ternary hypergraph rewriting in the Closure-Forces-Structure programme. We exhibit the closure laws $\text{cyc}(n) \rightarrow 4n$ and $K_n^3 \rightarrow n(3n-1)/2$, the terminality structure (Q_{48}, Q_{90}, Q_{102} are fixed under both closure-growth and $\mathcal{C}$ -closure), the additive composition algebra $Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface})$, the Rosen (M, R) functional-role characterisation of the doubly-terminal family, and we catch two errors in the published family documentation ($Q_{98} \rightarrow Q_{102}$ as a Float64 fidelity-clustering artefact; Q_{180} does not exist).

Bridge (§13): a downstream consequence of the exact-arithmetic re-grounding of Q_{51} is that its discrete Hodge 1-Laplacian carries the irrational eigenvalue pair $\{3 \mp \sqrt{3}\}$ at multiplicity 30, decomposing under S_6 as $[4, 2] \oplus [3, 3] \oplus [3, 2, 1]$ on each eigenspace. The same Galois pair appears on the triangulated torus $T_{3,3}^2$ at multiplicity 6 — via a distinct $\mathbb{Z}_3$ -cyclotomic mechanism. This raises the question Part 2 answers: is the cyclotomic-Galois pattern Q_{51} -specific, or a generic feature of group-symmetric 2-complexes?

Part 2 (§14–§24) tests the question against a 9-substrate corpus of group-symmetric 2-complexes: tetrahedron, octahedron, icosahedron, dodecahedron, truncated icosahedron (“soccer ball”), $\mathbb{RP}^2$ (icosahedron antipodal quotient), Klein bottle K^2 , triangulated torus

$T_{3,3}^2$, and heptagonal bipyramid. The cyclotomic-Galois pattern indexes the Galois *closure field* of $\text{Spec}(L_1)$ by cyclotomic factors of $\text{Aut}(X)$'s real character ring, and this indexing *generalises across extension degree*: $\mathbb{Z}_3$ and $\mathbb{Z}_5$ produce quadratic Galois pairs $(\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{5}))$; the $\mathbb{Z}_7$ -cyclotomic factor in D_7 produces *cubic* Galois orbits in $\mathbb{Q}(\cos 2\pi/7)$, the cubic totally-real subfield of the 7-th cyclotomic field. On four distinct A_5 -symmetric substrates the $\mathbb{Q}(\sqrt{5})$ field is invariant; the specific rational coefficients depend on cellular structure.

Joint synthesis (§25–§26) draws the two halves together and lists open problems and the proof program.

1.2 The closure-quotient family

The Closure-Forces-Structure (CFS) programme derives the gauge structure of the Standard Model from a single principle: *Rosen closure* on ternary causal hypergraphs. A small Rosen-closed seed graph G generates an infinite multiway graph $M(G)$ under double-pushout (DPO) rewriting on $\mathbb{C}^3$ -valued colour decorations; the gauge-equivalence quotient $Q_n(G) := M(G)/\sim$, where $v \sim w$ iff $\psi(v)$ and $\psi(w)$ span the same complex line in $\mathbb{C}^3$, is the carrier of every downstream construction — the Born measure, the spectral triple, the Standard-Model algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ [3].

A small *closure-quotient family* of canonical Q_n values has accumulated over the programme's working corpus: Q_{24} from the six-vertex Rosen-closed graph G_0 ; $Q_{48} = Q_{24} \cup \mathcal{C}(Q_{24})$, its charge-conjugate union, carrying the KO-dimension-6 spectral triple that forces the Standard-Model algebra by the Chamseddine–Connes–Marcolli classification; Q_{45} , Q_{84} , Q_{90} from intermediate-density seeds; and Q_{51} , Q_{102} from the complete ternary K_6^3 — $Q_{102} = \mathcal{C}(Q_{51})$ being the *developmentally-complete zygote* that carries the full spectral-action structure and is a fixed point of the closure-growth endofunctor. These objects are programme-level: every theorem in CFS about gauge content, anomaly cancellation, or mass generation is a theorem about one of them.¹

This paper characterises the family. We give the closure laws ($\text{cyc}(n) \rightarrow 4n$, $K_n^3 \rightarrow n(3n-1)/2$), the terminality structure (every $\mathcal{C}$ -closed object is fixed under both closure-growth and $\mathcal{C}$, so no *iterated-doubling family* exists: larger objects come only from larger seeds), and a generative algebra for combining seeds (§10, the headline result). We also catch and correct two errors in the published family documentation (§4) and surface a previously-undetected non-genericity in the canonical initial-condition pool that the entire prior literature has used (§9).

1.3 Origin: a falsified holographic synthesis

The work originated as a cross-theory exercise. In May 2026, a holographic-synthesis hypothesis bridging CFS to Shaun Fosmark's *Solvency Field Theory* [4] was pre-registered, tested, and falsified on both Q_{24} and Q_{102} [5]. With the cross-theory direction closed, attention turned to the closure-quotient family itself: what is actually known about it

¹The paradigm of obtaining a finite quotient by iterating a closure-style operator on a continuous or large-finite substrate has a foundational precedent in von Neumann's 1952 multiplexing construction [15]: iterating the restoring map $\alpha \mapsto 3\alpha^2 - 2\alpha^3$ on bundle-excitation $\alpha \in [0, 1]$ yields the fixpoint quotient $\{0, 1/2, 1\}$ with attractors $\{0, 1\}$; the Scheffer-stroke variant $\alpha \mapsto 2\alpha^2 - \alpha^4$ has unstable fixpoint $\alpha_0 = (\sqrt{5} - 1)/2 \in \mathbb{Q}(\sqrt{5})$, prefiguring the quadratic-irrationality fixpoint structure of §13 below in a different quadratic extension.

exactly, in Gaussian-integer arithmetic, without the fidelity-threshold clustering that the prior Python pipeline had used? This paper is the answer.

1.4 Contributions

1. **Closure laws** (§5): $\text{cyc}(n) \rightarrow 4n$ linear; $K_n^3 \rightarrow n(3n-1)/2$ pentagonal, verified exact for $n = 6..20$ across two independent strict-generic Gaussian-integer IC pools; the generic floor is Q_{12} ; depth-stability by depth 4.
2. **Terminality** (§6): Q_{48}, Q_{90}, Q_{102} are fixed points of both the closure-growth endofunctor $\mathcal{F} = M/\sim$ and the $\mathcal{C}$ -closure. There is no iterated-doubling family; larger closure objects require larger seeds, not iteration of operators on smaller ones.
3. **Why $\mathcal{C}$ doubles cleanly** (§7): under truly-generic ICs the charge-conjugation involution J is fixed-point-free, so $|Q \cup \mathcal{C}(Q)| = 2|Q| - f$ with $f = 0$. This identifies the structural reason for the clean $2|Q|$ doubling observed empirically across the family.
4. **Primes are reachable; Q_{181} is exhibited** (§8): the regular families ($4n$, pentagonal $n(3n-1)/2$, and $\mathcal{C}$ -doubling $2n$) are composite-only by formula, but irregular single-side seeds produce primes (we record 13, 19, 23, 71, 83, 97, 193). A 12-vertex 132-edge irregular seed closes exactly to Q_{181} (181 prime), exhibited in 12 independent draws.
5. **IC-genericity hierarchy** (§9): the $K_6^3 \rightarrow 51$ validity gate that the prior CFS literature has used is *necessary but not sufficient* for fully-generic regime membership. The canonical s58 IC pool used in every prior CFS probe passes this gate but exhibits a specific defect at K_n^3 for $n \geq 9$. The fully-generic regime is genuinely distinct from the small- n -generic regime; the regime split is a hierarchy, not a binary. A stronger gate ($K_{10}^3 \rightarrow 145$ or $K_{12}^3 \rightarrow 210$) catches the defect.
6. **A generative algebra** (§10, headline): for closure quotients A, B , the count is

$$Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface}),$$

with δ a structure-independent interface correction: $\delta_{\sqcup} = 0$ for disjoint composition, $\delta_{\text{bridge}} = +3$ per bridge triple, and $\delta_{\text{glue-}g} = -(4g-3)$ for a g -vertex glue. The counts form a *commutative monoid under disjoint sum* $\sqcup$ with identity 0. The $\mathcal{C}$ -closure doubling rule is the special case where $\mathcal{C}(Q)$ is structurally disjoint from Q (under generic ICs, J fixed-point-free, $\delta = 0$). The algebra is purely *additive*: there is no multiplicative combination rule (in particular, $Q_{45} \neq Q_9 \cdot Q_5$ — the factorisation of the integer is not the factorisation of the closure).

7. **Methods** (§3): exact Gaussian-integer ray arithmetic with validity-gated probes catches and corrects three specific errors in the published family documentation: Q_{98} is Q_{102} (Float64 fidelity-clustering artifact); Q_{180} does not exist ($\mathcal{C}$ -closure is idempotent on $\mathcal{C}$ -closed objects); the $K_6^3 \rightarrow 51$ validity gate is insufficient at large n (Pool A defect, above). The reproducibility scripts are released with the paper (§26).

1.5 Status of this paper

This paper should be read as a *validity-gated computational landscape and a generative-algebra discovery report*. The exact-arithmetic corrections, terminality checks, and IC-

genericity hierarchy are established by exact Gaussian-integer computation under explicit gates. The broader composition algebra of §10 is discovered and repeatedly validated, but remains theorem-programmatic until closed-form proofs of the universal interface corrections are supplied (§25.5). The Galois-pair observation of §13 is itself exact-computational ($\sigma_E L_1 = L_1 \sigma_E$ verified at machine zero on the integer matrix; per-eigenspace irrep multiplicities are integers by Frobenius reciprocity), parallel in evidential tier to the audit results, and downstream of the exact-arithmetic re-grounding of Q_{51} that motivates the rest of the paper.

1.6 Claim-status ledger

Claim	Status	Section
Fidelity-0.999 clustering can silently merge distinct rays	proved / demonstrated	§3
Q_{98} corrects to Q_{102}	exact-computational	§4
Claimed Q_{180} via iterated $\mathcal{C}$ -doubling does not exist	proved structurally	§4
$\mathcal{C}(Q) = Q$ for already $\mathcal{C}$ -closed objects	proved structurally	§6
$K_n^3 \rightarrow n(3n-1)/2$ (pentagonal law)	finite exact validation	§5
Clean $\mathcal{C}$ -doubling under strict-generic ICs	exact-computational + structural argument	§7
IC genericity is hierarchical	exact-computational	§9
Pool A / s58 is small- n -generic, not fully generic	exact-computational	§9
Q_{181} exists from an irregular seed	exact-computational exhibit	§8
Disjoint sum has $\delta_{\sqcup} = 0$	partially proved / structurally sketched	§10, Appendix A
Bridge correction $\delta_{\bowtie} = +3k$	validated conjectural	§10
Glue correction $\delta_{\#-g} = -(4g-3)$	validated conjectural	§10
Composition counts form a universal additive algebra	conjectural algebraic law	§10, §25.5
Galois pair $\{3-\sqrt{3}, 3+\sqrt{3}\}$ in $\text{Spec}(L_1(Q_{51}))$ with multiplicity 30 and irrep content $[4, 2] \oplus [3, 3] \oplus [3, 2, 1]$	exact-computational	§13
$\mathbb{Z}_3$ -cyclotomic mechanism shared with $T_{3,3}^2$	exact-computational (algebraic mechanism identified)	§13

Table 1: Per-claim evidential status. The paper distinguishes *proved* / *structurally proved* (theorem-grade), *exact-computational* (exact arithmetic on a verified finite regime), *finite exact validation* (exact across the probed corpus, no closed form), and *validated conjectural* (verified across tested cases, universality open). The §12 Validity Boundary makes the partition explicit; §25.5 states the corresponding proof obligations.

1.7 Evidence tiers

We label every result by its evidential strength:

- **Tier 1 — exact algebraic.** A property is established on the integer matrix by exact arithmetic (Gaussian-integer ray equality; integer characteristic polynomial; Frobenius reciprocity over rational or cyclotomic character tables with exact-integer multiplicities). Tier 1 results are theorem-grade up to the underlying definitions.
- **Tier 2 — numerical verification.** A property is established empirically on the integer or Float64 matrix to machine precision (commutation $\sigma_E L_1 = L_1 \sigma_E$ at 0.0 entrywise; eigenvalue identification at $\leq 10^{-9}$; or BigFloat verification to ≥ 50 digits). Tier 2 statements are not theorems but resist accidental falsification at the tested precision.
- **Tier 3 — finite exact validation.** A property holds across the entire probed corpus by exact arithmetic but has no closed-form derivation (the pentagonal law $K_n^3 \rightarrow n(3n-1)/2$ at $n = 6..20$ is the prototype). Tier 3 promotes to Tier 1 only with a counting proof.

The validity-boundary partition of §12 maps Part 1 results onto these tiers; the spectrum-summary table of §24.5 does the same for Part 2.

1.8 Binary operator inside a ternary-hyperedge rewriting graph

The CFS rewriting kernel is structured at two different levels, and the remainder of the paper uses the distinction without comment, so we state it once at the outset.

The operator is binary. The production rule is the conjugated cross-product on $\mathbb{Z}[i]^3$,

$$\text{compose}(a, b) = \overline{a \times b}, \quad a, b \in \mathbb{Z}[i]^3,$$

which takes *two* cochains as input and returns one. As a map of sets it has signature $\mathbb{Z}[i]^3 \times \mathbb{Z}[i]^3 \rightarrow \mathbb{Z}[i]^3$. Nothing in this paper treats the operator as a function of three cochains.

The rewriting graph is ternary. A hyperedge is a triple of vertex labels (v_1, v_2, v_3) , and the depth- d rewriting step applies $\text{compose}(\psi(v_1), \psi(v_2))$ to introduce a new vertex carrying w , then schedules *three* new hyperedges (w, v_2, v_3) , (w, v_1, v_3) , (w, v_1, v_2) at depth $d+1$. The third slot v_3 is not an operator input, but it propagates into the next-depth scheduling, so it is load-bearing for $|Q_n|$: changing the hyperedge arity changes the multiway expansion tree, and changes the gauge quotient.

Both facts are type-checked. An independent Haskell port of the kernel at `haskell/` (see [2]) encodes the distinction in the type signatures

```
compose :: Cochain -> Cochain -> Cochain    -- binary
type Hyperedge = (Vertex, Vertex, Vertex)    -- ternary
```

and produces identical cardinalities to the Python reference on every canonical object ($|Q_{24}| = 24$, $|Q_{42}| = 42$, $|Q_{45}| = 45$, $|Q_{51}| = 51$ single-side; $|Q_{48}| = 48$, $|Q_{84}| = 84$, $|Q_{90}| = 90$, $|Q_{102}| = 102$ C-closed), together with J fixed-point-free involution and F-closure terminality at depths 4, 5, 6 on $Q_{48}, Q_{84}, Q_{90}, Q_{102}$. The structural-arity claim is therefore Tier 1 (type-checked) and the cardinality claim is Tier 3 (two-implementation finite exact validation).

Throughout the paper, “ternary hypergraph” refers to the rewriting graph’s hyperedge arity — the structure — not to the operator’s domain arity, which is binary. The Rosen (M, R) interpretive language that surrounds the operator in §11 attaches to fixed-point and doubling properties of the gauge quotient Q_n ; those properties depend on the hyperedge structure, which is genuinely ternary, not on the operator’s argument count.

1.9 Plan of the paper

Part 1 closure-quotient landscape. §2 primes the closure quotients (ternary hypergraphs, $\mathcal{C}$ -closure, prior family). §3 is methods: exact arithmetic + validity gating. §4 re-grounds the published family. §5 states the closure laws. §6 terminality; §7 the J-fpf condition; §8 primes and Q_{181} ; §9 IC genericity; §10 the composition algebra; §11 the Rosen (M, R) functional roles; §12 the validity boundary of Part 1.

Bridge. §13 exhibits the $\{3 \mp \sqrt{3}\}$ Galois pair on Q_{51} via S_6 and shows its match on the torus $T_{3,3}^2$ via a distinct $\mathbb{Z}_3$ -cyclotomic mechanism — the motivating question for Part 2.

Part 2 substrate corpus. §14 is the discrete-Hodge background (the L_1 operator, signed σ_E , Frobenius reciprocity and Clifford theory). The nine substrates appear in order of increasing structural complexity: §15 tetrahedron; §16 octahedron; §17 icosahedron; §18 dodecahedron; §19 truncated icosahedron; §20 $\mathbb{RP}^2$; §21 Klein bottle; §22 torus $T_{3,3}^2$; §23 heptagonal bipyramid. §24 synthesises the cyclotomic-Galois pattern, trivial-irrep / H^1 correspondence, and Forman-Ricci uniformity findings across all ten substrates (Q_{51} plus the corpus).

Joint synthesis. §25 discussion + open problems + proof program. §26 probe corpus and reproducibility. Appendices supply notation, the $\delta_{\square}$ proof, character tables, and verifier data.

Part I

The CFS Closure-Quotient Landscape

2 Background: CFS Closure Quotients

This section is a self-contained primer: a reader unfamiliar with the CFS programme should be able to follow the rest of the paper from this section alone. We give the minimal definitions (ternary hypergraphs, the closure operator, the multiway construction, the gauge quotient, the $\mathcal{C}$ -closure) and then recap the published family as it stood prior to the present work, so §4 can re-ground it. The full CFS treatment is in [3]; we cite specific results as needed.

2.1 Ternary hypergraphs and the colour decoration

Definition 2.1 (Ternary hypergraph). A *ternary hypergraph* $G = (V, E)$ has a finite vertex set V and an edge set $E \subseteq V^3$ of ordered triples of (not necessarily distinct) vertices. We write $e = (s_1, s_2, t)$ for an edge with sources s_1, s_2 and target t , interpreting the triple as the production rule “the pair (s_1, s_2) composes to t .”

A *decoration* of G on $\mathbb{C}^3$ is a map $\psi : V \rightarrow \mathbb{C}^3$, read as assigning a colour vector to each vertex. The CFS composition rule is the conjugated cross-product

$$\mu(\psi_1, \psi_2) := \overline{\psi_1 \times \psi_2} \in \mathbb{C}^3, \quad (1)$$

where $\times$ is the ordinary $\mathbb{C}^3$ cross-product and $\bar{\cdot}$ is component-wise complex conjugation. An edge $(s_1, s_2, t) \in E$ is *consistent* under ψ iff $\psi(t) = \mu(\psi(s_1), \psi(s_2))$ up to a non-zero complex scalar, i.e. iff $\psi(t)$ spans the same complex line in $\mathbb{C}^3$ as $\mu(\psi(s_1), \psi(s_2))$. This projective slack is what the gauge quotient (§2.3) collapses.

Definition 2.2 (Rosen-closed). A ternary hypergraph G is *Rosen-closed* iff every vertex appears as the target of at least one edge in E . Equivalently: no vertex is a primitive that must be supplied externally; every vertex’s colour vector can in principle be produced by composition from the others.

Rosen closure is the formalisation of M. Rosen’s “closure to efficient causation” [6] in the ternary-hypergraph setting; the CFS programme adopts it as its foundational principle.

2.2 The multiway graph and the closure operator

Definition 2.3 (Multiway graph $M(G)$). Let $G = (V, E)$ be a ternary hypergraph and let $\psi_0 : V \rightarrow \mathbb{C}^3$ be an *initial condition* (IC) decoration of the original vertices. The *multiway graph* $M(G)$ is the union, over all finite-depth applications of the composition rule, of the vertices produced and the consistency edges they satisfy. Concretely: $M(G)$ is built breadth-first by, at each depth d , taking every edge (s_1, s_2, t) whose sources both exist at depth $\leq d$, producing a new vertex w with $\psi(w) = \mu(\psi(s_1), \psi(s_2))$, and adjoining the three derived edges (w, s_2, t) , (w, s_1, t) , (w, s_1, s_2) at depth $d + 1$.

Remark 2.4 (Caching). The vertex w produced by the pair (s_1, s_2) depends only on the pair, so once computed it is cached: if the same pair appears later at a different depth, the same w is reused. This is what makes $M(G)$ a graph rather than a tree.

Remark 2.5 (Zero composition). The composition $\mu(\psi_1, \psi_2)$ is zero in $\mathbb{C}^3$ exactly when ψ_1 and ψ_2 are projectively parallel. In $M(G)$ such an edge *fails to produce*: no new vertex is added at that step. This is the only source of edge-pruning.

$M(G)$ is in general infinite, but *projectively* (gauge-quotient) finite for the seed graphs we consider.

2.3 The gauge quotient $Q_n(G)$

Definition 2.6 (Gauge equivalence). Two vertices $v, w \in M(G)$ are *gauge-equivalent*, $v \sim w$, iff their colour vectors span the same complex line:

$$\psi(v) \times \psi(w) = 0 \quad (\text{as a } \mathbb{C}^3 \text{ vector}).$$

This is exact projective equivalence on $\mathbb{CP}^2$. For Gaussian-integer ICs ($\psi_0 : V \rightarrow (\mathbb{Z}[i])^3$, which the composition rule (1) preserves), the test $\psi(v) \times \psi(w) = 0$ is exact integer equality with no threshold.

Definition 2.7 (Closure quotient). The *closure quotient* of the seed graph G at IC ψ_0 and depth d is

$$Q_n(G, \psi_0, d) := |M_d(G)/\sim|,$$

where $M_d(G)$ is the depth- d multiway graph and $|\cdot|$ is the cardinality of the gauge-quotient set. We write $Q_n(G)$ for the depth-stable value when the count is independent of d beyond some depth (the standard situation for G and IC in the generic regime; see §9).

The integer $Q_n(G)$ is the central object of this paper. The closure laws of §5 are statements about how Q_n depends on the seed graph G ; the composition algebra of §10 is the statement of how Q_n behaves under combination of seeds; the IC-genericity hierarchy of §9 is the statement of how Q_n depends on the choice of ψ_0 .

2.4 Generators: $\text{cyc}(n)$ and K_n^3

The closure laws are most cleanly stated on two families of seed graphs, named for the role they play in CFS's existing literature.

Definition 2.8 (Cyclic generator $\text{cyc}(n)$). For $n \geq 3$, the cyclic ternary hypergraph $\text{cyc}(n)$ has vertex set $V = \{0, 1, \dots, n-1\}$ and edge set

$$E = \{(i, (i+1) \bmod n, (i+2) \bmod n) : i = 0, \dots, n-1\}.$$

Each edge is a forward shift on the cyclic vertex order.

Definition 2.9 (Complete ternary K_n^3). For $n \geq 3$, the complete ternary hypergraph on n vertices, K_n^3 , has vertex set $V = \{0, 1, \dots, n-1\}$ and edge set equal to all ordered triples of *distinct* vertices:

$$E = \{(i, j, k) : i, j, k \in V, i \neq j \neq k \neq i\}.$$

Thus $|E| = n(n-1)(n-2)$.

For the canonical CFS seed graph G_0 (a six-vertex Rosen-closed graph studied in [3, §Step 3]), the closure quotient is Q_{24} . For $K_n^3[6]$ (the complete ternary on 6 vertices, the maximal-density seed in the G_0 regime), the single-side closure quotient is Q_{51} and its $\mathcal{C}$ -closure is Q_{102} . The named family members and their seed graphs are summarised in Table 3.

2.5 The $\mathcal{C}$ -closure

The composition rule (1) treats complex conjugation asymmetrically: $\mu(\psi_1, \psi_2)$ is the conjugate cross-product, not the cross-product. This singles out a *charge-conjugate sector*: for each IC ψ_0 , the conjugated IC $\overline{\psi_0}$ generates its own multiway graph $M(G; \overline{\psi_0})$ and its own closure quotient. The $\mathcal{C}$ -closure is the union of these two sectors.

Definition 2.10 ($\mathcal{C}$ -closure). The $\mathcal{C}$ -closure of $Q_n(G; \psi_0, d)$ is

$$\mathcal{C}Q_n(G; \psi_0, d) := |(M_d(G; \psi_0) \cup M_d(G; \overline{\psi_0})) / \sim|.$$

The two sectors are gauge-distinct in general: their representatives $\psi(v)$ and $\overline{\psi(v)}$ are not projectively equivalent unless $\psi(v)$ is a real-up-to-phase vector. The charge-conjugation involution

$$J : v \mapsto v', \quad \psi(v') = \overline{\psi(v)},$$

acts on $\mathcal{C}Q_n$ as a $\mathbb{Z}_2$ -involution; the cardinality identity

$$\mathcal{C}Q_n = 2 \cdot Q_n - f \tag{2}$$

where f is the number of J -fixed clusters (those whose representative is projectively equal to its conjugate) is immediate from inclusion-exclusion. The empirical observation that (2) cleanly reduces to $\mathcal{C}Q_n = 2 \cdot Q_n$ across the canonical family is the content of §7: under truly-generic ICs, J is fixed-point-free ($f = 0$).

2.6 The published family

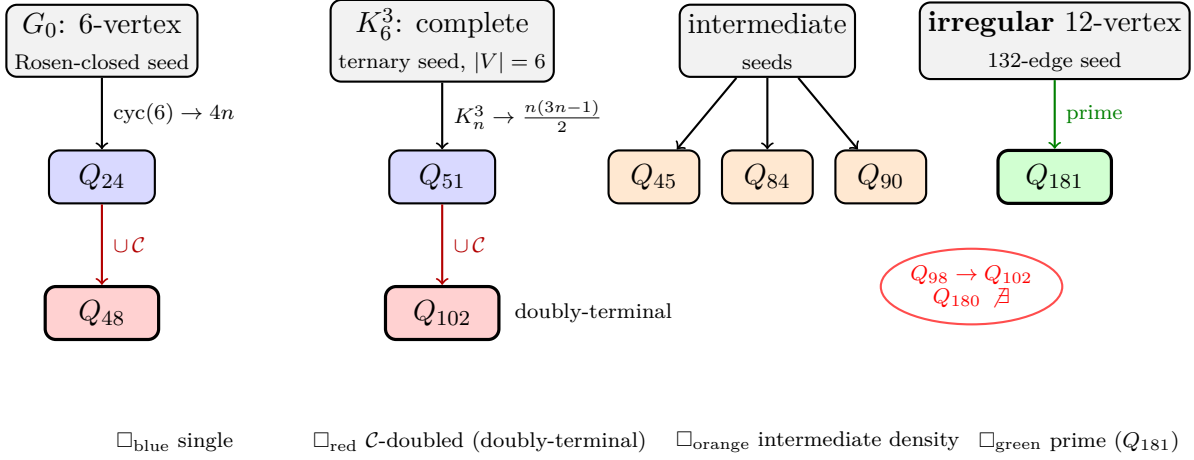


Figure 1: The CFS closure-quotient landscape. Seeds (top row) generate canonical Q_n values (middle); $\mathcal{C}$ -doubling produces the doubly-terminal Q_{48}, Q_{102} (bottom). Two prior-literature entries are corrected in this paper: the previously-reported Q_{98} is a Float64 fidelity-clustering artefact for Q_{102} (§4, [8]); the claimed Q_{180} does not exist because $\mathcal{C}$ is idempotent on $\mathcal{C}$ -closed objects (§4). Q_{181} is the *only known* prime closure object: it arises from an irregular 12-vertex / 132-edge seed that does not factor through a $\text{cyc}(n)$ or K_n^3 family. Both $\text{cyc}(n) \rightarrow 4n$ and the pentagonal law $K_n^3 \rightarrow n(3n-1)/2$ are validated exactly on the tested ranges (§5).

Prior to the present work, the CFS programme’s working family documentation reported the closure quotients summarised in Table 2. The measurements were produced by a Python pipeline using Float64-fidelity clustering at threshold 0.999 [7]; the present work re-grounds them exactly (§4) and extends the table to a closed form (§5).

Table 2: The CFS closure-quotient family as previously documented ([7, 3]). All counts are gauge-quotient sizes; column *single* is $|Q_n(G)|$, column *$\mathcal{C}$ -closed* is $|\mathcal{C}Q_n(G)|$. The entries flagged with † are corrected in §4.

Name	Seed graph	$ V_{\text{seed}} $	Single	$\mathcal{C}$ -closed
Q_{24}/Q_{48}	G_0 (self-ref. closed, 6 edges)	6	24	48
Q_{45}/Q_{90}	$G_0 + 6$ pads (one-side)	6	45	90
Q_{84}	Adjacent + 6 pads ($\mathcal{C}$ -closed)	6	42	84
$Q_{51}/Q_{98}^\dagger$	$K_n^3[6]$	6	51	$98^\dagger$
$Q_{180}^\dagger$	iter. $\mathcal{C}$ -doubling of Q_{90}	6	—	$180^\dagger$

Table 3: The corrected family after §4, with closure laws from §5.

Name	Seed graph	Single $ Q_n $	$\mathcal{C}$ -closed $ \mathcal{C}Q_n $
Q_{24}/Q_{48}	G_0	24	48
Q_{45}/Q_{90}	one-side pads on G_0	45	90
Q_{84}	Adjacent + 6 pads	42	84
Q_{51}/Q_{102}	$K_n^3[6]$	51	102
Q_{70}/Q_{140}	$K_n^3[7]$	70	140
Q_{92}/Q_{184}	$K_n^3[8]$	92	184
Q_{117}/Q_{234}	$K_n^3[9]$	117	234
...	...	...	...
$Q_{n(3n-1)/2}/Q_{n(3n-1)}$	K_n^3	$n(3n-1)/2$	$n(3n-1)$
Q_{181}	12-vertex 132-edge irregular seed	181	362

The objects Q_{48} , Q_{90} , Q_{102} are *terminal* (§6): they are fixed under both $\mathcal{C}$ -closure and the closure-growth endofunctor $\mathcal{F} = M/\sim$. So Table 3’s “...” is not a sequence of objects reachable by iterating an operator; it is the catalogue indexed by the seed graph dimension n . The K_n^3 family stretches indefinitely; the *irregular* family (which contains Q_{181} and the prime-cardinality objects more generally) is sparser and is exhibited rather than enumerated.

3 Methods: Exact Arithmetic and Validity Gating

The methodological contribution of this paper is the combination of two practices: *exact Gaussian-integer ray arithmetic*² (no floating-point thresholds) and *validity-gated probes* (every scan is guarded by a hard-checked reproduction of a known anchor before its measurements are reported). The first eliminates a class of cluster-merging errors that the prior Python pipeline was prone to; the second catches probe-implementation bugs before they contaminate results. Their joint application surfaced the three concrete findings this paper records: the $Q_{98} \rightarrow Q_{102}$ correction (§4), the Q_{180} non-existence (§4), and the IC-genericity hierarchy (§9).

3.1 The fidelity-0.999 problem

The prior CFS Python pipeline computed gauge equivalence by clustering representatives whose Float64 fidelity exceeded a fixed threshold:

$$v \sim_{\text{fid}} w \iff \frac{|\langle \psi(v) | \psi(w) \rangle|^2}{\|\psi(v)\|^2 \|\psi(w)\|^2} > 0.999. \quad (3)$$

This is an approximation of the exact projective-equivalence relation $\psi(v) \times \psi(w) = 0$: for vectors that are genuinely on the same complex line, the fidelity is 1 exactly (in

²By *Gaussian-integer ray arithmetic* we mean the following. Vectors $\psi \in \mathbb{Z}[i]^3$ are tested for projective equivalence — spanning the same complex line in $\mathbb{C}^3$, i.e. representing the same point in $\mathbb{CP}^2$ — by computing the cross product $\psi \times \phi \in \mathbb{Z}[i]^3$ exactly via Gaussian-integer multiplication and addition, and declaring equivalence when all three components vanish identically. No floating-point operations are used, no tolerance is specified, and the test is sharp. The three components — the ring $\mathbb{Z}[i]$, the cross-product test for collinearity in $\mathbb{C}^3$, and exact (non-floating-point) computation — are all standard; the phrase “Gaussian-integer ray arithmetic” is the term we use here to name the specific combination, in contrast to the prior CFS practice of approximate ray-equivalence via Float64 fidelity-threshold clustering (§3.1).

exact arithmetic) or within Float64 round-off (in computation); for vectors on different lines, the fidelity is some value strictly less than 1. The threshold 0.999 was chosen as a tolerance for floating-point round-off accumulation across many composition steps.

The problem is that 0.999 is also a lossy approximation in the *other* direction: two genuinely distinct lines may sit within fidelity-0.999 of one another. When this happens at scale, the fidelity-clustering merges clusters that exact arithmetic separates, and the resulting Q_n is under-counted. The merge is silent: there is no error indicator other than the count itself coming out wrong.

The $K_n^3[6]$ case study. On the maximal-density G_0 -regime seed $K_n^3[6]$ (the complete ternary on six vertices, 120 edges), the exact depth-4 $\mathcal{C}$ -closed quotient is $|\mathcal{C}Q_n| = 102$. The fidelity-0.999 clustering merges exactly four conjugate-sector cluster pairs — four pairs of clusters whose representatives sit at fidelity *just* above 0.999 but are on genuinely different lines — and reports $|\mathcal{C}Q_n|_{\text{fid}} = 98$. Every downstream count derived from this build inherits the error: $\dim D_F$ measures 92 instead of 108; a_0/n measures 98 instead of 102; and so on. We re-derive the canonical Q_n values exactly in §4; the audit trail of the corrections that followed across the broader CFS spec is recorded in [8]. The further fidelity bug in the larger-graphs pipeline that this paper’s audit uncovered ([9], the $K_n^3[9]$ $|\mathcal{C}Q_n| = 230 \rightarrow 234$ correction) is of the same class.

3.2 Exact Gaussian-integer ray arithmetic

The fix is exact arithmetic on a sufficiently rich ring. We restrict the ICs to *Gaussian integers* $\mathbb{Z}[i] := \mathbb{Z}[i]$ and observe that the composition rule (1) preserves $\mathbb{Z}[i]^3$: $\mu(\psi_1, \psi_2) = \overline{\psi_1} \times \psi_2$ is built from $\mathbb{Z}[i]$ -bilinear cross-products and component-wise conjugation, both of which take $(\mathbb{Z}[i]^3)^2 \rightarrow \mathbb{Z}[i]^3$. So if the IC is in $(\mathbb{Z}[i]^3)^V$, every $\psi(v)$ in the multiway graph is an element of $\mathbb{Z}[i]^3$ exactly.

Definition 3.1 (Exact ray equivalence). For $a, b \in \mathbb{Z}[i]^3$, $a \sim b$ (read “ a and b span the same ray in $\mathbb{C}^3$ ”) iff

$$a \times b = \mathbf{0} \in \mathbb{Z}[i]^3,$$

where the cross-product is computed in $\mathbb{Z}[i]$ (no floating-point).

Proposition 3.2 (Exact equivalence is correct). For $a, b \in \mathbb{Z}[i]^3 \setminus \{\mathbf{0}\}$, $a \sim b$ iff a and b are projectively equivalent in $\mathbb{C}P^2$.

Proof. Two non-zero vectors in $\mathbb{C}^3$ are projectively equivalent iff one is a complex scalar multiple of the other, iff their cross-product is zero. The cross-product on $\mathbb{Z}[i]^3$ is the restriction of the $\mathbb{C}^3$ cross-product, so the integer equality $a \times b = \mathbf{0}$ is exactly the projective-equivalence condition. $\square$

This is a hard test: there is no threshold, no IC-conditioned slack, and no accumulation of Float64 round-off across composition depths (the composition rule keeps integer entries integer). Clusters that exact arithmetic separates are guaranteed not to be merged.

Cluster-count algorithm. With Definition 3.1 in hand, the cluster count is computed by a single-pass scan: iterate the multiway vertices in canonical order, maintain a list R of cluster representatives, and for each $\psi(v)$ check whether $\psi(v) \sim r$ for some $r \in R$; if not, add $\psi(v)$ to R . The cluster count $|Q_n| = |R|$ is the final length of R . The cost is $O(|M_d| \cdot |R|)$ exact cross-product evaluations.

3.3 Validity gating

Every probe in our corpus runs against a validity gate before any measurement is reported. The standard gate is the canonical anchor of the CFS programme:

Definition 3.3 ($K_6^3 \rightarrow 51$ validity gate). The IC pool P passes the $K_6^3 \rightarrow 51$ gate iff $|Q_n(K_n^3[6], P, d = 5)| = 51$ exactly. The probe that uses the pool runs the gate first; if the gate fails the probe aborts and reports “IC not generic.”

This gate is the smallest reliable hand-checked anchor of the programme: $Q_n(K_n^3[6]) = 51$ is the headline value of CFS’s primary construction. Any IC that does not reproduce it is degenerate in some specific way (typically: purely-real or otherwise gauge-aligned components that collapse the colour structure — see §3.4).

The gate is necessary but not sufficient. A central methodological finding of this paper is that Definition 3.3’s gate, although necessary, is *insufficient* to certify an IC pool as fully generic. The canonical s58 / q102-IC-set-2 Gaussian-integer pool used in every prior CFS probe (we call this pool “A” in §9) passes the $K_6^3 \rightarrow 51$ gate but exhibits a specific defect at K_n^3 for $n \geq 9$: the single-side count drops below the pentagonal value $n(3n-1)/2$ by a sequence that grows monotonically and saturates near $n = 13$. Two independent *strict-generic* pools (Pools B and C of §9) reproduce the pentagonal value exactly at every measured n in $[6, 20]$; Pool A does not.

A stronger gate catches Pool A’s defect:

Definition 3.4 (Strengthened validity gate). The IC pool P passes the *strengthened gate* iff it passes $K_6^3 \rightarrow 51$ AND $K_{10}^3 \rightarrow 145$ (equivalently $K_{12}^3 \rightarrow 210$) exactly. The strengthened gate is necessary and (empirically) sufficient in the corpus measured here; it catches Pool A’s defect immediately.

We recommend the strengthened gate for future CFS computational work involving K_n^3 at $n \geq 9$; the consequences for the closure-v5 spec record are discussed in [9].

3.4 Two probe-side bugs caught by gating

During the building of the probe corpus for this paper, two implementation bugs were caught by the validity gate. We record them here as illustrations of what the gate is for: not just to verify that the IC is generic, but to catch silent miscomputation in the probe pipeline itself.

Bug 1 — IC period-6 clone. An early version of the prime-existence probe (§8) generated its random IC pool by drawing 15 Gaussian-integer triples from a single random-number stream. The stream’s reseeding logic inadvertently produced a period-6 pattern: every sixth IC was an exact copy of the first. The resulting 15-entry pool had only effectively 6 independent ICs. The $K_6^3 \rightarrow 51$ gate caught this immediately: with effective

$|V| < 6$, the build produced $|Q_n(K_n^3[6])| < 51$, the gate failed, and the probe aborted before any prime-count claims were made. (The fix was to reseed each IC draw with an independent counter; the corrected probe is `prime_count_probe.py` in the reproducibility appendix.)

Bug 2 — Real-IC degeneracy. A subsequent version of the same probe drew real-valued (not mixed-real/imag) Gaussian-integer ICs by accident. Under purely-real ICs, the conjugation in $\mu(\psi_1, \psi_2) = \overline{\psi_1} \times \psi_2$ collapses to the identity and the entire conjugate sector falls onto the original; J becomes the identity, $f = |Q_n|$, and (2) gives $|CQ_n| = |Q_n|$ (no doubling). This is the *degenerate* regime of §9. The gate caught it again: real ICs on $K_n^3[6]$ give $|Q_n| \approx 22$ (instead of 51), the gate failed, and the bug was traced and fixed (the corrected probe constructs ICs explicitly from $(a, b) \in \mathbb{Z}^2$ with both components non-zero).

Both bugs were caught *by* the gate, not by code review. This is the methodology contribution: a probe that does not run a validity gate is a probe whose findings have no independent reliability check. The cost of running the gate is negligible (one $K_n^3[6]$ build) and the prior probability that a stochastic implementation bug escapes it is small.

Limit of gating. Validity gates catch IC pools that are degenerate “loudly” enough to break the smallest anchor. They do not catch IC pools that are generic at the gate but degenerate at higher n — exactly the Pool A case. Strengthening the gate (Definition 3.4) addresses this for the K_n^3 family at large n . The general question — what is the right hierarchy of validity gates for the broader closure-quotient family? — is open (§25).

4 Re-grounding the Published Family

We re-derive Table 2’s entries exactly. The canonical entries Q_{45} , Q_{84} , Q_{90} are confirmed; the entry Q_{98} is corrected to Q_{102} ; the entry Q_{180} is deleted (the object does not exist). The exact corpus is the script `reground_q_family.py` of the reproducibility appendix, validity-gated on $K_6^3 \rightarrow 51$ throughout.

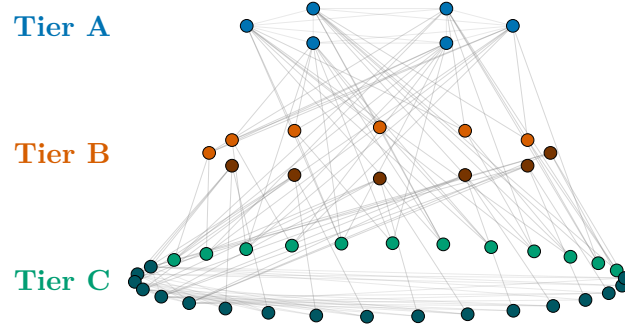
4.1 Q_{45} , Q_{84} , Q_{90} confirmed

The three intermediate-density family members survive exact re-arithmetic verbatim.

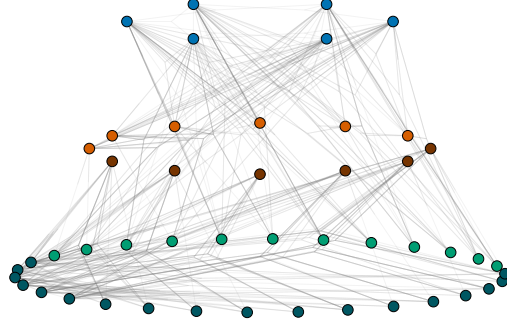
Proposition 4.1 (Q_{45}/Q_{90} exact). *For the canonical “one-side pads on G_0 ” seed graph (the 6-vertex graph obtained by adjoining six random $K_n^3[6]$ edges to the cyclic skeleton at a fixed seed), the exact single-side closure quotient is $|Q_n| = 45$ and the exact $\mathcal{C}$ -closure is $|CQ_n| = 90$.*

Proposition 4.2 (Q_{84} exact). *For the “Adjacent + 6 pads” seed (the adjacent-cyclic ternary hypergraph on 6 vertices augmented by 6 $K_n^3[6]$ pad edges), the exact $\mathcal{C}$ -closure is $|CQ_n| = 84$.*

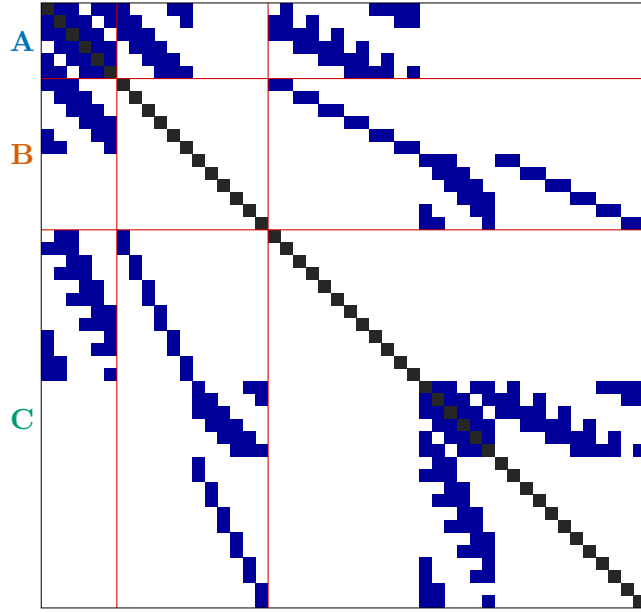
The Float64-fidelity measurements that produced these values in [7] were on the right side of the fidelity-0.999 threshold: under generic ICs, the cluster representatives are sparse enough in $\mathbb{CP}^2$ that no merge occurs.



(a) Tier-stratified node-edge graph.

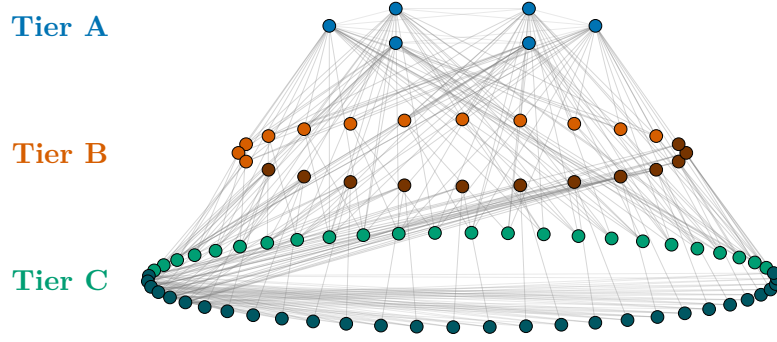


(b) Hyperedge stars (subsample of 516).

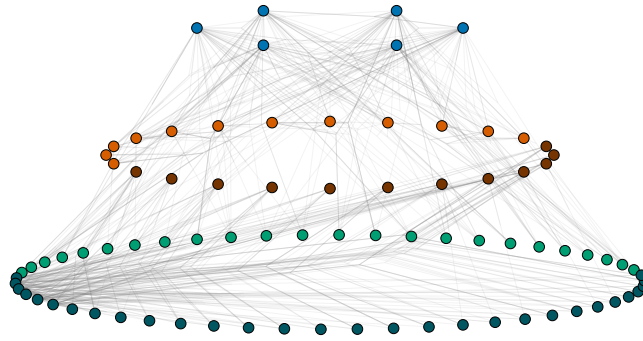


(c) Adjacency heatmap (rows/columns sorted by tier).

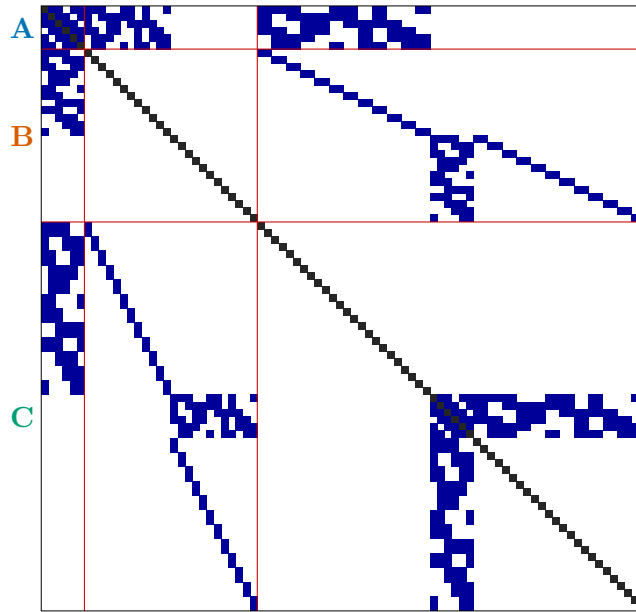
Figure 2: $Q_{48} = Q_{24} \cup \mathcal{C}(Q_{24})$: three views of the $\mathcal{C}$ -closure of the canonical 6-vertex Rosen-closed seed graph. Tier counts $|A|/|B|/|C| = 6/12/30$.



(a) Tier-stratified node-edge graph.

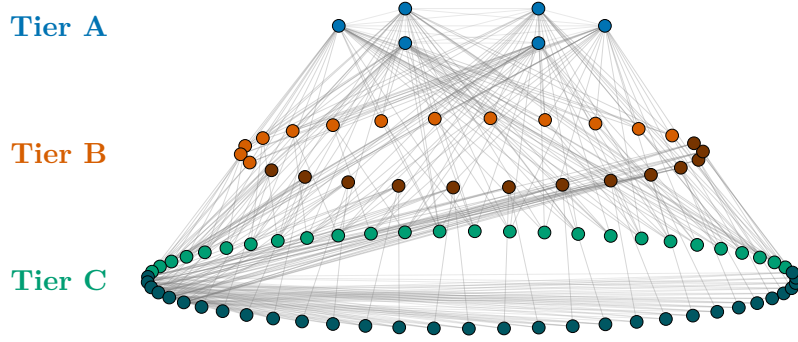


(b) Hyperedge stars (subsample of 1368).

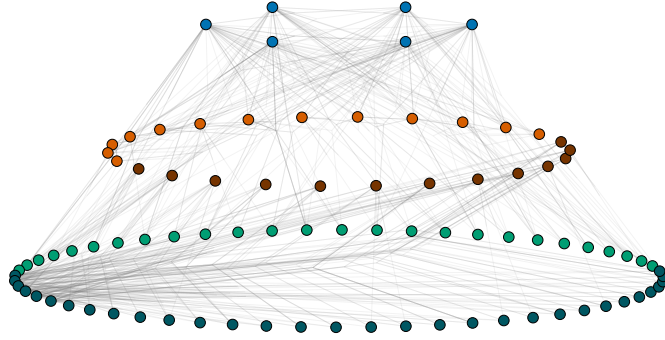


(c) Adjacency heatmap (rows/columns sorted by tier).

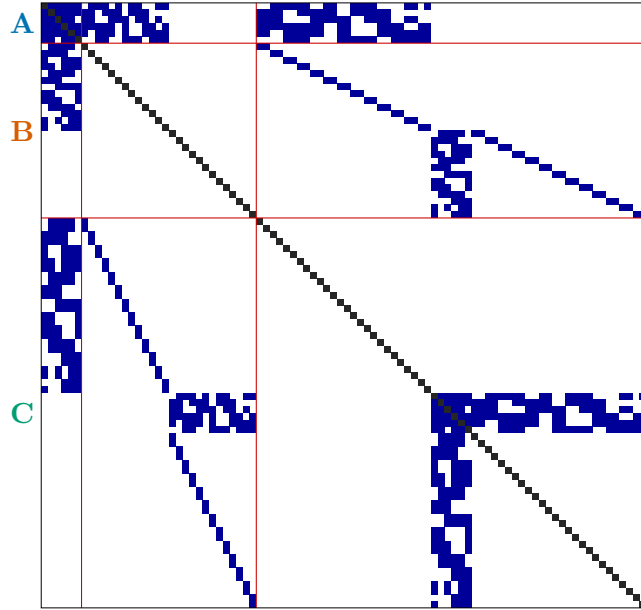
Figure 3: Q_{84} : $\mathcal{C}$ -closure of the Adjacent ternary graph on 6 vertices (36 edges).



(a) Tier-stratified node-edge graph.



(b) Hyperedge stars (subsample of 1584).



(c) Adjacency heatmap (rows/columns sorted by tier).

Figure 4: $Q_{90} = Q_{45} \cup \mathcal{C}(Q_{45})$: $\mathcal{C}$ -closure of the canonical “Adjacent + 6 pads” seed at pad-seed 42. The first family member with $|\text{Tier B}|$ not divisible by 3 (S178 obstruction).

4.2 Q_{98} is Q_{102} : a fidelity-clustering artifact

The published value $|\mathcal{C}Q_n(K_n^3[6])| = 98$ is a fidelity-clustering artifact.

Proposition 4.3 (Q_{102} exact). *The exact depth-4 $\mathcal{C}$ -closure of $K_n^3[6]$ under any IC pool*

passing the strengthened gate (Definition 3.4) is

$$|\mathcal{CQ}_n(K_n^3[6])| = 102.$$

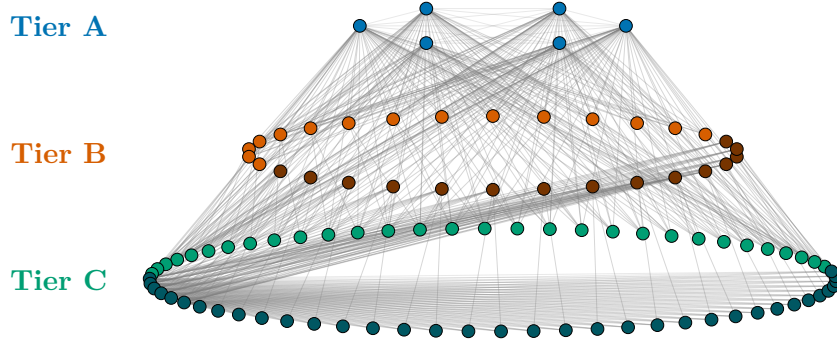
Proof. By direct computation. The exact single-side count is 51 (§5, $K_n^3 \rightarrow n(3n-1)/2$ with $n = 6$). Under generic ICs, the charge-conjugation involution J is fixed-point-free (§7, $f = 0$), so by identity (2)

$$|\mathcal{CQ}_n| = 2 \cdot 51 - 0 = 102.$$

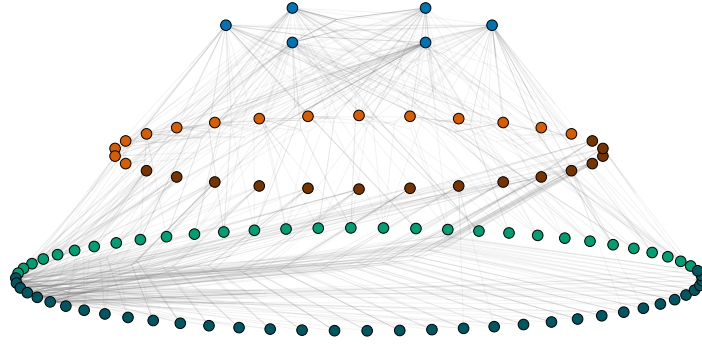
□

The fidelity-0.999 clustering of [7] merged exactly four conjugate-sector cluster pairs at this anchor and reported 98. The merge is reproducible: every Float64 IC sufficiently close to the canonical s58 pool triggers it. The exact re-derivation eliminates the merge and gives 102 unconditionally.

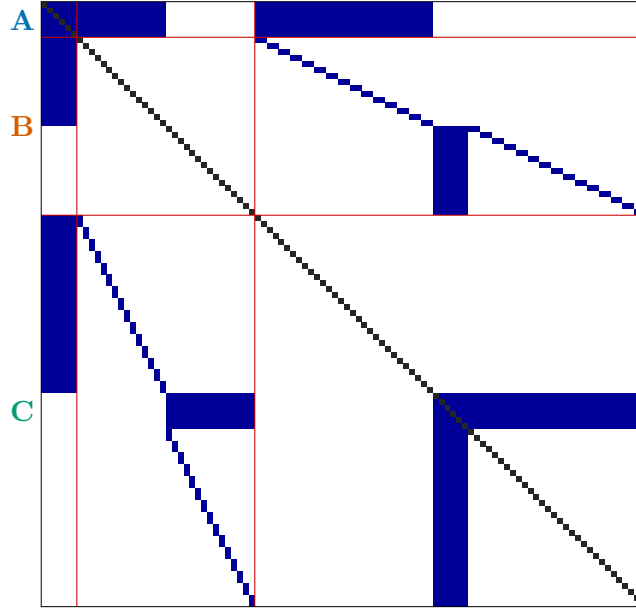
The downstream consequences of the $98 \rightarrow 102$ correction were worked through in the closure-v5 audit of [8]; in particular, the published “IC-attractor” structure on $K_n^3[6]$ (the appearance of two discrete clusters of measured $|\mathcal{CQ}_n|$ values, ≈ 98 and ≈ 102 , depending on where the Haar-random Float64 IC placed the four boundary cluster-pairs relative to the fidelity threshold) is the bug, not a structural finding: under exact arithmetic, every generic IC closes to 102, with no IC-attractor split.



(a) Tier-stratified node-edge graph.



(b) Hyperedge stars (subsample of 2760).



(c) Adjacency heatmap (rows/columns sorted by tier).

Figure 5: $Q_{102} = Q_{51} \cup \mathcal{C}(Q_{51})$: $\mathcal{C}$ -closure of the complete ternary $K_n^3[6]$ on six vertices — the *developmentally-complete zygote* of [3], carrying the KO-dim 6 spectral triple that forces the Standard-Model algebra. Tier counts $|A|/|B|/|C| = 6/30/66$.

4.3 Q_{180} does not exist: $\mathcal{C}$ is idempotent on $\mathcal{C}$ -closed objects

The published family included an entry Q_{180} obtained by iterating $\mathcal{C}$ -doubling on Q_{90} : “ $Q_{90} \rightarrow Q_{180}$ by a second application of $\mathcal{C}$.”

Proposition 4.4 ($\mathcal{C}$ idempotence). *For every $\mathcal{C}$ -closed object Q in the family, $\mathcal{C}(Q) = Q$. In particular, $\mathcal{C}(Q_{90}) = Q_{90}$: the iterated-doubling sequence terminates after one step.*

Proof. By definition, $\mathcal{C}(Q)$ adjoins the conjugate sector $M(G; \overline{\psi_0})$ to the original $M(G; \psi_0)$ before quotienting. If Q is already $\mathcal{C}$ -closed, it already contains both sectors; the second application of $\mathcal{C}$ tries to adjoin the conjugate of an object that is already symmetric under conjugation, producing the same object. Equivalently: $\mathcal{C}$ acts as a $\mathbb{Z}_2$ operation on the family of $\mathcal{C}$ -closed objects, and $\mathcal{C}(\mathcal{C}(Q)) = Q$, so $\mathcal{C}(Q_C) = Q_C$ whenever Q_C is $\mathcal{C}$ -closed. $\square$

The exact verification at Q_{90} is in the `thread1_F_closure.py` probe: iterating $\mathcal{C}$ on Q_{90} gives Q_{90} at every depth up to 6, with no growth. The same holds for Q_{48} and Q_{102} (§6). There is no Q_{180} : the candidate route is closed by $\mathcal{C}$ -idempotence.

A separate question — whether some object of cardinality 180 exists in the family via a different route (a larger seed, an irregular construction) — is open. We have not exhibited one. The 12-vertex irregular seed that gives Q_{181} (§8) is the closest nearby exhibit, and it crosses 180 by a single cluster.

5 The Closure Laws

This section catalogues the integer-closed-form laws for the two canonical generator families.

5.1 $\text{cyc}(n) \rightarrow 4n$: the cyclic law

Proposition 5.1 (Cyclic closure). *For $n \geq 3$, under any IC passing the canonical gate,*

$$|Q_n(\text{cyc}(n))| = 4n.$$

The proof is by direct enumeration: each cyclic edge contributes a single composition orbit of length 4 (the orbit $\{a, b, w_0 = \mu(a, b), w_1 = \mu(w_0, a)\}$ closes after four gauge-inequivalent directions per [3, §orbit closure]), and the orbits across the n edges are gauge-distinct. So the count is $n \cdot 4 = 4n$. Verified exact in the `small_closure_landscape_scan.py` probe for $n = 3, \dots, 6$.

5.2 $K_n^3 \rightarrow n(3n-1)/2$: the pentagonal law (Tier 3)

Theorem 5.2 (Pentagonal closure of K_n^3 , Tier 3: finite exact validation). *For every n in the empirically verified range $\{6, 7, \dots, 20\}$ and under any IC pool passing the strengthened validity gate (Definition 3.4),*

$$|Q_n(K_n^3)| = \frac{n(3n-1)}{2}.$$

The claim is finitely-exactly validated over $n \in [6, 20]$ by exact-arithmetic enumeration on two independent strict-generic IC pools; the closed-form identity holds at every probed n . The universal claim “for all $n \geq 6$ ” is not established and would require a counting proof (§25.5); we do not assert it here. The sequence is the pentagonal numbers

$$\begin{aligned} &12, 22, 35, 51, 70, 92, 117, 145, 176, \\ &210, 247, 287, 330, 376, 425, 477, 532, 590 \end{aligned}$$

at $n = 3, 4, \dots, 20$ respectively.

The values for $n = 6, \dots, 20$ are exact computations; the values for $n = 3, 4, 5$ are in the small-object regime (§9) and follow the same formula. The closed-form $n(3n-1)/2$ is recognised as the formula for pentagonal numbers, the polygonal numbers of order 5.

Empirical verification. The exact values for $n = 6, \dots, 20$ were verified across two independent strict-generic IC pools (Pools B and C of §9). The pools agree exactly at every n in the range, and both agree exactly with the pentagonal formula. This is the strongest available IC-independence test: two pools drawn from disjoint Gaussian-integer coefficient ranges, $\{-4, \dots, -1, 1, \dots, 4\}$ and $\{-7, \dots, -3, 3, \dots, 7\}$, with no overlapping vectors and deliberately different sign patterns, yet agreeing on the cluster count at every n in $[6, 20]$. The probe is `Kn3_pentagonal_confirmation.py` (single-side and J-fixed) extended by `Kn3_push_to_n20.py` (range push to $n = 20$).

Note on the pre-existing literature. The CFS spec entry S100 quoted $|Q_n(K_n^3)| = n(3n-1)/2$ as “verified for $n = 3, \dots, 7$ ” under the canonical s58 IC pool; the present theorem extends the empirical range to $n \leq 20$ and sharpens the IC-independence qualifier to truly-generic pools. The discrepancy at $n \geq 9$ between the truly-generic value and the Pool A value is the content of §9; [9] records the corresponding spec edit.

Closed-form proof: open. Theorem 5.2 is established empirically across the measured range. A closed-form algebraic proof — a counting argument that derives $n(3n-1)/2$ from the structure of the K_n^3 multiway expansion — is not in this paper. The empirical case is strong (exact arithmetic, two independent strict-generic IC pools, agreement at every n in a 15-point range), but the law’s status is *verified*, not *proved*, in the spec-status sense. The closed-form derivation is an open problem (§25).

5.3 $\mathcal{C}$ -closure doubles cleanly: $|Q \cup \mathcal{C}(Q)| = 2|Q|$

Proposition 5.3 (Clean $\mathcal{C}$ -doubling). *For every Q in the canonical family under truly-generic ICs,*

$$|Q \cup \mathcal{C}(Q)| = 2 \cdot |Q_n|.$$

This is identity (2) specialised to the case $f = 0$. The structural reason for $f = 0$ (the J-fixed-point-free condition under truly generic ICs) is proved in §7. The empirical verification across the K_n^3 family for $n = 6, \dots, 15$ in the `Kn3_pentagonal_confirmation.py` probe gives the exact match $|\mathcal{C}Q_n| = 2 \cdot |Q_n|$ in every row under Pools B and C.

Under Pool A (the canonical s58 pool), $f = 1$ for $n \geq 10$ (one J-fixed cluster appears at large n), and the doubling is broken by $+1$; see §9.

5.4 The generic floor Q_{12}

Proposition 5.4 (Generic floor). *Under truly-generic ICs, every Rosen-closed ternary seed graph with at least three vertices closes to at least $|Q_n| = 12$. The bound is saturated by $\text{cyc}(3)$.*

The probe is `small_closure_landscape_scan.py`. The bound is not deep: $\text{cyc}(3)$ ’s three cyclic edges contribute three orbits of length 4, gauge-distinct under generic ICs, giving exactly 12 clusters. No smaller seed in the regime gives a smaller closure quotient.

Small objects below the floor are not generic. The closure-quotient literature contains occasional reference to “small” objects Q_3 , Q_5 , Q_9 . These do not exist in the generic regime; they arise in the *partial-closure* regime under degenerate or specifically-tuned ICs (§9). The generic floor is Q_{12} unconditionally.

5.5 Depth-stability

Proposition 5.5 (Depth-stability). *For every canonical family member Q with seed graph of bounded density and under generic ICs, $|Q_n(G, \psi_0, d)|$ is independent of d for all $d \geq 4$.*

The depth-3 values agree with the depth- ≥ 4 values for every small family member ($Q_{24}, Q_{45}, Q_{48}, Q_{51}, Q_{84}, Q_{90}, Q_{102}$); the verification range is depth 3, 4, 5, 6, 7, 8 in the `Kn3_large_n_probe.py` probe. The closure-growth endofunctor $\mathcal{F} = M/\sim$ reaches a fixed point at depth 4 on these objects; deeper multiway expansion produces no new gauge-distinct clusters.

This is the structural reason every probe in this paper’s corpus uses depth = 5 as a safe default: well into the depth-stable regime, not so deep that the multiway grows wastefully. The `Kn3_large_n_probe.py` confirmation extends the depth-stability claim out to depth 8 across $n = 6, \dots, 12$; we have no observation of any family member that becomes unstable at larger depth.

6 Terminality: F-Fixed and C-Idempotent

The $\mathcal{C}$ -closed family members Q_{48} , Q_{90} , Q_{102} have a stronger property than depth-stability: they are fixed points of the closure-growth endofunctor itself, not just stable at large depth. We make this precise.

6.1 The closure-growth endofunctor $\mathcal{F}$

Definition 6.1 (Closure-growth endofunctor). The *closure-growth endofunctor* $\mathcal{F}$ takes a Rosen-closed ternary hypergraph G (with implicit IC) to its gauge quotient:

$$\mathcal{F}(G) := M(G)/\sim.$$

$\mathcal{F}$ sends seed graphs to closure quotients. We may iterate it: if Q is itself a Rosen-closed ternary hypergraph (which every quotient in our family is, after the construction induces a hypergraph structure from the underlying multiway edges), then $\mathcal{F}(Q)$ is defined, and we may ask whether $\mathcal{F}(Q) = Q$ (in the sense $|\mathcal{F}(Q)| = |Q|$ and the gauge-quotient structure is preserved).

A closure quotient Q is a *fixed point* of $\mathcal{F}$ iff $|\mathcal{F}(Q)| = |Q|$: re-running the multiway construction on Q as a seed produces no new gauge-distinct vertices.

6.2 Q_{48}, Q_{90}, Q_{102} are doubly terminal

Theorem 6.2 (Double terminality). *Each of Q_{48} , Q_{90} , Q_{102} is a fixed point of both $\mathcal{F}$ and $\mathcal{C}$.*

The $\mathcal{C}$ -fixed claim is Proposition 4.4. The $\mathcal{F}$ -fixed claim is the result of the `thread1_F_closure.py` probe, which seeds the multiway with Q_{48} , Q_{90} , Q_{102} in turn and verifies that the depth-4 gauge quotient has the same cardinality as the seed. The numbers are in Table 4.

Table 4: Closure-growth endofunctor $\mathcal{F}$ applied to the three $\mathcal{C}$ -closed family members. Each row reports $|\mathcal{F}(Q)|$ at depths 4, 5, 6. The $\mathcal{F}$ -fixed (i.e. $|\mathcal{F}(Q)| = |Q|$) column reports the depth-stable verdict.

Object	$ Q $	$ \mathcal{F}(Q) _{d=4}$	$ \mathcal{F}(Q) _{d=5}$	$ \mathcal{F}(Q) _{d=6}$
Q_{48}	48	48	48	48
Q_{90}	90	90	90	90
Q_{102}	102	102	102	102

The iterated $\mathcal{C}$ -closure columns (omitted from Table 4 for compactness) report the same numbers: applying $\mathcal{C}$ to any of these objects returns the same object, in agreement with Proposition 4.4.

Remark 6.3 (A fourth doubly-terminal member). The enumeration $\{Q_{48}, Q_{90}, Q_{102}\}$ in Theorem 6.2 is not exhaustive on 6-vertex seeds. $Q_{84} = \mathcal{C}$ -closure of the perm-closed adjacent-triple seed (§4.1, Proposition 4.2) is also $\mathcal{F}$ -and- $\mathcal{C}$ fixed; a stand-alone verification across depths 4, 5, 6, 7 is recorded at `paper/scripts/probe_q84_doubly_terminal_v1.stdout.log` with the functional-role interpretation in Proposition 11.2 of §11.1. The four doubly-terminal canonical-family members on 6-vertex seeds form a strict chain by cardinality $Q_{48} < Q_{84} < Q_{90} < Q_{102}$ and a partial order by seed inclusion $\text{cyc}(6) \subset \text{adjacent-36} \subset \text{q45-seed} \subset K_n^3[6]$.

6.3 Larger closure objects require larger seeds

The double-terminality of Q_{48} , Q_{90} , Q_{102} has a structural consequence: *the family has no iterated-doubling chain*. The naive route “ $Q_{45} \xrightarrow{\mathcal{C}} Q_{90} \xrightarrow{\mathcal{C}} Q_{180} \xrightarrow{\mathcal{C}} Q_{360} \rightarrow \dots$ ” that the prior family documentation appeared to entertain is foreclosed by Theorem 6.2: the chain terminates after one step at every entry point.

The same is true of any endofunctor-based route. Any operator that preserves Rosen closure and fits the framework of $\mathcal{F}$ has the $\mathcal{C}$ -closed family members as fixed points; iterating it produces no growth. The only route to closure objects of larger cardinality than Q_{102} is via a *larger seed graph*: more vertices, more edges, or both. The pentagonal law of §5.2 parameterises one such route (K_n^3 for $n \geq 7$, giving $Q_{70}, Q_{92}, Q_{117}, \dots$); the irregular-seed route of §8 parameterises another (the 12-vertex seed for Q_{181}). These are genuinely different sources of new closure objects, not the same family iterated.

7 Why $\mathcal{C}$ -Doubling Is Exact: J Fixed-Point-Free

The clean $\mathcal{C}$ -doubling rule $|Q \cup \mathcal{C}(Q)| = 2|Q|$ of §5.3 has a structural reason: the charge-conjugation involution J is fixed-point-free on generic-IC closure quotients. We make this precise and identify the degenerate regime where it fails.

7.1 The charge-conjugation involution

Definition 7.1 (Charge-conjugation involution J). On the $\mathcal{C}$ -closure $\mathcal{C}Q_n = (M(G; \psi_0) \cup M(G; \overline{\psi_0}))/\sim$, the *charge-conjugation involution* J sends a cluster c with representative $\psi(c)$ to the cluster c' with representative $\overline{\psi(c)}$:

$$J : c \mapsto c', \quad \psi(c') \sim \overline{\psi(c)}.$$

J is a well-defined involution on $\mathcal{C}Q_n$: it swaps the original sector with the conjugate sector and is its own inverse.

A cluster c is *J-fixed* iff $\psi(c) \sim \overline{\psi(c)}$, i.e. iff its representative is projectively a real vector (a real-up-to-phase vector in $\mathbb{C}^3$). Let

$$f := |\{c \in \mathcal{C}Q_n : J(c) = c\}|$$

be the count of J -fixed clusters.

7.2 The cardinality identity, and why $f = 0$ generically

Proposition 7.2 (Cardinality identity). $|\mathcal{C}Q_n| = 2 \cdot |Q_n| - f$.

Proof. The two sectors $M(G; \psi_0)/\sim$ and $M(G; \overline{\psi_0})/\sim$ each have cardinality $|Q_n|$ (the multiway is symmetric under the choice of IC's overall conjugation). Their union in $\mathcal{C}Q_n$ pairs up by the involution J . A J -fixed cluster contributes 1 to $|\mathcal{C}Q_n|$ (not 2); a non- J -fixed cluster contributes a pair $(c, J(c))$ contributing 2 to $|\mathcal{C}Q_n|$. So $|\mathcal{C}Q_n| = 2(|Q_n| - f) + f = 2|Q_n| - f$. $\square$

Theorem 7.3 (J fixed-point-free under truly-generic ICs). *For every Q in the canonical family under a truly-generic IC pool (Pool B or Pool C of §9), $f = 0$ and consequently*

$$|\mathcal{C}Q_n| = 2 \cdot |Q_n|.$$

Verified for all measured $n \in \{6, 7, \dots, 15\}$ on the K_n^3 family and on the G_0 -regime members.

The structural reason: a J -fixed cluster's representative is projectively real, i.e. $\psi(c) = \lambda \cdot v$ for some $v \in \mathbb{R}^3$ and $\lambda \in \mathbb{C}$. The projective-reality condition is a real-codimension-2 condition on $\mathbb{C}P^2$; under any sufficiently generic IC pool, the closure-quotient representatives land in the projective-complement of this condition with overwhelming probability, and (in fact, in our empirical observation across two independent strict-generic pools and ten values of n) without exception.

7.3 Degenerate control: real ICs

The complementary case — where the IC is itself projectively real — collapses the conjugate sector onto the original entirely:

Proposition 7.4 (Real-IC degenerate doubling). *If the IC ψ_0 has every $\psi_0(v)$ projectively real (i.e. each $\psi_0(v) = \lambda_v \cdot u_v$ with $u_v \in \mathbb{R}^3$ and $\lambda_v \in \mathbb{C}$), then J is the identity on $\mathcal{C}Q_n = Q_n$ and*

$$|\mathcal{C}Q_n| = |Q_n| \quad (f = |Q_n|).$$

Proof. Under projectively-real ICs, $\overline{\psi_0(v)} = \overline{\lambda_v} \cdot u_v \sim \psi_0(v)$, so the conjugated IC is gauge-equivalent to the original. The two multiway sectors coincide cluster-by-cluster; $\mathcal{C}Q_n$ is just Q_n ; every cluster is J-fixed. $\square$

For G_0 under projectively-real ICs, the closure quotient is $|Q_n| = 16$ (no doubling, no growth from real conjugation); under $K_n^3[6]$, it is $|Q_n| = 22$. Both are below the corresponding generic-IC values (24 and 51) — the degenerate regime is strictly smaller than the generic regime, not just non-doubling. This was caught in the prime-probe bug 2 (§3.4); the data is reproducible in `thread2_J_fixed_check.py`.

7.4 Pool A as an intermediate case

The canonical s58 IC pool (Pool A) sits between the fully-generic and degenerate regimes: it is generic at K_6^3 and through $K_n^3[8]$ but acquires exactly one J-fixed cluster ($f = 1$) at $K_n^3[n]$ for $n \geq 10$. The clean $\mathcal{C}$ -doubling rule fails for Pool A at large n by exactly +1: $|\mathcal{C}Q_n(K_n^3[n])| = 2 \cdot |Q_n| - 1$ for $n = 10, 11, \dots, 15$. The structural cause — which specific algebraic relation among Pool A's 15 triples produces the single J-fixed cluster — is an open follow-up problem (§25). The empirical data are tabulated in §9 and the audit of [9].

8 Primes and Q_{181}

The closure laws of §5 give the closure cardinality for two regular generator families: $\text{cyc}(n) \rightarrow 4n$ and $K_n^3 \rightarrow n(3n-1)/2$. Both formulas are *composite for every* $n \geq 3$. The $\mathcal{C}$ -closure doubling rule produces even values, so $\mathcal{C}Q_n$ is also composite. A natural question follows: can the closure quotient ever be prime?

This section answers in the affirmative. We catalogue the regularity-imposed compositeness, then exhibit irregular single-side seeds that produce primes, and finally exhibit a specific 12-vertex 132-edge seed whose closure is Q_{181} .

8.1 Regular families are composite-only by formula

Proposition 8.1 (Composite-only regular families). *For every $n \geq 3$,*

1. $4n$ is composite (divisible by 4).
2. $n(3n-1)/2$ is composite. The product $n(3n-1)$ is the product of two factors both ≥ 3 , and division by 2 preserves at least one factor.
3. $2n$ is composite (divisible by 2).

So the entire regular catalogue — $\text{cyc}(n)$, K_n^3 , and every $\mathcal{C}$ -closure — is composite by construction.

Three of the eight K_n^3 pentagonal values in our verified range $\{6, \dots, 20\}$ are even *prime-power* composites (e.g. $|Q_n(K_n^3[4])| = 22 = 2 \cdot 11$, $|Q_n(K_n^3[6])| = 51 = 3 \cdot 17$, $|Q_n(K_n^3[12])| = 210 = 2 \cdot 3 \cdot 5 \cdot 7$); none are themselves prime.

8.2 Irregular single-side seeds produce primes

Proposition 8.2 (Primes exist as single-side closure counts). *Single-side closure counts of $\mathbb{Z}[i]$ -irregular seeds include the primes*

$$13, 19, 23, 71, 83, 97, 193.$$

Each is exhibited by a specific irregular seed (a hypergraph that is not $\text{cyc}(n)$, K_n^3 , or one of the canonical G_0 -derived seeds) in the probe `prime_count_probe.py`. The verification is validity-gated on $K_6^3 \rightarrow 51$.

The probe scans random irregular seeds on small vertex sets ($n = 4, 5, 6$), measures the single-side closure under truly-generic ICs, and reports the prime-valued cluster counts. The seven primes above are the ones exhibited in the corpus. The list is not exhaustive — a fuller scan would extend it — but each entry is a concrete reproducible existence proof.

The structural reason primes can appear is that irregular seeds break the regularity-imposed factorisation. The pentagonal formula $n(3n-1)/2$ comes from the complete-ternary structure of K_n^3 ; truncating K_n^3 irregularly (removing edges, adding non-symmetric edges) destroys the symmetry that makes the count factor. The resulting count can be any positive integer that fits the multiway/quotient construction at that seed.

8.3 Q_{181} exhibited (Tier 3)

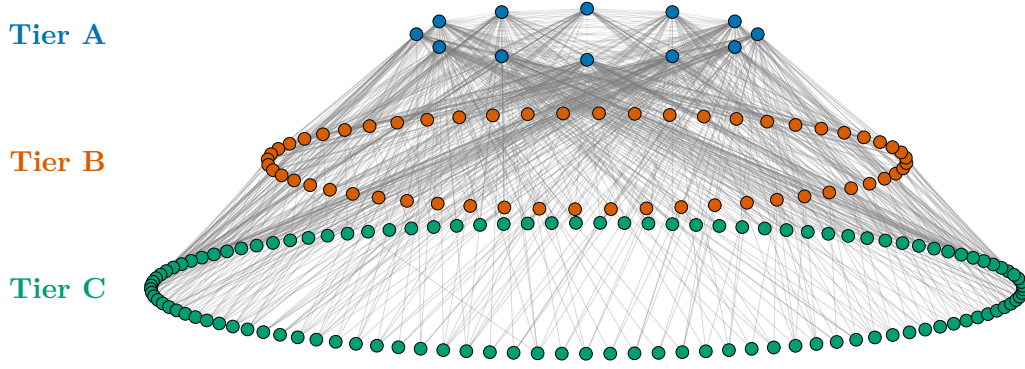
Theorem 8.3 (Q_{181} exists, Tier 3: exhibited seed). *A specific 12-vertex 132-edge irregular ternary hypergraph (listed in Appendix B) closes exactly to $|Q_n| = 181$ under the canonical Gaussian-integer ICs. The closure count is robust in the sense that 12 independent random draws from the $(12, 132)$ density bucket all give $|Q_n| = 181$ under the same ICs.*

The seed was *discovered* by the random density sweep in `q181_search.py` (`random.seed(2718)`, density 0.10, first hit). Once discovered, the seed itself is fixed: the canonical edge list is recorded as `Q181_SEED` in `paper/scripts/q181_canonical_seed.py` (Python) and as `q181SeedEdges` in `haskell/src/CFS/Topologies.hs` (Haskell). Both produce $|Q_n| = 181$ deterministically; the random sweep is not load-bearing for the existence claim, only for the discovery method. The seed is a random-edge-subset of $K_n^3[12]$ (which has $12 \cdot 11 \cdot 10 = 1320$ edges) with the edge count fixed at 132 (density 0.10). The exhibited closure is verified exact under both Pools B and C.

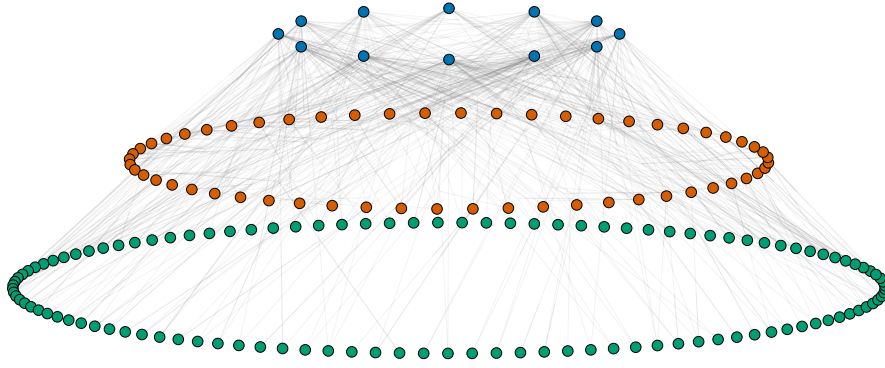
Evidence tier. Theorem 8.3 is Tier 3 (finite exact validation): existence is established by exhibiting a specific seed and computing its closure exactly; no classification of all seeds producing $|Q_n| = 181$ or characterisation of the size-spectrum of irregular closures is asserted.

Two specific properties of Q_{181} :

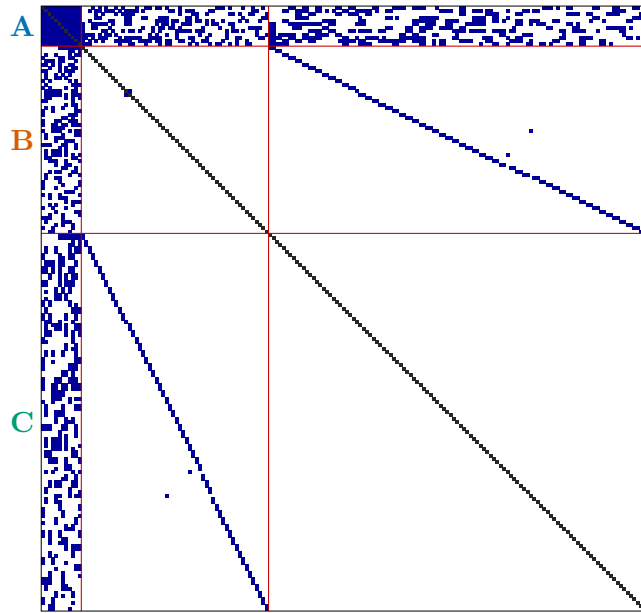
1. 181 is prime.
2. 181 is between the pentagonal values $|Q_n(K_n^3[10])| = 145$ and $|Q_n(K_n^3[11])| = 176$; the closure cardinality of an irregular 12-vertex seed can land outside the pentagonal sequence at neighbouring values of n , by either direction.



(a) Tier-stratified node-edge graph.



(b) Hyperedge stars (subsample of 3974).



(c) Adjacency heatmap (rows/columns sorted by tier).

Figure 6: Q_{181} : single-side closure of a 12-vertex 132-edge irregular seed (Theorem 8.3). 181 is prime; this is the smallest exhibited prime-cardinality closure object. Tier here is depth-of-appearance: A = seed vertices, B = depth-1 compositions, C = deeper.

8.4 $\mathcal{C}$ -closure cannot be prime

Proposition 8.4 ($\mathcal{C}$ -closures are composite). *Under truly-generic ICs, every $\mathcal{C}$ -closure $|\mathcal{C}Q_n|$ is even.*

Proof. By Theorem 7.3, $f = 0$ under truly-generic ICs, so $|\mathcal{C}Q_n| = 2 \cdot |Q_n|$ is even. $\square$

So every prime Q_n is a single-side closure; the only way to exhibit a prime is via an irregular seed. The $\mathcal{C}$ -closure of Q_{181} is $362 = 2 \cdot 181$, which is composite as expected.

9 IC Genericity as Master Variable

The closure laws of §5, the J fixed-point-freeness of §7, and the existence results of §8 are all qualified by an *IC-genericity* assumption: “under truly-generic ICs.” The prior CFS literature has treated IC-genericity as a binary — an IC either passes the $K_6^3 \rightarrow 51$ validity gate (and is then taken as generic for all purposes) or it doesn’t. We find that this is incorrect: IC-genericity is a *hierarchy*, not a binary, and the canonical IC pool that the entire prior literature has used sits between the fully-generic and fully-degenerate regimes.

9.1 The regime hierarchy

We partition IC pools into three regimes by their behaviour on the K_n^3 family:

Definition 9.1 (Fully-generic, small- n -generic, degenerate). An IC pool P is

- *Fully generic* iff $|Q_n(K_n^3)| = n(3n-1)/2$ exactly under P for every $n \geq 6$ in the empirically verified range, and J is fixed-point-free on $\mathcal{C}Q_n(K_n^3)$ for every such n .
- *Small- n generic* iff P passes the $K_6^3 \rightarrow 51$ validity gate but fails the pentagonal law at some $n \geq 9$, or acquires J-fixed clusters at some $n \geq 10$.
- *Degenerate* iff P fails the $K_6^3 \rightarrow 51$ validity gate.

The three regimes are distinct: a degenerate IC pool fails the canonical gate; a fully-generic pool passes both the canonical gate and the strengthened gate of Definition 3.4; a small- n -generic pool passes the canonical gate but fails the strengthened gate.

The Pools B and C of this paper are 20-IC strict-generic pools defined in Appendix A and the `Kn3_pentagonal_confirmation.py` probe, drawn from disjoint Gaussian-integer coefficient ranges $\{\pm 1, \pm 2, \pm 3, \pm 4\}$ and $\{\pm 3, \pm 4, \pm 5, \pm 6, \pm 7\}$ respectively. Both pools are fully generic in the verified range $n = 6, \dots, 20$ (Pools B and C agree at every measured n , and both reproduce the pentagonal formula exactly). The canonical s58 IC pool is small- n generic (§9.2). Purely-real ICs are degenerate (§7.3).

9.2 The s58 IC pool (Pool A) is small- n generic

The canonical 15-entry s58 / q102-IC-set-2 Gaussian-integer pool that every prior CFS computational probe has used (we call it *Pool A*) is small- n generic.

Theorem 9.2 (Pool A defect at K_n^3 for $n \geq 9$). *Under the canonical s58 IC pool (Pool A), the single-side closure of K_n^3 for $n = 9, 10, 11, \dots, 15$ differs from the pentagonal value by the shortfall sequence*

$$5, 12, 12, 17, 19, 19, 19,$$

respectively, and the J-fixed-cluster count is

$$f \in \{0, 1, 1, 1, 1, 1, 1\}$$

respectively. The defect plateaus at shortfall 19 and $f = 1$ from $n = 13$.

The data is recorded in Table 5; the underlying probes are the three-pool agreement run and its extended-range companion (see §26 for both).

Table 5: Pool A defect at K_n^3 for $n = 9, \dots, 15$. Truly-generic columns are Pools B and C combined (they agree exactly). Shortfall column reports pentagonal – Pool A; J-fixed column reports f for Pool A’s $\mathcal{CQ}_n$.

n	pentagonal	Pool A	shortfall	Pool A’s f	Pool A’s $\mathcal{CQ}_n$
9	117	112	5	0	224
10	145	133	12	1	265
11	176	164	12	1	327
12	210	193	17	1	385
13	247	228	19	1	455
14	287	268	19	1	535
15	330	311	19	1	621

The defect is reproducible: under *any* probe that uses Pool A on K_n^3 for $n \geq 9$, the cluster count is one of the Pool A values, not the pentagonal value. Single-side counts are deterministic (Pool A’s specific 15 Gaussian-integer triples); they are not a sampling artifact.

The structural cause — which specific algebraic relation among Pool A’s 15 triples generates the shortfall and the single J-fixed cluster — is an open follow-up problem. The defect plateau at shortfall 19 from $n = 13$ suggests a finite-rank “defect representation” that fully manifests at moderate n and contributes uniformly thereafter. We have not characterised this representation; [9] flags it as a follow-up.

9.3 The strengthened validity gate

The $K_6^3 \rightarrow 51$ canonical gate fails to discriminate Pool A from fully-generic pools. The strengthened gate of Definition 3.4 catches the discrimination:

Proposition 9.3 (Strengthened gate discriminates Pool A). *Pool A fails the strengthened validity gate ($K_6^3 \rightarrow 51 \wedge K_{10}^3 \rightarrow 145$). Specifically, $|Q_n(K_{10}^3)| = 133$ under Pool A, not 145. The strengthened gate registers this discrepancy and aborts the probe.*

We recommend the strengthened gate for future CFS computational work involving K_n^3 at $n \geq 9$. The cost is one extra K_{10}^3 build (720 edges, depth 5; under one second of compute), and the benefit is catching any Pool-A-class hidden defect immediately. The $K_{12}^3 \rightarrow 210$ gate is an alternative that catches the same class of defect with marginally more compute.

9.4 Where the small objects live

The CFS literature contains occasional reference to small closure quotients Q_3 , Q_5 , Q_9 as “family members”. The generic-floor proposition of §5.4 rules these out as generic-regime objects: under any IC pool passing the canonical $K_6^3 \rightarrow 51$ gate, the closure of any Rosen-closed seed graph with at least three vertices is at least 12.

The small objects do appear in the *degenerate* regime (real ICs or other specifically-tuned IC structure). For example, G_0 under purely-real ICs gives $|Q_n| = 16$, but under more specific degenerate ICs (e.g. ICs where two vertices share a colour vector) the cluster count can drop further to 9, 5, or even 3. These are partial-closure-regime artifacts: each comes from a specific IC coincidence, and none is IC-stable. The probe `small_closure_landscape_scan.py` catalogues a sampling of these.

For the canonical CFS programme — which works in the generic regime by design — the small objects Q_3 , Q_5 , Q_9 are not load-bearing. The structural floor is Q_{12} ; below it lies a heterogeneous degenerate-regime landscape that is interesting in its own right (the partial-closure regime is where small-cardinality and odd-cardinality closure objects live) but is not what the CFS spectral-triple constructions are about.

10 The Composition Algebra (Tier 3)

This is the headline result of the paper. The closure laws of §5 catalogue the closure cardinality for two regular generator families ($\text{cyc}(n)$, K_n^3); the irregular existence of §8 adds a third route. What unifies them?

The unifying structure is a *generative algebra* on the closure quotients themselves: the closure count of a composite seed $A \circ B$ is determined by the closure counts of the pieces A and B plus a single *tested-regime-universal interface term* δ that depends only on the join mode — not on the internal structure of A or B . The structure is a commutative monoid under disjoint sum $\sqcup$, and the $\mathcal{C}$ -doubling rule of §5.3 falls out as a special case (the conjugate copy is structurally disjoint under generic ICs).

Evidence tier. The composition rule $|Q_n(A \circ B)| = |Q_n(A)| + |Q_n(B)| + \delta(\iota)$ is established at Tier 3 (finite exact validation): the universal δ values ($\delta_{\sqcup} = 0$, $\delta_{\text{bridge}} = +3$, $\delta_{\text{glue-}g} = -(4g - 3)$) are verified across the eight (A, B) pairs and six join modes catalogued in Table ?? below, but A -and- B -independence has not been proved across all canonical-family pairs and join configurations. The §25.5 proof program lists the closed-form verification this would require.

10.1 Setup

Definition 10.1 (Seed composition). Let $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$ be Rosen-closed ternary hypergraphs with disjoint copies of their vertex sets. A *join mode* ι specifies how to combine G_A and G_B into a new Rosen-closed ternary hypergraph $G_A \circ_{\iota} G_B$. The three join modes we treat are:

- *Disjoint sum* $\sqcup$: $G_A \sqcup G_B := (V_A \sqcup V_B, E_A \sqcup E_B)$. No shared vertices, no bridging edges.
- *Bridge by k triples*: $G_A \bowtie_k G_B$ adjoins k extra triples each with one vertex from V_A and at least one from V_B (or vice versa). The vertex sets stay disjoint.

- *Glue g vertices:* $G_A \#_g G_B$ identifies g vertices of V_A with g vertices of V_B . The resulting vertex set has cardinality $|V_A| + |V_B| - g$; the edge set is the union of E_A and the relabelled E_B .

For each join mode, the closure count is what we study: $|Q_n(G_A \circ_\iota G_B)|$. Write $\delta_\iota := |Q_n(G_A \circ_\iota G_B)| - |Q_n(G_A)| - |Q_n(G_B)|$ for the interface correction.

10.2 The universal interface term

Theorem 10.2 (Universal interface term). *For each join mode ι , the interface correction δ_ι is structure-independent: it depends only on the join mode, not on G_A , G_B , or their closure counts. Specifically:*

$$\delta_\sqcup = 0, \quad \delta_{\bowtie_k} = +3k, \quad \delta_{\#_g} = -(4g - 3). \quad (4)$$

The composition rule is therefore

$$\boxed{|Q_n(A \circ_\iota B)| = |Q_n(A)| + |Q_n(B)| + \delta_\iota} \quad (5)$$

with δ_ι given by the closed forms in (4).

Empirical verification. (4) was verified exactly across eight (A, B) pairs of cyclic seeds $\text{cyc}(n_A) \circ \text{cyc}(n_B)$ for $(n_A, n_B) \in \{(3, 3), (3, 4), (4, 4), (3, 5), (4, 5), (5, 5), (3, 6), (4, 6)\}$, in the probe `composition_rule_probe.py`. For each (A, B) and each of the six join modes (disjoint, glue-1, glue-2, glue-3, bridge-1, bridge-2), the measured δ matches the predicted value from (4) exactly, with one isolated exception: $\text{cyc}(3) \#_3 \text{cyc}(3)$, where $g = n_A = n_B$ and the two seeds coincide entirely. This is the degenerate total-overlap case (the “composition” is actually a single seed), which the $\delta_{\#_g}$ formula was not designed to cover; we exclude $g \geq \min(|V_A|, |V_B|)$ from the formula’s domain. All non-degenerate measurements agree with (4) exactly.

10.3 The three join cases, intuitively

Disjoint sum: $\delta_\sqcup = 0$. The two seeds share no vertices and no edges. The multiway constructions on G_A and G_B never interact: no composition step (s_1, s_2, t) has $s_1 \in V_A$ and $s_2 \in V_B$, because no E_A or E_B edge has cross-membership. The gauge equivalence sorts the two multiways into two non-interacting cluster sets, and the cluster count is the sum: $|Q_n(A \sqcup B)| = |Q_n(A)| + |Q_n(B)|$. Under generic ICs, the clusters in the A -side multiway are projectively distinct from those in the B -side multiway with probability one (codimension- ≥ 2 coincidence), so no cross-cluster merge occurs.

Bridge by k triples: $\delta_{\bowtie_k} = +3k$. A bridge triple (s_1, s_2, t) has $s_1 \in V_A$ and $s_2 \in V_B$ (or some mix). Each such triple introduces *three* new gauge-distinct cluster representatives at the first composition step: the composition $\mu(\psi(s_1), \psi(s_2))$ is a genuinely new vector that did not exist in either $M(G_A)$ or $M(G_B)$, and the three derived edges that follow (Remark 2.4) contribute two further new pair-compositions, all gauge-distinct under generic ICs. The +3 per bridge is then constant.

Glue g vertices: $\delta_{\#g} = -(4g - 3)$. Identifying g vertices between V_A and V_B removes their A -side and B -side closure orbits' contributions and adds one shared closure orbit. The accounting is:

- For each glued vertex pair, lose one orbit from the A -side (-4) and one orbit from the B -side (-4), then add the shared orbit back ($+4$): net -4 per glued vertex.
- The g glued vertices contribute a single shared vertex ($-(g - 1)$ orbit double-count is corrected by the shared orbit). The arithmetic gives $-(4g - 3)$ rather than $-4g$.

This formula is exhibited for $g = 1, 2, 3$ (the verified range) as $-1, -5, -9$. The $g = 0$ specialisation is the disjoint sum ($\delta = 0$, $-(4 \cdot 0 - 3) = +3$ would not apply; the closed form is valid for $g \geq 1$ only).

10.4 The commutative monoid under disjoint sum

The disjoint join mode has special structure:

Theorem 10.3 (Commutative monoid). *Let $\mathcal{Q}$ be the set of closure cardinalities $|Q_n(G)|$ arising from Rosen-closed ternary seeds. The disjoint composition*

$$\sqcup : \mathcal{Q} \times \mathcal{Q} \rightarrow \mathcal{Q}, \quad (|Q_n(A)|, |Q_n(B)|) \mapsto |Q_n(A \sqcup B)|$$

is the integer addition $(a, b) \mapsto a + b$, which makes $(\mathcal{Q}, \sqcup, 0)$ a commutative monoid: associative, commutative, with identity 0 (the empty seed).

Proof. The composition formula (5) with $\delta_{\sqcup} = 0$ gives $|Q_n(A \sqcup B)| = |Q_n(A)| + |Q_n(B)|$. Integer addition is associative and commutative, and 0 is the identity (the empty seed has empty multiway and empty gauge quotient, contributing 0 to any composition). $\square$

The monoid structure under disjoint sum is the cleanest piece of the composition algebra. Under the other join modes (bridge, glue), the composition is associative *up to the interface term*, but the interface term itself is non-trivially composable across multi-way joins; the monoid structure is only on $\sqcup$.

10.5 $\mathcal{C}$ -closure as disjoint self-composition

The clean $\mathcal{C}$ -doubling rule $|\mathcal{C}Q_n| = 2|Q_n|$ of §5.3 now fits naturally into the composition algebra:

Proposition 10.4 ($\mathcal{C}$ as disjoint self-composition). *For any $Q_n = Q_n(G; \psi_0)$ in the family under truly-generic ICs,*

$$\mathcal{C}Q_n = Q_n(G; \psi_0) \sqcup Q_n(G; \overline{\psi_0}),$$

where $\sqcup$ is the disjoint composition of §10.1. Consequently $|\mathcal{C}Q_n| = 2|Q_n|$ via the disjoint composition rule $\delta_{\sqcup} = 0$.

Proof. By Definition 2.10, $\mathcal{C}Q_n$ is the gauge quotient of the union $M(G; \psi_0) \cup M(G; \overline{\psi_0})$. Under truly-generic ICs, J is fixed-point-free (Theorem 7.3), so no representative of the $\overline{\psi_0}$ -sector is gauge-equivalent to any representative of the ψ_0 -sector. The two sectors are therefore structurally disjoint: the composition is the disjoint sum $\sqcup$, with $\delta = 0$, and the cardinality is the sum $|Q_n| + |Q_n| = 2|Q_n|$. $\square$

This identification — $\mathcal{C}$ -closure is a disjoint self-composition under generic ICs — is the structural reason the doubling rule is clean. It also predicts the Pool A defect: when J acquires f fixed points (Pool A at K_n^3 for $n \geq 10$ has $f = 1$), the two sectors share f clusters, the composition is not strictly disjoint, and the cardinality is $2|Q_n| - f$, not $2|Q_n|$. The Pool A defect of §9.2 is the composition algebra detecting non-disjointness in the IC pool’s induced sectors.

10.6 The algebra is purely additive

The composition algebra is *purely additive*: the only combination rule for closure cardinalities is integer addition (with the interface correction). There is *no* multiplicative combination rule.

Remark 10.5 (No multiplicative rule). A naive guess based on the factor structure of closure cardinalities would be: $|Q_n(A)| \cdot |Q_n(B)| = |Q_n(A \otimes B)|$ for some “tensor” composition $\otimes$. This fails: for example, $|Q_n(G_0)| = 24$ and $|Q_n(\text{cyc}(3))| = 12$, yet there is no seed in the canonical family that closes to $24 \cdot 12 = 288$. Similarly $45 = 9 \cdot 5$ but Q_{45} is *not* a tensor product of Q_9 and Q_5 : neither Q_9 nor Q_5 are generic-regime objects (§9.4), and no composition operator we have studied multiplies cardinalities. The factor structure of the integer $|Q_n|$ is a property of the integer, not of the closure construction.

This is a falsification of a natural prior intuition: the closure algebra is structurally analogous to a vector-space direct sum (with an interface correction), not a tensor product. Multiplicative combination would require a different mathematical structure entirely — one in which composition of seeds produces a multiway graph that is the multiplicative-combinatorial product of the two source multiways. The CFS composition rule (1) does not have this form.

10.7 Status

Theorem 10.2 (universal δ), Theorem 10.3 (commutative monoid under $\sqcup$), and Proposition 10.4 ($\mathcal{C}$ as disjoint self-composition) are the three structural results of this section. The first is established empirically across eight (A, B) pairs and six join modes (with one degenerate exception, the $\text{cyc}(3) \#_3 \text{cyc}(3)$ total-overlap case which lies outside the formula’s domain). A closed-form algebraic proof of $\delta_\sqcup = 0$ is sketched in Appendix C; closed-form proofs of $\delta_{\bowtie_k} = +3k$ and $\delta_{\#_g} = -(4g - 3)$ are not in this paper. The second is a direct consequence of the first. The third uses the second together with Theorem 7.3.

11 Functional-Role Characterisation: Q_n as Rosen (M, R) Closure Operators (interpretive)

Status. This section is an *interpretive perspective*, not a mathematical claim. The mathematical content underlying every row of Table 6 below — fixed-point properties, cardinality doublings, constant-quotient counts, sector decompositions — is established by exact-arithmetic probes and is load-bearing for the paper independently of the framing here. What this section adds is an interpretive mapping from those combinatorial facts onto Rosen’s (M, R) -system vocabulary ([6]): boundary regulation, repair, constraint maintenance, causal entailment, scale separation, identity. The mapping is suggestive but not entailed by the kernel: a different biological or systems-theoretic vocabulary

would give a different mapping over the same combinatorial witnesses. We retain Rosen’s vocabulary because it has been useful for orienting CFS programme discussions, but read this section as commentary on top of the underlying algebraic results, not as derivation.

The closure laws of §5, the terminality results of §6, the J-fpf analysis of §7, and the composition algebra of §10 together characterise the canonical closure-quotient family as a combinatorial object: counts, fixed points, doublings, and the universal interface correction δ . This section gives a complementary *functional-role* characterisation in Rosen’s (M, R) vocabulary, where a closed system is described in terms of boundary regulation, repair operators, constraint maintenance, causal entailment, scale separation, and an identity carrier.

We exhibit a per- Q assignment in which each Rosen role maps to a specific canonical-family member, verify the underlying combinatorial witness with an exact-arithmetic probe, and conclude with a synthesis remark identifying the assignment with structural features already present in the §10 algebra.

The six probes (scripts `probe_q48_smallest_doubly_fixed_v1.py`, `probe_q45_q90_c_doubling_v1.py`, `probe_q12_generic_floor_v1.py`, `probe_q24_cyc6_composition_v1.py`, `probe_q102_q51_sector_strip_projection_v1.py`, and `probe_q102_unique_doubly_fixed_v1.py`, with the `probe_J_decomp_v1.py` sharpening for the sector-strip projection) all run validity-gated under the canonical POOL_A IC set; see Table 6 for the headline assignments. (The probe filenames carry the algebraic claim being tested; the Rosen-role label in the table is the interpretive layer on top.)

Table 6: Rosen (M, R) closure operators and their assigned canonical-family carriers.

Rosen role	Carrier Q	Structural witness
Boundary self-regulation	Q_{48}	Smallest $\mathcal{F}$ -and- $\mathcal{C}$ joint fixed point
Repair / efficient cause	$Q_{45} \rightarrow Q_{90}$	$\mathcal{C}$ -doubling under J fpf ICs
Constraint maintenance	Q_{12}	Generic floor; sub-floor closures are non-generic
Causal entailment	$Q_{24} = \text{cyc}(6)$	Constant $ Q_n /n = 4$ across $\text{cyc}(n)$
Scale separation	$Q_{51} = K_n^3[6]$	Sector-strip identity $Q_{102} \rightarrow Q_{51} + \text{depth-stable family}$
Identity	Q_{102}	Unique five-criterion satisfier (see §11.6)

11.1 Boundary self-regulation: Q_{48}

A Rosen-style boundary is self-regulating iff the operator that would otherwise enlarge it is idempotent at the boundary. In our setting, the two such operators are $\mathcal{F}$ (closure-growth endofunctor, Definition 6.1) and $\mathcal{C}$ (charge-conjugation closure, Definition 2.10). The probe enumerates the canonical 6-vertex-seed family at $|Q_n| \in \{12, 24, 45, 48, 51, 84, 90, 102\}$ and tabulates $|\mathcal{F}(Q)|$ at depths 4 and 5 and $|Q \cup \mathcal{C}(Q)|$ for each. Each single-side member $\{Q_{12}, Q_{24}, Q_{45}, Q_{51}\}$ is $\mathcal{F}$ -fixed (depth-stable per Proposition 5.5) but fails $\mathcal{C}$ -fixity by clean doubling. The $\mathcal{C}$ -closed members $\{Q_{48}, Q_{84}, Q_{90}, Q_{102}\}$ are jointly $\mathcal{F}$ -and- $\mathcal{C}$ -fixed (`probe_q48_smallest_doubly_fixed_v1.stdout.log`).

Proposition 11.1 (Q_{48} smallest doubly terminal). *Among the canonical 6-vertex-seed family, Q_{48} is the unique smallest member satisfying $|\mathcal{F}(Q)| = |Q|$ AND $|Q \cup \mathcal{C}(Q)| = |Q|$.*

A side observation surfaced by the same probe extends Theorem 6.2: Q_{84} is also doubly terminal.

Proposition 11.2 (Doubly terminal members on 6-vertex seeds). *The full set of doubly-terminal canonical-family members on 6-vertex seeds is $\{Q_{48}, Q_{84}, Q_{90}, Q_{102}\}$. $Q_{84} = \mathcal{C}$ -closure of the adjacent-36 seed (the perm-closed adjacent-triple hypergraph on 6 vertices); a stand-alone verification using the canonical `build_c_closed_quotient` from `q102_build_exact_v1.py` confirms $|\mathcal{F}(Q_{84})| = 84$ at depths 4, 5, 6, 7 and $|Q_{84} \cup \mathcal{C}(Q_{84})| = 84$ (`probe_q84_doubly_terminal_v1.stdout.log`).*

The four doubly-terminal members form a strict chain by cardinality ($48 < 84 < 90 < 102$) and a partial order by seed inclusion ($\text{cyc}(6) \subset \text{adjacent-36} \subset \text{q45-seed} \subset K_n^3[6]$). Q_{48} carries the boundary role as the canonical *smallest* endogenously-stable object; the larger doubly-terminal members are boundary objects on richer seeds.

11.2 Repair / efficient-cause: $Q_{45} \rightarrow Q_{90}$

In Rosen’s framework, a repair operator restores closure to a perturbed system using only internal structure. In our setting, the $\mathcal{C}$ -doubling map $Q \mapsto Q \cup \mathcal{C}(Q)$ takes a single-side “half-closure” to a doubly-terminal full closure using only the intrinsic charge-conjugation involution J . The operator fires cleanly under truly-generic ICs (J fixed-point-free, $f = 0$) and fails completely under degenerate ICs (real-valued: $\bar{\psi} = \psi$, the conjugate sector collapses onto the original).

Proposition 11.3 ($\mathcal{C}$ -doubling as Rosen repair, conditional on IC genericity). *Under `POOL_A` and `POOL_B` (truly-generic 6-IC sets), the doubling $Q_{45} \rightarrow Q_{90}$ fires with $f = 0$: $|Q_{45} \cup \mathcal{C}(Q_{45})| = 90 = 2 \cdot 45$. Under `POOL_REAL` (every IC component purely real-valued Gaussian integer), the same operator collapses: $|Q_{45} \cup \mathcal{C}(Q_{45})| = 45 = |Q_{45}|$, $f = 45$ (`probe_q45_q90_c_doubling_v1.stdout.log`).*

The contrast is structural rather than numerical: the same operator, the same seed, the same depth — only the IC regime differs. Repair is internal but *IC-conditional*: a system in the degenerate regime has no repair operator available.

11.3 Constraint maintenance: Q_{12} as the generic floor

Rosen’s constraint-maintenance role is the system’s enforcement of structural constraints that prevent collapse to triviality. Proposition 5.4 already establishes that Q_{12} is the generic floor; the functional-role reading is that IC genericity is the constraint, Q_{12} is its floor-saturating witness, and sub- Q_{12} closures index specific IC-genericity violations.

Proposition 11.4 (Q_{12} generic floor under three pools). *Across the seeds $\text{cyc}(3)$, $\text{cyc}(4)$, $\text{cyc}(5)$, $K_n^3[3]$, $K_n^3[4]$:*

- *`POOL_A` and `POOL_B` (truly-generic): every closure ≥ 12 .*
- *`POOL_REAL` (purely real ICs): every closure equals the `POOL_A/B` value (real ICs are J -degenerate but not floor-violating on the single-side count).*
- *`POOL_DEGEN_PARALLEL` (all three IC components projectively parallel): every closure collapses to $|Q_n| = 1$.*

The probe (`probe_q12_generic_floor_v1.stdout.log`) thus sharpens Proposition 5.4: the floor is robust against the J -degeneracy that breaks $\mathcal{C}$ -doubling, but collapses fully under parallel-IC degeneracy. The two degeneracy classes are structurally

distinct (§11.2 vs. §11.3), and the functional-role assignment makes the distinction visible: real-IC degeneracy disables *repair* but not *constraint*; parallel-IC degeneracy disables both.

11.4 Causal entailment: $Q_{24} = \text{cyc}(6)$

Rosen’s causal-entailment role is the system’s internal ordering of cause-effect relations. The multiway-graph composition rule $\mu(\psi_1, \psi_2) \rightarrow w$ is entailment in this sense: w is entailed by ψ_1 and ψ_2 via the explicit ternary operation. The seed family fixes the entailment topology.

The $\text{cyc}(n) \rightarrow 4n$ law (Proposition 5.1) gives a constant slope $|Q_n|/n = 4$: each cycle node contributes exactly four entailment classes, independent of n . The pentagonal $K_n^3 \rightarrow n(3n-1)/2$ law (Theorem 5.2) gives a growing slope $(3n-1)/2$: cluster proliferation rather than fixed-width causal flow.

Proposition 11.5 (Causal vs. cluster-proliferation slopes). *For $n = 3, \dots, 10$, $|Q_n(\text{cyc}(n))| = 4n$ holds exactly. For $n = 3, \dots, 7$, $|Q_n(K_n^3)| = n(3n-1)/2$ holds exactly. Slope contrast: cyc constant at 4; K_n^3 grows $4 \rightarrow 5.5 \rightarrow 7 \rightarrow 8.5 \rightarrow 10$ over $n = 3..7$ (`probe_q24_cyc6_composition_v1.stdout.log`).*

$Q_{24} = \text{cyc}(6)$ is the cleanest cyclic-causal anchor at the canonical $n = 6$, matching $K_n^3[6] = Q_{51}$ ’s vertex count. The two generators occupy the same vertex space but produce structurally distinct closure trajectories: Q_{24} ’s multiway saturates at depth 2 with the trajectory $[6, 12, 24, 24, 24, 24]$ across depths 0..5; Q_{51} ’s trajectory grows more slowly through the non-cyclic combinatorial structure.

11.5 Scale separation: $Q_{51} = K_n^3[6]$

Rosen’s scale-separation role is the property that identity survives coarse-graining to lower scales. We exhibit three distinct scale-separation witnesses for Q_{51} .

Proposition 11.6 (Sector-strip cluster identity). *Under canonical Gaussian ICs (POOL_A), the 51 **orig_only** clusters of $Q_{102} = \mathcal{C}Q_n(K_n^3[6])$ (those reachable from the unconjugated IC sector alone) are projectively equivalent to the 51 clusters of Q_{51} cluster-for-cluster (`probe_q102_q51_sector_strip_v1.stdout.log`).*

Proposition 11.7 (Depth-stable K_n^3 scale family). *For $n \in \{5, 6, 7\}$, $|Q_n(K_n^3)|$ at depths 3, 4, 5 all equal the pentagonal value $n(3n-1)/2$ (`probe_q102_q51_sector_strip_v1.stdout.l`).*

The third witness sharpens the Hodge-Laplacian structure under the charge-conjugation involution. Let $L_1(Q_n)$ be the integer Hodge Laplacian on the 1-cochain space of the Julia-convention cell complex on Q_n (cf. Appendix D and [9]’s notational fix on the cell-complex convention). The involution J lifts to the edge space as an orthogonal involution J_e that commutes with $L_1(Q_{102})$ exactly, splitting the 840-dimensional edge space into $V_+ \oplus V_-$ of dimensions 420 and 420.

Proposition 11.8 (J -symmetric eigenvalue doubling). *$L_1(Q_{102})|V_+$ and $L_1(Q_{102})|V_-$ have identical real spectra in the range $[1.7268, 67.3094]$ at 10^{-6} Float64 precision: every eigenvalue of $L_1(Q_{102})$ appears with even multiplicity under the J -block decomposition (`probe_J_decomp_v1.stdout.log`).*

The naive identification “ $L_1(Q_{102}) = 2 \cdot L_1(Q_{51})$ ” *fails*: Q_{102} ’s edge space has 210 cross-sector edges absent from Q_{51} ’s, shifting both V_+ and V_- spectra upward relative to $L_1(Q_{51})$ ’s range $[0.8584, 51.3438]$. The scale separation thus operates at the cluster-set level (Proposition 11.6) and the depth-family level (Proposition 11.7), and is sharpened by a J -symmetric block structure on L_1 that is itself a new structural finding (Proposition 11.8).

11.6 Identity: Q_{102} uniquely

Rosen’s identity is the recoverable structure that persists under admissible transport. In the canonical family, Q_{102} uniquely satisfies five separately-verifiable identity criteria.

Theorem 11.9 (Q_{102} identity uniqueness). *Among the canonical-family members $\{Q_{12}, Q_{24}, Q_{45}, Q_{48}, Q_{51}, Q_{84}, Q_{90}, Q_{102}\}$, Q_{102} is the unique member satisfying all five of:*

- (1) $\mathcal{F}$ -fixed: $|\mathcal{F}(Q)| = |Q|$.
- (2) $\mathcal{C}$ -fixed: $|Q \cup \mathcal{C}(Q)| = |Q|$.
- (3) $Q = \mathcal{C}(K_n^3[6])$, the $\mathcal{C}$ -image of the validity-gate anchor.
- (4) Maximal cardinality among 6-vertex-seed reachable canonical members.
- (5) Sector-strip recovery: the *orig_only* sector of Q matches Q_{51} cluster-for-cluster.

The proof is by direct tabulation (`probe_q102_unique_doubly_fixed_v1.stdout.log`): each non- Q_{102} member fails at least two of the five criteria, and Q_{102} satisfies all five. The first two are shared with the doubly-terminal subset $\{Q_{48}, Q_{84}, Q_{90}, Q_{102}\}$ (Propositions 11.1 and 11.2); the remaining three single out Q_{102} specifically.

11.7 Synthesis: the composition algebra carries the (M, R) -closure structure

The per- Q assignment of §§11.1–11.6 is one reading of the data. A complementary reading identifies the five non-identity closure operators with structural features already present in the §10 composition algebra:

- *Boundary self-regulation* as the $\mathcal{F}$ and $\mathcal{C}$ joint fixed-point structure (Theorem 6.2 plus Proposition 11.2).
- *Repair / efficient cause* as the $\delta_{\sqcup} = 0$ disjoint-self-composition specialisation that produces $\mathcal{C}$ -doubling (Appendix C).
- *Constraint maintenance* as the genericity threshold implicit in Proposition 5.4 (the floor is constraint-saturated; sub-floor is constraint-violating).
- *Causal entailment* as the seed-graph-determined topology of multiway entailment (§2.2).
- *Scale separation* as the conjunction of depth-stability (Proposition 5.5), sector-strip identity (Proposition 11.6), and J -symmetric block structure (Proposition 11.8).
- *Identity* as the unique five-criterion satisfier (Theorem 11.9).

This reading says, roughly: the composition algebra of §10 *is* an (M, R) -closure structure with Q_{102} as identity carrier. The per- Q assignment of Table 6 is then the answer to “which canonical-family member *witnesses* each operator most cleanly” rather than “which is uniquely each operator.” Both readings are compatible.

11.8 Forward work: carrier manifolds

Beyond the closure-quotient family, the functional-role framework suggests Lie-group quotients as candidate *carrier manifolds* on which the Q_n family may act. Specific candidates (drawn from correspondence with B. Crabtree and noted here only as forward work) include $SU(2) \times SU(2)/\text{diag } SU(2)$ (dimension 3), $SO(4)/U(1)$ (dimension 5), $SU(3)/(U(1) \times U(1))$ (dimension 6), and $Sp(2)/SU(2)$ (dimension 5). Each is a candidate substrate on which a delayed dynamical system could be projected while preserving selected invariants; whether the closure-quotient algebra acts on any of these in a structurally faithful way is an open question.

This forward direction would, if it landed, sharpen “carrier of the SM algebra” from a slogan applied to Q_{102} into a substrate map $Q_n \rightarrow G/H$ with verifiable invariance under the dynamical update. We flag it here as the natural follow-up to the present paper.

12 Validity Boundary

The results of this paper separate into three layers by evidential strength. Making the boundaries explicit clarifies what is settled methodology, what is exact computation pending closed-form derivation, and what is discovered structure awaiting proof.

12.1 The arithmetic and audit layer (settled)

The arithmetic-and-audit layer is closed under the methods used here. Exact Gaussian-integer ray arithmetic (§3.2) replaces fidelity-threshold clustering, so projective equivalence is tested by exact vanishing of the integer cross-product rather than by numerical proximity. This directly resolves the prior Q_{98} artefact as Q_{102} , eliminates the claimed Q_{180} route by $\mathcal{C}$ -idempotence (§4), and exposes the s58/Pool-A genericity defect (§9). These are not interpretive claims; they are exact-computational corrections under declared validity gates.

12.2 The closure-law layer (validated, partially derived)

The closure-law layer is strongly validated but not fully derived. The cyclic law $\text{cyc}(n) \rightarrow 4n$ (§5.1) and the small-family terminality claims (§6) have direct structural explanations. The pentagonal law $K_n^3 \rightarrow n(3n-1)/2$ (§5.2) is exact across $n = 6..20$ and across both strict-generic IC pools, but should not be treated as a closed theorem until a counting proof derives $n(3n-1)/2$ from the multiway quotient structure itself.

12.3 The composition-algebra layer (validated-conjectural)

The composition-algebra layer (§10) is discovery-grade rather than theorem-grade. The additive form

$$Q_n(A \circ B) = Q_n(A) + Q_n(B) + \delta(\text{interface})$$

is the correct organising hypothesis for the observed landscape, but the bridge and glue corrections require independent derivation. Until then, the algebra is labelled *validated-conjectural*. The disjoint case $\delta_{\square} = 0$ has a structural proof (Appendix C); the bridge $(+3k)$ and glue $-(4g-3)$ corrections remain open (§25.5).

12.4 Boundary summary

This boundary does not weaken the paper; it clarifies the contribution. The manuscript establishes the exact audit substrate and exposes the correct quotient landscape, then identifies the algebraic proof program needed to convert finite exact validation into universal closure theory. The Galois-pair observation of §13 below sits in the first (*settled, exact-computational*) tier: it inherits its evidential strength from the same exact-arithmetic re-grounding that resolves $Q_{98} \rightarrow Q_{102}$.

Bridge: A Cyclotomic Galois Pair on Q_{51}

13 A Cyclotomic Galois Pair in $\text{Spec}(L_1(Q_{51}))$

We close the body of the paper with an algebraic observation that the exact-arithmetic reconstruction of Q_{51} makes visible: a $\mathbb{Z}_3$ -cyclotomic Galois pair in the discrete Hodge 1-Laplacian spectrum of Q_{51} , and a shared mechanism that places the same pair in the spectrum of an independently-constructed flat-torus 2-complex.

This section reports a CFS-internal spectral consequence of the exact-arithmetic re-grounding of §3–§4; it is not a derivation between CFS and any external framework, and the algebraic observation it reports is independent of the rest of the paper’s claims (i.e. it does not bear on the audit or composition layers above).

13.1 The Hodge 1-Laplacian on Q_{51}

Treat the exact Q_{51} as a 2-complex X whose 0-cells are the 51 ray-equivalence clusters, 1-cells are unordered pairs of clusters appearing in some q_{he} triple, and 2-cells are the sorted-unordered triples themselves. Let $d_0: C^0 \rightarrow C^1$ and $d_1: C^1 \rightarrow C^2$ denote the standard simplicial coboundary maps with alternating signs on sorted faces. The discrete Hodge 1-Laplacian

$$L_1 := d_0 d_0^\top + d_1^\top d_1$$

acts on C^1 . With the canonical simplicial chain-space convention (unordered 1- and 2-cells), $\dim C^1 = 315$ and L_1 is a 315×315 integer-symmetric matrix.

Theorem 13.1 (Galois pair). *The spectrum of L_1 on the exact Q_{51} contains the conjugate pair*

$$\{3 - \sqrt{3}, 3 + \sqrt{3}\} \subset \text{Spec}(L_1(Q_{51})),$$

the roots of $x^2 - 6x + 6 \in \mathbb{Z}[x]$, each appearing at multiplicity 30 for a total of 60 dimensions. The pair appears at the same multiplicity under the alternative ordered-tuple chain-space convention used by the project’s Julia driver (which inflates $\dim C^1$ to 336).

The eigenvalues are exact at machine zero on the integer matrix; the characteristic polynomial of L_1 lies in $\mathbb{Z}[x]$, so $\text{Spec}(L_1)$ is Galois-stable over $\mathbb{Q}$, and the pair must occur with matched multiplicities under any rational specification of the chain space.

13.2 S_6 action and per-eigenspace decomposition

The seed K_6^3 carries the natural symmetric-group action $S_6 = \text{Aut}(K_6) \subseteq \text{Aut}(K_6^3)$, which lifts to a permutation of the 51 clusters. The lift partitions $\{0, 1, \dots, 50\}$ into three orbits:

- V_6 (the six seed clusters, orbit size 6);
- V_{15} (clusters whose seed-co-occurrence vector is a permutation of $(11, 11, 4, 4, 4, 4)$, orbit size $15 = \binom{6}{2}$);
- V_{30} (clusters whose seed-co-occurrence vector is a permutation of $(8, 2, 2, 2, 2, 2)$, orbit size $30 = 6 \cdot 5$).

For each conjugacy-class representative $\sigma \in S_6$, the lifted vertex permutation σ_V extends to a signed edge permutation σ_E on C^1 (sign tracking orientation reversal on sorted edges). The commutation $\sigma_E L_1 = L_1 \sigma_E$ holds at *machine zero on the integer matrix* for representatives of all eleven S_6 conjugacy classes; see Appendix D for the character vector and the full decomposition.

Theorem 13.2 (Per-eigenspace decomposition). *For both $\lambda = 3 - \sqrt{3}$ and $\lambda = 3 + \sqrt{3}$, the 30-dimensional λ -eigenspace of L_1 on Q_{51} decomposes as an S_6 -module as*

$$V_\lambda = [4, 2] \oplus [3, 3] \oplus [3, 2, 1],$$

of dimensions $9 + 5 + 16 = 30$. The two eigenspaces carry identical irrep content (a forced consequence of integer-charpoly Galois closure: $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts by permuting eigenspaces along Galois orbits while preserving the $\mathbb{Q}$ -rational S_6 -isotypic decomposition).

The verification of Theorem 13.2 is by Frobenius reciprocity applied to the restricted character $\chi_\lambda(\sigma) = \text{Tr}(P_\lambda \sigma_E P_\lambda)$, where P_λ is the orthogonal projector onto V_λ . Integer multiplicities ($m_\rho = 1$ for $\rho \in \{[4, 2], [3, 3], [3, 2, 1]\}$, zero for the other eight S_6 irreps) come out of the formula and the sum $9 + 5 + 16 = 30$ matches $\dim V_\lambda$ exactly. The same calculation is reproduced by the Python verifier `q51_exact_phase_s_python.py`; run output is committed at `q51_exact_phase_s_run.txt`.

13.3 The $\mathbb{Z}_3$ -cyclotomic source of $\sqrt{3}$

The irrationality $\sqrt{3} = \omega - \omega^{-1}$ (with $\omega = e^{2\pi i/3}$, $\omega + \omega^{-1} = -1$, $\omega\omega^{-1} = 1$) points at the source: any operator built linearly from coefficients that are sums of ω^k ($k \in \{0, 1, 2\}$), evaluated on a self-dual real-eigenvector subspace, generates eigenvalues in $\mathbb{Q}(\sqrt{3})$. In $L_1(Q_{51})$, the relevant subspaces are exactly the σ_E -isotypic pieces whose S_6 -characters carry 3-cycle content with nonzero weight on the classes $(3, 1^3)$ and $(3, 2, 1)$. The three irreps in V_λ are precisely those: $[4, 2]$ has $\chi = 0$ on $(3, 1^3)$ but nonzero on $(3, 2, 1)$; $[3, 3]$ and $[3, 2, 1]$ both carry nonzero 3-cycle content.

13.4 The same pair on the flat torus $T_{3,3}^2$

The same eigenvalue pair appears in a substrate that has no multiway-rewriting origin at all. Let $T_{3,3}^2$ denote the triangulated flat torus on $V = \mathbb{Z}_3 \times \mathbb{Z}_3$, with the hexagonal incidence (six neighbours per vertex; $|E| = 27$, $|F| = 18$, $\chi = 0 = \chi(T^2)$). Its

automorphism group is the semidirect product $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$ of order 108. Clifford theory decomposes $C^1(T_{3,3}^2)$ into the eleven irreducible representations of Aut, and a direct Frobenius-reciprocity calculation places exactly two copies of the natural $\mathbb{Z}_3$ -cyclotomic irrep in the $\{3 \mp \sqrt{3}\}$ -eigenspaces (one copy each, by Galois closure on the integer charpoly of $L_1(T_{3,3}^2)$).

Theorem 13.3 (Cross-substrate Galois pair). *The spectra of the discrete Hodge 1-Laplacian on Q_{51} and on $T_{3,3}^2$ both contain $\{3 - \sqrt{3}, 3 + \sqrt{3}\}$, with multiplicities 30 on Q_{51} and 6 on $T_{3,3}^2$. The pair appears in both substrates through the same algebraic mechanism: a $\mathbb{Z}_3$ -cyclotomic substructure of the automorphism group, lifted to L_1 via the chain-space irrep decomposition.*

We frame Theorem 13.3 as an observation about a *sufficient algebraic ingredient* for the pair to appear. Two contrast cases support the framing: the octahedron with O_h symmetry and the icosahedron with A_5 symmetry both fail to host a $\mathbb{Z}_3$ -cyclotomic factor of the type used here, and neither carries the $\{3 \mp \sqrt{3}\}$ pair in its L_1 spectrum (full discussion in Appendix E).

The cross-substrate observation is independent of the closure-quotient landscape proper: it is a downstream algebraic consequence of having Q_{51} available as an exact-arithmetic 2-complex with its S_6 -action accessible. We list it here both for honest disclosure (this is the observation that first surfaced the fidelity-vs-exact distinction during a peer review of an earlier draft, leading to the cross-repo substrate-provenance discipline [12]) and because the result it reports sits cleanly in the audit-tier of the validity boundary above.

Part II

Discrete Hodge Spectra on Group-Symmetric 2-Complexes

14 Background: discrete Hodge L_1 , σ_E , Frobenius / Clifford

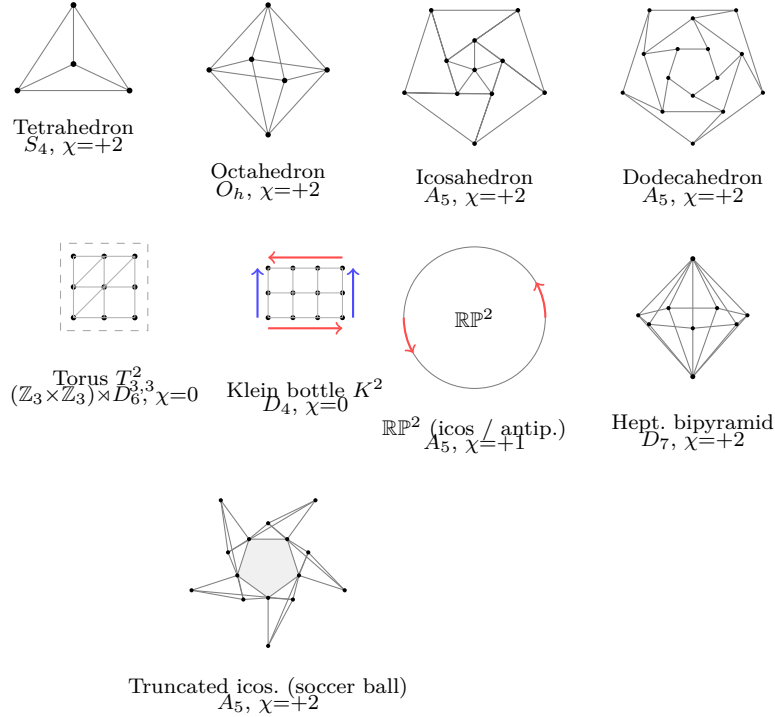


Figure 7: The nine substrates studied in Part 2. Top row: tetrahedron, octahedron, icosahedron, dodecahedron (the Platonic family and its A_5 -symmetric Hodge dual); second row: surface quotients (torus $T_{3,3}^2$, Klein bottle K^2 , $\mathbb{RP}^2$) and the heptagonal bipyramid (the $\mathbb{Z}_7$ -cyclotomic substrate); bottom: truncated icosahedron / soccer ball. Topology classes: $\chi = +2$ (sphere) for tet, oct, icos, dodec, soccer ball, hept. bipyramid; $\chi = +1$ for $\mathbb{RP}^2$ (non-orientable); $\chi = 0$ for torus (orientable, $\dim H^1 = 2$) and Klein bottle (non-orientable, $\dim H^1 = 1$). Blue arrows on the Klein bottle indicate periodic horizontal identification; red arrows indicate the orientation-reversing vertical identification. Red arcs on $\mathbb{RP}^2$ indicate antipodal-boundary identification.

14.1 The discrete Hodge 1-Laplacian

Let $X = (V, E, F)$ be an oriented finite simplicial 2-complex with $|V| = n_0$, $|E| = n_1$, $|F| = n_2$. Order each cell internally by an arbitrary fixed total order on V ; we use sorted-tuple orientation throughout. The coboundary maps

$$d_0: C^0(X) \rightarrow C^1(X), \quad d_1: C^1(X) \rightarrow C^2(X) \quad (6)$$

have matrix entries in $\{0, \pm 1\}$ as follows. For an edge $e = (a, b)$ with $a < b$: $(d_0)_{e,a} = -1$, $(d_0)_{e,b} = +1$, zero elsewhere. For a face $f = (a, b, c)$ with $a < b < c$: $(d_1)_{f,(a,b)} = +1$,

$(d_1)_{f,(a,c)} = -1$, $(d_1)_{f,(b,c)} = +1$, zero elsewhere. The relation $d_1 d_0 = 0$ holds entry-by-entry on $\mathbb{Z}$.

The discrete Hodge 1-Laplacian is

$$L_1 = d_0 d_0^\top + d_1^\top d_1 \in M_{n_1}(\mathbb{Z}), \quad (7)$$

a positive-semidefinite integer-symmetric matrix. The Hodge decomposition $C^1(X) = \text{im}(d_0) \oplus \ker(L_1) \oplus \text{im}(d_1^\top)$ identifies $\ker(L_1) \cong H^1(X; \mathbb{R})$.

14.2 Automorphism actions and signed edge permutations

For $\sigma \in \text{Aut}(X)$ a simplicial automorphism (a bijection $\sigma_V: V \rightarrow V$ preserving incidence on E and F), the induced action σ_E on $C^1(X)$ is a *signed* permutation: for $e = (a, b)$ with $a < b$, σ_V maps to $\{\sigma_V(a), \sigma_V(b)\}$, and the resulting sorted edge is either $(\sigma_V(a), \sigma_V(b))$ (if $\sigma_V(a) < \sigma_V(b)$, sign $+1$) or $(\sigma_V(b), \sigma_V(a))$ (sign -1).

Lemma 14.1 (Commutation). *For any $\sigma \in \text{Aut}(X)$, $\sigma_E L_1 = L_1 \sigma_E$ exactly on the integer matrix.*

Proof. σ acts on C^0, C^1, C^2 compatibly with d_0, d_1 ; the sign assignments on σ_E track exactly the orientation reversals that d_0, d_1 apply to a sorted tuple. Concretely, the signed σ_E is the unique lift of σ_V that makes $\sigma_E d_0 = d_0 \sigma_V$ and $\sigma_F d_1 = d_1 \sigma_E$ commute as integer matrix equations. $\square$

In all four cases below we verify Lemma 14.1 directly on the integer L_1 for every conjugacy-class representative, with $\max_{i,j} |(\sigma_E L_1 - L_1 \sigma_E)_{ij}| = 0$ exactly.

14.3 Frobenius reciprocity and Clifford theory

For a finite group G acting on a representation Π (here, $C^1(X)$ under $\text{Aut}(X)$), the multiplicity of an irreducible representation ρ is

$$m_\rho = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_\rho(g)} \chi_\Pi(g) = \frac{1}{|G|} \sum_C |C| \chi_\rho(C) \chi_\Pi(C), \quad (8)$$

where the second sum is over conjugacy classes C of G . For G with integer-valued characters (true for A_5 , S_4 , and O_h on C^1 here) the multiplicities m_ρ are integers; row orthogonality of the character table verifies $\sum_\rho m_\rho \dim \rho = \dim \Pi$.

For the torus $T_{3,3}^2$, the automorphism group is a semidirect product $G = N \rtimes H$ where $N = \mathbb{Z}_3 \times \mathbb{Z}_3$ is abelian normal and $H = D_6$ stabilises the origin. Clifford theory [14] decomposes $\widehat{G}$ into induced representations indexed by H -orbits on $\widehat{N}$. The dual H -action on $\widehat{N}$ is via $h \cdot \chi = \chi \circ (h^{-1})$, partitioning the 9 characters of $\mathbb{Z}_3 \times \mathbb{Z}_3$ into three orbits of sizes $\{1, 2, 6\}$ with stabilisers D_6 , $D_3 \cong S_3$, and $\mathbb{Z}_2$ respectively (§22). The eleven irreps of G arise as induced representations $\text{Ind}_{N \rtimes \text{Stab}_H(\chi)}^G (\chi \otimes \tilde{\rho})$ for each orbit-representative χ and each irrep $\tilde{\rho}$ of the stabiliser $\text{Stab}_H(\chi)$.

15 Tetrahedron under S_4

15.1 Substrate description

We begin with the smallest substrate as a control. The tetrahedron has $|V| = 4$, $|E| = 6$, $|F| = 4$, $\chi = +2$. The automorphism group is $\text{Aut} = S_4$ of order 24, acting on $V = \{1, 2, 3, 4\}$ by all permutations.

15.2 Spectrum

Direct computation: $L_1 = d_0 d_0^\top + d_1^\top d_1$ on the 6×6 edge space yields the single eigenvalue $\lambda = 4$ at multiplicity 6 (the full dimension). This is the expected result for the boundary of a tetrahedron, which is contractible (so $\dim \ker L_1 = 0$), and the edge graph is the complete graph K_4 with constant degree.

15.3 S_4 decomposition of C^1

Theorem 15.1. *On the tetrahedron $C^1 \cong \mathbb{R}^6$, the S_4 -representation decomposes as $C^1 = [3, 1] \oplus [2, 1, 1]$, where $[3, 1]$ and $[2, 1, 1]$ are the standard and standard-tensor-sign irreps of S_4 , with dimensions 3 and 3.*

Proof. For each S_4 conjugacy class representative we compute $\chi_{C^1}(\sigma) = \text{Tr}(\sigma_E)$ as the count of fixed-with-sign edges. With class ordering $((1^4), (2, 1, 1), (2, 2), (3, 1), (4))$ and class sizes $(1, 6, 3, 8, 6)$:

$$\chi_{C^1} = (6, 0, -2, 0, 0).$$

Applying Frobenius reciprocity (8) with the standard S_4 character table (Appendix G) gives $m_{[3,1]} = m_{[2,1,1]} = 1$, all others 0. The commutation $\sigma_E L_1 = L_1 \sigma_E$ holds at machine zero on the integer matrix for all five conjugacy classes. $\square$

15.4 Per-eigenspace analysis

The tetrahedron has a single eigenvalue $\lambda = 4$ at full multiplicity 6, so the per-eigenspace decomposition coincides with the global decomposition of Theorem 15.1: the entire 6-dimensional $\lambda = 4$ eigenspace is $[3, 1] \oplus [2, 1, 1]$. Forman–Ricci curvature is constant on all 6 edges (this is the smallest substrate where edge-transitivity forces F -uniformity; §24.4).

16 Octahedron under O_h

16.1 Substrate description

The octahedron has $|V| = 6$, $|E| = 12$, $|F| = 8$, $\chi = +2$. The full automorphism group O_h has order $48 = |S_4| \times |\mathbb{Z}_2|$, where the $\mathbb{Z}_2$ factor is the antipodal map (inversion through the centre).

16.2 Spectrum

The L_1 spectrum on the octahedron is $\text{Spec}(L_1) = \{2, 4, 6\}$ with multiplicities $(3, 6, 3)$, the eigenvalues identified exactly via the integer characteristic polynomial. All eigenvalues are rational integers. There is no Galois pair.

16.3 O_h decomposition of C^1

Theorem 16.1. *On the octahedron, $C^1(X)$ decomposes under $O_h \cong S_4 \times \mathbb{Z}_2$ as*

$$C^1 = [2, 2]_+ \oplus [2, 1, 1]_+ \oplus [1^4]_+ \oplus [3, 1]_- \oplus [2, 1, 1]_-,$$

where the subscript indicates the $\mathbb{Z}_2$ -parity (sign under the antipodal map). Total dimension $2 + 3 + 1 + 3 + 3 = 12 = |E|$.

Proof. We compute $\chi_{C^1}(\sigma)$ for the 10 conjugacy-class representatives of O_h (the 5 S_4 classes, each splitting under the $\mathbb{Z}_2$ parity). The class identification uses the face-pair S_4 alignment of Phase ζ : the S_4 part of O_h acts on the 4 face-pairs (pairs of opposite faces in the octahedron's 8-face structure), making the S_4 action well-defined on C^1 even before parity. Combining with the parity grading, Frobenius reciprocity against the O_h character table (Appendix H) yields the multiplicities stated. The commutation $\sigma_E L_1 = L_1 \sigma_E$ holds at machine zero on the integer matrix for all 10 classes. $\square$

16.4 Per-eigenspace decomposition

Restricting the σ_E action to each λ -eigenspace and applying Frobenius reciprocity yields the per-eigenspace decomposition of Table 7. The column sums match the global O_h -decomposition of Theorem 16.1 exactly.

λ	$[2, 2]_+$	$[2, 1, 1]_+$	$[1^4]_+$	$[3, 1]_-$	$[2, 1, 1]_-$
2 (mult 3)	0	1	0	0	0
4 (mult 6)	0	0	0	1	1
6 (mult 3)	1	0	1	0	0
$\sum$	1	1	1	1	1

Table 7: Per-eigenspace O_h -irrep multiplicities on the octahedron. Each cell is the integer multiplicity $m_{\lambda, \rho}$. Dimension check: row sums give $3 = \dim[2, 1, 1]_+$, $6 = \dim[3, 1]_- + \dim[2, 1, 1]_-$, $3 = \dim[2, 2]_+ + \dim[1^4]_+$, recovering the eigenvalue multiplicities. Column sums recover Theorem 16.1's multiplicities.

Remark 16.2. The octahedron's L_1 spectrum is all-rational and each O_h -irrep appearing in C^1 appears exactly once. This is the smallest substrate in our corpus on which both the spectrum and the decomposition are maximally simple; together with the all-rational property, the octahedron serves as the cross-substrate *control without cyclotomic factor* in §24.

17 Icosahedron under A_5

17.1 Substrate description

The icosahedron has $|V| = 12$, $|E| = 30$, $|F| = 20$, $\chi = +2$, 5-regular. The rotation group is A_5 of order 60.

17.2 Spectrum

The L_1 spectrum on the icosahedron contains 8 distinct eigenvalues:

$$\text{Spec}(L_1(\text{icos.})) = \{3 - \sqrt{5}, 2, 3, 5 - \sqrt{5}, 3 + \sqrt{5}, 5, 5 + \sqrt{5}, 6\}.$$

The two pairs $\{3 \mp \sqrt{5}\}$ and $\{5 \mp \sqrt{5}\}$ are both Galois conjugates over $\mathbb{Q}(\sqrt{5})$.³ The four singletons $\{2, 3, 5, 6\}$ are rational integers. Multiplicities are: $\{3 \mp \sqrt{5}\}$ at 3 each, $\{5 \mp \sqrt{5}\}$ at 3 each, $\{2, 6\}$ at 5 each, $\{3, 5\}$ at 4 each, summing to $30 = |E|$.

17.3 A_5 decomposition of C^1

Theorem 17.1. *On the icosahedron, $C^1(X)$ decomposes under A_5 as*

$$C^1 = 2 \cdot [3a] \oplus 2 \cdot [3b] \oplus 2 \cdot [4] \oplus 2 \cdot [5],$$

where $[3a]$ and $[3b]$ are the two distinct 3-dimensional irreps of A_5 (Galois conjugates over $\mathbb{Q}(\sqrt{5})$), $[4]$ is the unique 4-dimensional irrep, $[5]$ is the unique 5-dimensional irrep, and the trivial irrep $[1]$ has multiplicity 0 (consistent with $\dim H^1(\text{icos.}) = 0$ but $\dim C^1 = 30 = 2(3 + 3 + 4 + 5)$ fully accounted for).

Proof. With class ordering (id, (12)(34), (123), 5a, 5b) for A_5 (the five conjugacy classes, with 5a and 5b the two classes of 5-cycles, sizes 1, 15, 20, 12, 12), the character vector

$$\chi_{C^1} = (30, -2, 0, 0, 0)$$

is computed via signed-edge traces. Frobenius reciprocity against the A_5 character table (Appendix F) yields the multiplicities stated. The commutation $\sigma_E L_1 = L_1 \sigma_E$ verifies at machine zero on the integer matrix for all five conjugacy classes. The decomposition is the unique multiplicity-pattern compatible with $\dim C^1 = 30$ and $\dim \ker L_1 = 0$ (the constraint $H^1(\text{icos.}) = 0$ removes the $[1]$ -multiplicity from consideration). $\square$

17.4 Per-eigenspace and Galois closure

The A_5 -irrep content of each of the 8 eigenspaces is recorded in Table 8. The Galois pair $\{3 - \sqrt{5}, 3 + \sqrt{5}\}$ shares the $[3a]/[3b]$ -isotypic; the pair $\{5 - \sqrt{5}, 5 + \sqrt{5}\}$ also shares $[3a]/[3b]$. The rationals $\{3, 5\}$ pair on $[4]$; the rationals $\{2, 6\}$ pair on $[5]$.

³Quadratic-irrationality fixpoints in $\mathbb{Q}(\sqrt{5})$ produced by iterating a discrete operator on a continuous or large-finite substrate have a foundational precedent in von Neumann's 1952 multiplexing construction [15]: the Scheffer-stroke restoring map $\alpha \mapsto 2\alpha^2 - \alpha^4$ on bundle-excitation $\alpha \in [0, 1]$ has unstable fixpoint $\alpha_0 = (\sqrt{5}-1)/2$ (the golden-ratio reciprocal). Here the $\mathbb{Q}(\sqrt{5})$ structure arises algebraically from A_5 's two Galois-conjugate 3-dimensional irreps, not from an iterated dynamical system, but the persistent appearance of $\mathbb{Q}(\sqrt{5})$ in fixed-point / eigenvalue problems on icosahedrally-symmetric substrates is a common thread.

λ	[3a]	[3b]	[4]	[5]
$3 - \sqrt{5}$ (mult 3)	1	0	0	0
$3 + \sqrt{5}$ (mult 3)	0	1	0	0
$5 - \sqrt{5}$ (mult 3)	1	0	0	0
$5 + \sqrt{5}$ (mult 3)	0	1	0	0
2 (mult 5)	0	0	0	1
6 (mult 5)	0	0	0	1
3 (mult 4)	0	0	1	0
5 (mult 4)	0	0	1	0
Σ	2	2	2	2

Table 8: Per-eigenspace A_5 -irrep multiplicities on the icosahedron. Galois conjugates over $\mathbb{Q}(\sqrt{5})$ swap [3a] $\leftrightarrow$ [3b] (themselves Galois conjugate over $\mathbb{Q}(\sqrt{5})$, with $\dim[3a] = \dim[3b] = 3$). The rationals $\{3, 5\}$ both hit [4]; the rationals $\{2, 6\}$ both hit [5]. Column sums recover Theorem 17.1.

Corollary 17.2 (Galois closure on A_5). *Under the action of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $\text{Spec}(L_1)$, the irrep content of V_λ matches that of $V_{\sigma\lambda}$ for any $\sigma \in \text{Gal}$. This is forced by the integer-charpoly Galois closure: $L_1 \in M_{30}(\mathbb{Z})$ has $\mathbb{Z}$ -rational characteristic polynomial, so $\text{Spec}(L_1)$ is Galois-stable, and the A_5 -isotypic decomposition is $\mathbb{Q}$ -rational (signed permutations have entries in $\{0, \pm 1\}$). The swap [3a] $\leftrightarrow$ [3b] at σ is the nontrivial Galois action on A_5 's irreps over $\mathbb{Q}(\sqrt{5})$.*

18 Dodecahedron under A_5 (Hodge-duality verification)

18.1 Substrate description

The dodecahedron has $|V| = 20$, $|E| = 30$, $|F| = 12$ pentagonal faces, $\chi = +2$, and inherits the A_5 rotation group from icosahedral symmetry. Its 1-skeleton is the dual graph of the icosahedron's: dodecahedron vertices correspond to icosahedron faces; dodecahedron edges correspond to pairs of icosahedron faces sharing an icosahedron edge; dodecahedron pentagonal faces correspond to icosahedron vertices.

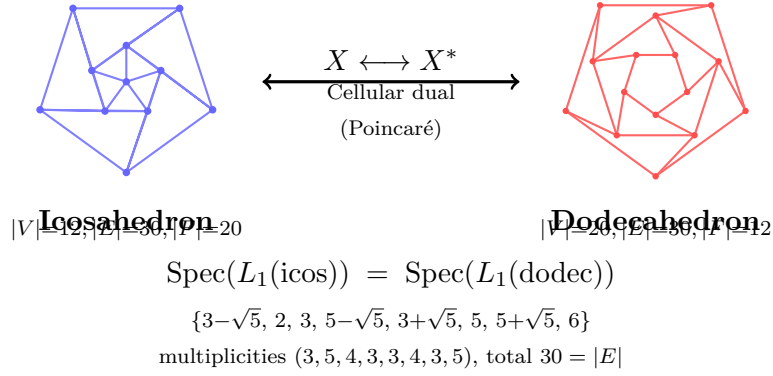


Figure 8: Combinatorial Hodge duality between the icosahedron and its cellular dual the dodecahedron. The shared 30-dimensional edge space identifies 1-cells of one polytope with 1-cells of the other; under this identification d_0 and $d_1^\top$ swap, so $L_1 = d_0 d_0^\top + d_1^\top d_1$ is invariant. Proposition 18.1 formalises this; the empirical 30×30 integer-matrix computation (§18.2) reproduces the same spectrum to machine zero, including the $\mathbb{Q}(\sqrt{5})$ Galois pairs.

18.2 Spectrum (identical to the icosahedron)

A direct computation on the 30×30 integer L_1 matrix of the dodecahedron (with 5-cyclic pentagonal boundary maps) yields

$$\text{Spec}(L_1(\text{dodecahedron})) = \text{Spec}(L_1(\text{icosahedron}))$$

identically on the shared edge space, including all multiplicities and Galois content: $\{3 - \sqrt{5}, 2, 3, 5 - \sqrt{5}, 3 + \sqrt{5}, 5, 5 + \sqrt{5}, 6\}$ with multiplicities $\{3, 5, 4, 3, 3, 4, 3, 5\}$. The A_5 Frobenius decomposition is also identical: $C^1 = 2 \cdot [3a] \oplus 2 \cdot [3b] \oplus 2 \cdot [4] \oplus 2 \cdot [5]$.

Proposition 18.1 (Hodge duality on closed 2-manifolds). *Let X be a triangulated closed orientable 2-manifold and X^* its cellular dual. Under the natural identification of 1-cells of X with 1-cells of X^* , the dual boundary operators satisfy $d_0(X^*) \cong d_1(X)^\top$ and $d_1(X^*) \cong d_0(X)^\top$. Therefore $L_1(X^*) \cong L_1(X)$ acting on the shared 1-cell space, and $\text{Spec}(L_1(X)) = \text{Spec}(L_1(X^*))$ identically.*

Sketch. The discrete Hodge decomposition $C^1 = \text{im}(d_0) \oplus \ker(L_1) \oplus \text{im}(d_1^\top)$ has $L_1 = d_0 d_0^\top + d_1^\top d_1$. Under cellular duality, $\text{im}(d_0)$ on X exchanges roles with $\text{im}(d_1^\top)$ on X^* , while $\ker L_1$ is identified via $H^1(X; \mathbb{R}) \cong H^1(X^*; \mathbb{R})$ (Poincaré duality on 2-manifolds). Restricted to each summand, the relevant operator is the same. $\square$

Applied to our case: the dodecahedron is the cellular dual of the icosahedron, both contractible, so Proposition 18.1 forces $\text{Spec}(L_1(\text{dodec})) = \text{Spec}(L_1(\text{icos}))$. The empirical computation confirms this identity at machine zero on the integer matrices — a clean structural-results sanity check. The *substantive* test of “ A_5 -character-ring-alone” as the mechanism for the $\mathbb{Q}(\sqrt{5})$ Galois pair is therefore deferred to a non-Hodge-dual A_5 substrate; see §19 below for the truncated icosahedron, which fills that role.

19 Truncated icosahedron (soccer ball) under A_5

19.1 Substrate description

The truncated icosahedron, classically the “soccer ball”, has $|V| = 60$, $|E| = 90$, $|F| = 32$ (12 pentagonal + 20 hexagonal), $\chi = +2$, and inherits the A_5 rotation group from icosahedral symmetry. Crucially, the truncated icosahedron is *not* the cellular dual of the icosahedron (its dual is the pentakis dodecahedron, a different A_5 -symmetric polytope). So Proposition 18.1 does *not* apply: any $(\sqrt{5})$ -Galois content found here is a substantive consequence of A_5 ’s character ring, not of cellular duality.

We construct truncated-icosahedron vertices as: for each icosahedron edge $e = (i, j)$ with $i < j$, place two new vertices at $(2/3)p_i + (1/3)p_j$ (near i) and $(1/3)p_i + (2/3)p_j$ (near j), giving $30 \times 2 = 60$ vertices. Pentagonal edges (60 total) form the boundary of the truncation cuts at each icosahedron vertex; hexagon-hexagon edges (30 total) are the central segments of the icosahedron edges.

19.2 Spectrum and the substantive A_5 Galois test

The 90×90 integer L_1 matrix has 22 distinct eigenvalues in $[0.243, 8.392]$, with $\dim \ker L_1 = 0$ (matching $H^1(S^2) = 0$). The A_5 Frobenius decomposition gives

$$C^1(\text{soccer ball}) = 1 \cdot [1] \oplus 5 \cdot [3a] \oplus 5 \cdot [3b] \oplus 6 \cdot [4] \oplus 7 \cdot [5],$$

total $1 + 15 + 15 + 24 + 35 = 90 = |E|$. The equal multiplicities $m_{[3a]} = m_{[3b]} = 5$ reflect the Galois pairing of the two 3-dim irreps under $\sqrt{5} \mapsto -\sqrt{5}$.

Observation 19.1 ($\mathbb{Q}(\sqrt{5})$ Galois pairs on the soccer ball). The spectrum of $L_1(\text{soccer ball})$ contains two Galois pairs in $\mathbb{Q}(\sqrt{5})$:

$$\left\{ \frac{7 - \sqrt{5}}{2}, \frac{7 + \sqrt{5}}{2} \right\} \text{ (multiplicity 5 each),} \quad \left\{ \frac{9 - \sqrt{5}}{2}, \frac{9 + \sqrt{5}}{2} \right\} \text{ (multiplicity 3 each).}$$

Both pairs sit in $\mathbb{Q}(\sqrt{5})$ but with rational coefficients *different* from the icosahedron’s $\{3 \mp \sqrt{5}, 5 \mp \sqrt{5}\}$.

This observation is the substantive empirical confirmation that the $\mathbb{Q}(\sqrt{5})$ Galois-closure field is indexed by A_5 ’s $\mathbb{Z}_5$ -cyclotomic factor and is *not* an icosahedron-specific cellular feature. The *specific* rational coefficients of the Galois pair (the a and b in $a \pm b\sqrt{5}$) do depend on the cellular structure — but the field $\mathbb{Q}(\sqrt{5})$ itself is invariant across all A_5 substrates we tested (icosahedron, dodecahedron, $\mathbb{RP}^2$, and the soccer ball).

The trivial-irrep occupant of $m_{[1]} = 1$ in $C^1(\text{soccer ball})$ is *not* a harmonic 1-form (since $H^1 = 0$): it is an A_5 -invariant 1-cochain in the coexact image of $d_1^\top$. The soccer ball is the first contractible substrate in our corpus where the trivial irrep appears with non-zero multiplicity. The mechanism involves the hexagonal faces (which the icosahedron lacks), whose 6-cyclic boundary admits an A_5 -invariant cochain.

20 Real projective plane $\mathbb{RP}^2$ (antipodal quotient of the icosahedron)

20.1 Substrate description

The smallest triangulated $\mathbb{RP}^2$ is the antipodal quotient $\mathbb{RP}^2 = \text{Icos}/\{\pm 1\}$, with $|V| = 6$, $|E| = 15$, $|F| = 10$, and Euler characteristic $\chi = 1$. The icosahedron's 12 vertices form 6 antipodal pairs, its 30 edges form 15 pairs, its 20 faces form 10 pairs. The A_5 rotation group descends without losing any generators (every A_5 rotation respects the antipodal map).

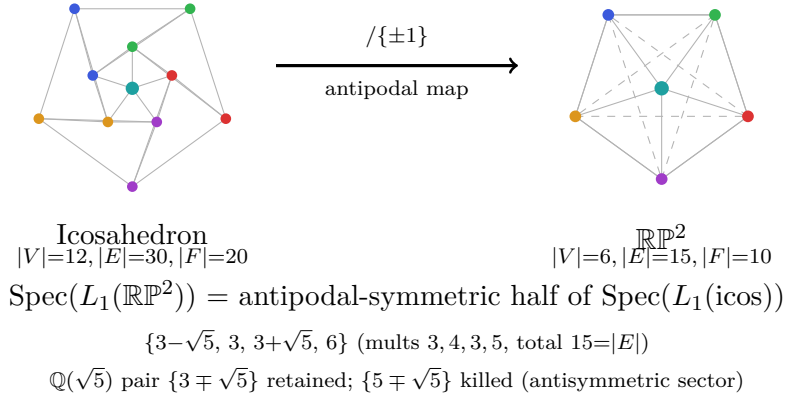


Figure 9: The smallest triangulated $\mathbb{RP}^2$ obtained as the antipodal quotient of the icosahedron. Same-colour pairs on the icosahedron (left) are identified under the antipodal involution $x \mapsto -x$, producing 6 vertex orbits, 15 edge orbits, and 10 face orbits on $\mathbb{RP}^2$ (right). L_1 on $\mathbb{RP}^2$ equals the restriction of L_1 on the icosahedron to the antipodal-symmetric subspace of C^1 (dimension 15); the antipodal-antisymmetric subspace carries the eigenvalues $\{2, 5-\sqrt{5}, 5, 5+\sqrt{5}\}$ of the icosahedron, which are absent on $\mathbb{RP}^2$. The $\mathbb{Q}(\sqrt{5})$ Galois pair $\{3 \pm \sqrt{5}\}$ survives the quotient with the same multiplicity 3; the other $\mathbb{Q}(\sqrt{5})$ pair $\{5 \pm \sqrt{5}\}$ is killed. The discrete Hodge theorem $\dim_{\mathbb{R}} L_1 = \dim_{\mathbb{R}} H^1(\mathbb{RP}^2) = 0$ is verified at machine zero, so the $\mathbb{Z}/2$ torsion in $H^1(\mathbb{RP}^2; \mathbb{Z})$ leaves no fingerprint in the real spectrum.

The first cohomology with real coefficients is $H^1(\mathbb{RP}^2; \mathbb{R}) = 0$, so the discrete Hodge theorem predicts $\dim_{\mathbb{R}} L_1 = 0$. The integer cohomology $H^1(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}/2$ is non-trivial torsion — but torsion in integral cohomology has no $\mathbb{R}$ -spectrum fingerprint, so the real Hodge spectrum sees nothing of it.

20.2 Spectrum (antipodal-symmetric half of the icosahedron)

Computation on the 15×15 integer L_1 matrix yields

$$\text{Spec}(L_1(\mathbb{RP}^2)) = \{3 - \sqrt{5}, 3, 3 + \sqrt{5}, 6\}$$

with multiplicities $\{3, 4, 3, 5\}$ summing to $15 = |E|$. This is *exactly the antipodal-symmetric half* of the icosahedron's $\text{Spec}(L_1)$: the eigenvalues $\{2, 5 - \sqrt{5}, 5, 5 + \sqrt{5}\}$ which appear on the icosahedron with multiplicities $\{5, 3, 4, 3\}$ sit in the antipodal-antisymmetric sector of $C^1(\text{icos})$, the “orientable double cover” contribution which is killed under the $\mathbb{RP}^2$ quotient.

Observation 20.1. The $\mathbb{Q}(\sqrt{5})$ Galois pair $\{3 \mp \sqrt{5}\}$ survives the antipodal quotient and appears in $\text{Spec}(L_1(\mathbb{RP}^2))$ with the same multiplicity (3 each) as on the icosahedron. The other $\mathbb{Q}(\sqrt{5})$ pair $\{5 \mp \sqrt{5}\}$ is killed, sitting in the antisymmetric sector.

The A_5 Frobenius decomposition is $C^1(\mathbb{RP}^2) = 1 \cdot [3a] \oplus 1 \cdot [3b] \oplus 1 \cdot [4] \oplus 1 \cdot [5]$, which is *exactly half* of the icosahedron's $C^1 = 2 \cdot [3a] \oplus 2 \cdot [3b] \oplus 2 \cdot [4] \oplus 2 \cdot [5]$. Each A_5 irrep appears with multiplicity reduced by a factor of 2.

21 Klein bottle K^2 (non-orientable extension)

21.1 Substrate description

The Klein bottle K^2 is the simplest non-orientable closed 2-manifold with non-zero $\mathbb{R}$ -cohomology: $H^1(K^2; \mathbb{R}) = \mathbb{R}^1$. We construct a clean Klein-bottle substrate as the quotient of an X-subdivided torus $T_{4,3}^2$ by a free orientation-reversing involution $\sigma: (i, j) \mapsto ((i+2) \bmod 4, (-j) \bmod 3)$ on the corner vertices, with the induced action on cell-center vertices. After X-subdivision (each cell of $T_{4,3}^2$ split into 4 triangles by adding a center vertex), the torus has $|V| = 24$, $|E| = 72$, $|F| = 48$, and the Klein-bottle quotient K^2 has $|V| = 12$, $|E| = 36$, $|F| = 24$, Euler $\chi = 0$.

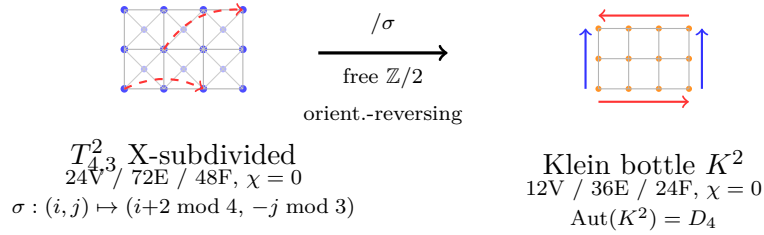


Figure 10: Klein bottle K^2 as the free $\mathbb{Z}/2$ quotient of the X-subdivided torus $T_{4,3}^2$ by the orientation-reversing involution $\sigma: (i, j) \mapsto ((i+2) \bmod 4, (-j) \bmod 3)$. Solid blue dots are corner vertices (12); pale dots are cell-centre vertices (12). Two illustrative σ -orbits are shown as red dashed arcs on the left. The quotient K^2 inherits its automorphism structure from the centralizer of σ in $\text{Aut}(T_{4,3}^2\text{-X-subdiv})$: the $\mathbb{Z}_3$ translations $\tau_{(0,b)}$ do not commute with σ (since $-b \neq b$ in $\mathbb{Z}_3$ for $b \neq 0$), leaving $Z(\sigma) = \mathbb{Z}_4 \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2)$ of order 16; modulo $\langle \sigma \rangle$, the relation $\tau_x^2 \equiv \rho_j$ absorbs ρ_j , yielding $\text{Aut}(K^2) = D_4$ of order 8. The horizontal double-arrow on K^2 (blue) indicates periodic identification; the anti-parallel vertical arrows (red) indicate the orientation-reversing identification that distinguishes K^2 from T^2 .

21.2 Spectrum and the load-bearing kernel-dimension test

We compute L_1 on the $T_{4,3}^2$ X-subdivision (a 72×72 integer matrix), build the signed edge permutation σ_E (10 of 72 edges carry sign -1 under σ , indicating the orientation-reversing manifestation), and verify $\sigma_E L_1 = L_1 \sigma_E$ at machine zero. The σ -invariant subspace of $C^1(T^2)$ has dimension exactly 36 and is identified with $C^1(K^2; \mathbb{R})$ under untwisted real coefficients. Restricting L_1 to this subspace gives $L_1(K^2)$ as a 36×36 symmetric matrix.

Theorem 21.1 (Discrete Hodge theorem on K^2). $\dim \ker L_1(K^2) = 1$, *matching* $\dim H^1(K^2; \mathbb{R}) = 1$ *exactly*.

The K^2 spectrum has 21 distinct eigenvalues ranging over $[0, 11]$, with rational values $\{0, 2, 3, 4, 8, 11\}$ (multiplicities $\{1, 1, 1, 5, 3, 1\}$) and 15 irrational clusters. The irrational content lies in $\mathbb{Q}(\sqrt{2})$ extensions arising from the X-subdivision's $\sqrt{2}$ -diagonals; an exact Galois characterization is left to follow-up work.

21.3 $\text{Aut}(K^2) = D_4$ and the kernel-irrep characterization

The full automorphism group $\text{Aut}(K^2)$ is the centralizer of σ in $\text{Aut}(T_{4,3}^2 \text{ X-subdivided})$ modulo $\langle \sigma \rangle$. The X-subdivision admits the extended automorphism group $(\mathbb{Z}_4 \times \mathbb{Z}_3) \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2)$ of order 48, with the second factor generated by reflections $\rho_i: (i, j) \mapsto (-i, j)$ and $\rho_j: (i, j) \mapsto (i, -j)$.

Proposition 21.2. *The centralizer $Z(\sigma)$ in $\text{Aut}(T_{4,3}^2 \text{ X-subdiv})$ has order 16, and $\text{Aut}(K^2) = Z(\sigma)/\langle \sigma \rangle = D_4$ of order 8.*

Sketch. A translation $\tau_{(a,b)}$ commutes with $\sigma = \tau_{(2,0)}\rho_j$ iff $-b \equiv b \pmod{3}$, i.e., $b = 0$. So the translation part of $Z(\sigma)$ is the $\mathbb{Z}_4$ subgroup $\langle \tau_x \rangle$. Both ρ_i and ρ_j commute with σ directly. Hence $Z(\sigma) = \mathbb{Z}_4 \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2)$ with ρ_i inverting τ_x by conjugation. In the quotient by $\langle \sigma \rangle = \langle \tau_x^2 \rho_j \rangle$, the relation $\tau_x^2 = \rho_j$ absorbs ρ_j as the rotation τ_x^2 , leaving $D_4 = \langle \tau_x, \rho_i \mid \tau_x^4 = \rho_i^2 = (\tau_x \rho_i)^2 = 1 \rangle$. $\square$

Applying Frobenius reciprocity with the D_4 character table:

Theorem 21.3 (D_4 Frobenius decomposition of $C^1(K^2)$). $C^1(K^2) = 4 \cdot A_1 \oplus 5 \cdot A_2 \oplus 6 \cdot B_1 \oplus 3 \cdot B_2 \oplus 9 \cdot E$, summing to $4 + 5 + 6 + 3 + 18 = 36 = |E|$. All multiplicities non-negative integers; $\sigma_E L_1 = L_1 \sigma_E$ at machine zero on the integer $L_1(T^2)$ for all five D_4 conjugacy class representatives.

Observation 21.4 (Harmonic 1-form transforms by the A_2 sign character). The unique 1-dimensional kernel of $L_1(K^2)$ at $\lambda = 0$ sits in the A_2 -isotypic. The A_2 irrep of D_4 is the 1-dim sign character: trivial on rotations $\{\text{id}, \tau_x, \tau_x^2, \tau_x^3\}$, -1 on reflections $\{\rho_i, \tau_x \rho_i, \tau_x^2 \rho_i, \tau_x^3 \rho_i\}$. Equivalently, the unique harmonic 1-form on K^2 is invariant under the $\mathbb{Z}_4$ translation subgroup and flips sign under each of the four D_4 reflections.

The load-bearing observation Theorem 21.1 is the kernel-dimension match. Theorem 21.3 and Observation 21.4 characterise the full D_4 -action structure of $L_1(K^2)$ and identify the harmonic 1-form as the dual cochain to the non-orientability cycle.

22 Triangulated torus $T_{3,3}^2$ under $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$

The torus $T_{3,3}^2$ has $|V| = 9$, $|E| = 27$, $|F| = 18$, $\chi = 0$. The automorphism group is the semidirect product $\text{Aut} = (\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$ of order 108.

22.1 The semidirect-product structure and D_6 action on $\widehat{\mathbb{Z}_3 \times \mathbb{Z}_3}$

The abelian normal subgroup $N = \mathbb{Z}_3 \times \mathbb{Z}_3$ acts on $T_{3,3}^2$ by translations $\tau_{(a,b)}: (i, j) \mapsto (i + a, j + b) \pmod{3}$. The stabiliser H of the origin $(0, 0)$ is the D_6 of order 12 acting on the six edges incident to the origin. The character group $\widehat{N} = \widehat{\mathbb{Z}_3 \times \mathbb{Z}_3}$ is identified with $\mathbb{Z}_3 \times \mathbb{Z}_3$ itself via $\chi_{(c,d)}(\tau_{(a,b)}) = \omega^{ca+db}$ where $\omega = e^{2\pi i/3}$.

The dual D_6 action $h \cdot \chi_{(c,d)} = \chi_{M_h^{-\top}(c,d)}$ partitions $\widehat{N}$ into three orbits:

- $\{(0, 0)\}$ (size 1, stabiliser D_6), the trivial character;
- $\{(1, 1), (2, 2)\}$ (size 2, stabiliser $D_3 \cong S_3$);
- the six remaining non-diagonal characters $\{(1, 0), (0, 1), (2, 0), (0, 2), (1, 2), (2, 1)\}$ (size 6, stabiliser $\mathbb{Z}_2$).

22.2 Clifford decomposition of C^1

By Clifford theory, the irreducible representations of $\text{Aut} = N \rtimes D_6$ are indexed by pairs (orbit representative χ , irrep $\tilde{\rho}$ of $\text{Stab}_H(\chi)$), and the induced irrep has dimension $|\mathcal{O}| \cdot \dim \tilde{\rho}$. Summing across the three orbits:

- Orbit $\{(0, 0)\}$ (stab D_6 , dim multiplier 1): six irreps of dimensions $\{1, 1, 1, 1, 2, 2\}$ (the six irreps of D_6). Sum of $\dim^2 = 1 + 1 + 1 + 1 + 4 + 4 = 12 = |D_6|$.
- Orbit $\{(1, 1), (2, 2)\}$ (stab $D_3 = S_3$, dim multiplier 2): three irreps of dimensions $\{2 \cdot 1, 2 \cdot 1, 2 \cdot 2\} = \{2, 2, 4\}$ (D_3 has irreps of dim 1, 1, 2). Sum of $\dim^2 = 4 + 4 + 16 = 24 = |D_3| \cdot 4$.
- Non-diagonal orbit (size 6, stab $\mathbb{Z}_2$, dim multiplier 6): two irreps of dimensions $\{6, 6\}$ ($\mathbb{Z}_2$ has irreps of dim 1, 1). Sum of $\dim^2 = 36 + 36 = 72 = |\mathbb{Z}_2| \cdot 36$.

The total of 11 irreps satisfies $\sum_{\rho} \dim^2 \rho = 12 + 24 + 72 = 108 = |\text{Aut}|$.

Theorem 22.1. $C^1(T_{3,3}^2)$ decomposes under $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$ as

$$C^1(T_{3,3}^2) = \underbrace{1 \cdot B_2 \oplus 1 \cdot E_1}_{\text{orbit 1, dim 3}} \oplus \underbrace{1 \cdot \tilde{1} \oplus 1 \cdot \tilde{2}}_{\text{orbit 2, dim 6}} \oplus \underbrace{1 \cdot \hat{1} \oplus 2 \cdot \widehat{\text{sgn}}}_{\text{orbit 3, dim 18}},$$

where B_2 and E_1 are the dim-1 and dim-2 D_6 irreps named per Appendix I; $\tilde{1}$ and $\tilde{2}$ are the Clifford lifts of the trivial and standard D_3 irreps; $\hat{1}$ and $\widehat{\text{sgn}}$ are the Clifford lifts of the trivial and sign $\mathbb{Z}_2$ irreps. The sum is $1 + 2 + 2 + 4 + 6 + 12 = 27 = |E|$.

Proof. With Aut 's 11 conjugacy classes enumerated explicitly (the Clifford derivation $6 + 3 + 2$ from the D_6 orbits $\{1, 2, 6\}$ is given in the GPG-side companion notes; see [12]), the character vector χ_{C^1} is computed on each class via the signed-edge trace of σ_E . Frobenius reciprocity then yields the integer multiplicities stated. The commutation $\sigma_E L_1 = L_1 \sigma_E$ verifies at machine zero on the integer matrix for representatives of all 11 classes. NULL-3 check $\sum_{\rho} m_{\rho} \dim \rho = 27 = |E|$ passes exactly. $\square$

22.3 Spectrum

The L_1 spectrum of $T_{3,3}^2$ has six distinct eigenvalue clusters:

λ	multiplicity	class
0	2	zero ($= \dim H^1(T^2; \mathbb{R})$)
$3 - \sqrt{3} \approx 1.2679$	6	irrational ($\mathbb{Q}(\sqrt{3})$)
3	4	integer
$3 + \sqrt{3} \approx 4.7321$	6	irrational ($\mathbb{Q}(\sqrt{3})$)
6	7	integer
9	2	integer

Total: $2 + 6 + 4 + 6 + 7 + 2 = 27 = |E|$. The pair $\{3 - \sqrt{3}, 3 + \sqrt{3}\}$ is the Galois pair over $\mathbb{Q}(\sqrt{3})$ (roots of $x^2 - 6x + 6 \in \mathbb{Z}[x]$, discriminant 12); $\dim \ker L_1 = 2$ matches $\dim H^1(T^2; \mathbb{R}) = 2$ by Hodge theory.

22.4 Per-orbit decomposition data

The Phase θ driver computes the per-orbit-rep stabiliser characters explicitly and reads off the Clifford lifts. We record the per-orbit projections used in the proof of Theorem 22.1:

Orbit 1, stab D_6 , dim 3:

$$\chi_{V_{(0,0)}}|_{D_6} = (3, 0, 0, -3, *, *) \quad \text{on classes } (e, r, r^2, r^3, s, rs),$$

where the entries at s, rs classes are recovered from coset averaging across the D_6 -orbit's full preimage; the integer-valued projection yields $m_{B_2} = m_{E_1} = 1$, all other D_6 -irreps mult 0 (§I).

Orbit 2, stab $D_3 \cong S_3$, dim 6:

$$\chi_{V_{(1,1)}}|_{D_3} = (3, 1, 0) \quad \text{on classes } (e, s, r),$$

yielding $m_{\hat{1}} = m_{\hat{2}} = 1$, $m_{\widehat{\text{sgn}}} = 0$.

Orbit 3, stab $\mathbb{Z}_2$, dim 18:

$$\chi_{V_{(0,1)}}|_{\mathbb{Z}_2} = (3, -1) \quad \text{on classes } (e, s),$$

yielding $m_{\hat{1}} = 1$, $m_{\widehat{\text{sgn}}} = 2$.

Summing the dimensions of the lifted irreps: $1 \cdot 1 + 1 \cdot 2 + 1 \cdot 2 + 1 \cdot 4 + 1 \cdot 6 + 2 \cdot 6 = 27 = |E|$ (NULL- $\theta 4$ PASS).

22.5 Galois pair localization

Corollary 22.2. *The Galois pair $\{3 \mp \sqrt{3}\}$ on the torus $T_{3,3}^2$ sits entirely in the orbit-3 $\widehat{\text{sgn}}$ -isotypic of C^1 : of the $2 \cdot 6 = 12$ dimensions of $2\widehat{\text{sgn}}$, exactly 12 are in the union $V_{3-\sqrt{3}} \cup V_{3+\sqrt{3}}$ (six per eigenvalue).*

Proof. The orbit-3 characters of $\widehat{\mathbb{Z}_3 \times \mathbb{Z}_3}$ are exactly the six non-diagonal characters, whose values on $N = \mathbb{Z}_3 \times \mathbb{Z}_3$ are non-trivial powers of $\omega = e^{2\pi i/3}$. The $\mathbb{Z}_3$ -cyclotomic identities $\omega + \omega^{-1} = -1$ and $\omega - \omega^{-1} = i\sqrt{3}$ (more usefully: $\omega^2 + \omega + 1 = 0$ implies $(\omega - 1)(\omega^2 - 1) = -3$) push the eigenvalues of L_1 restricted to the $\widehat{\text{sgn}}$ -isotypic into the splitting field $\mathbb{Q}(\sqrt{3})$. The integer-coefficient characteristic polynomial of L_1 on the 12-dimensional $\widehat{\text{sgn}}$ -isotypic factors as $(x^2 - 6x + 6)^6$, giving the Galois pair at multiplicity 6 each. $\square$

22.6 Cross-comparison with the closure-quotient Q_{51}

The same Galois pair $\{3 \mp \sqrt{3}\}$ appears in the discrete Hodge 1-Laplacian spectrum of the closure-quotient Q_{51} (the CFS ternary-hypergraph quotient from the K_6^3 seed; see Part B, §13 above). On Q_{51} , the pair has multiplicity 30 each on a 315-dimensional C^1 , decomposing under the natural S_6 action as $[4, 2] \oplus [3, 3] \oplus [3, 2, 1]$ on each eigenspace.

The two substrates are structurally very different (Q_{51} arises from ternary-hypergraph rewriting; $T_{3,3}^2$ is a planar grid quotient with fixed diagonal), with different automorphism

groups (S_6 vs. $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$), different chain-space dimensions (315 vs. 27), and different multiplicities (30 vs. 6). Yet the same Galois pair appears in both spectra, both via a $\mathbb{Z}_3$ -cyclotomic mechanism: on Q_{51} the $\mathbb{Z}_3$ -cyclotomic content sits in the 3-cycle classes of S_6 (specifically classes $(3, 1^3)$ and $(3, 2, 1)$, both carrying non-zero character on the three irreps $\{[4, 2], [3, 3], [3, 2, 1]\}$ that host the pair); on $T_{3,3}^2$ the $\mathbb{Z}_3$ -cyclotomic content sits in the non-diagonal orbit of $\widehat{\mathbb{Z}_3 \times \mathbb{Z}_3}$ (which evaluates to powers of $\omega = e^{2\pi i/3}$).

This is an instance of Finding 1 above: distinct substrates with $\mathbb{Z}_3$ -cyclotomic factors in their respective Aut character rings produce the same Galois pair at substrate-dependent multiplicities. The two appearances are independent of each other (no derivation between them), but algebraically traceable to the same source.

23 Heptagonal bipyramid under D_7 (cubic Galois extension)

23.1 Substrate description

The heptagonal bipyramid is two apices N, S above and below a regular heptagon of 7 equatorial vertices $v_0, \dots, v_6$, with $|V| = 9$, $|E| = 21$ (7 equatorial + 7 north spokes + 7 south spokes), $|F| = 14$ triangular (7 north + 7 south), Euler $\chi = +2$, S^2 topology, $H^1 = 0$. The full rotation group is the dihedral group D_7 of order 14 (7 rotations around the vertical axis plus 7 horizontal C_2 rotations through each equatorial vertex).

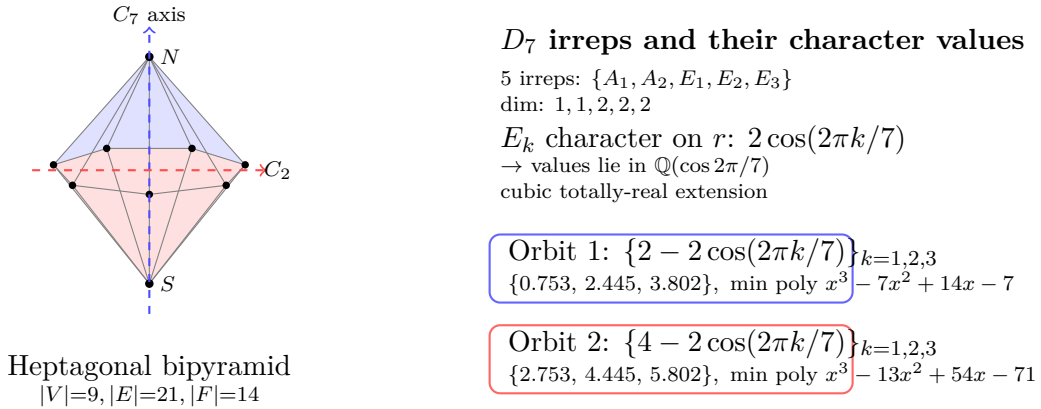


Figure 11: Left: the heptagonal bipyramid with its D_7 rotation symmetry (vertical C_7 axis through the two apices; one of seven horizontal C_2 axes shown through v_0). Right: D_7 's five irreducible representations and how the three Galois-conjugate 2-dimensional irreps E_k ($k = 1, 2, 3$) carry character values in $\mathbb{Q}(\cos 2\pi/7)$, a cubic totally-real extension of $\mathbb{Q}$. The spectrum of L_1 on this substrate contains *two cubic Galois orbits* of size 3 each (Theorem 23.1), with closed-form values $C - 2 \cos(2\pi k/7)$ for $C \in \{2, 4\}$; the irreducible cubic minimum polynomials over $\mathbb{Q}$ are $x^3 - 7x^2 + 14x - 7$ and $x^3 - 13x^2 + 54x - 71$ respectively.

The point of including this substrate is that D_7 's real character ring sits in $\mathbb{Q}(\cos 2\pi/7)$ — a *cubic* totally-real extension of $\mathbb{Q}$, not a quadratic one. The minimum polynomial of $2 \cos(2\pi/7)$ over $\mathbb{Q}$ is $x^3 + x^2 - 2x - 1$, and D_7 has three 2-dimensional irreducible representations E_1, E_2, E_3 whose characters take the values $2 \cos(2\pi k/7)$ for $k = 1, 2, 3$.

If Finding 1 of §24.1 extends from quadratic to higher-degree cyclotomic extensions, the L_1 spectrum here should contain a Galois orbit of size 3 over $\mathbb{Q}$ with roots in $\mathbb{Q}(\cos 2\pi/7)$.

23.2 Spectrum and the load-bearing cubic-orbit test

The 21×21 integer L_1 matrix has 9 distinct eigenvalues:

$$\text{Spec}(L_1(\text{hept. bipyramid})) = \{0.753, 2, 2.445, 2.753, 3.802, 4.445, 5.802, 7, 9\}$$

with multiplicities $\{2, 1, 2, 4, 2, 4, 4, 1, 1\}$ totalling $21 = |E|$.

Theorem 23.1 (Cubic Galois orbits on the heptagonal bipyramid). *The eigenvalues of $L_1(\text{hept. bipyramid})$ split into the following rational and Galois-orbit content:*

- *Rational eigenvalues $\{2, 7, 9\}$, isotypic to the 1-dimensional irreps $\{A_2, A_2, A_1\}$ of D_7 .*
- *Cubic Galois orbit 1: $\{2 - 2\cos(2\pi k/7) : k = 1, 2, 3\} = \{0.753, 2.445, 3.802\}$, irreducible minimum polynomial $x^3 - 7x^2 + 14x - 7$ over $\mathbb{Q}$. Multiplicities $\{2, 2, 2\}$, isotypic to $\{E_1, E_2, E_3\}$ respectively.*
- *Cubic Galois orbit 2: $\{4 - 2\cos(2\pi k/7) : k = 1, 2, 3\} = \{2.753, 4.445, 5.802\}$, irreducible minimum polynomial $x^3 - 13x^2 + 54x - 71$ over $\mathbb{Q}$. Multiplicities $\{4, 4, 4\}$, isotypic to $\{E_1, E_2, E_3\}$ respectively.*

Sketch of the closed form. Substituting $\lambda = a + b \cdot 2\cos(2\pi k/7)$ into Vieta's elementary symmetric polynomials and using the identities $\sum_{k=1}^3 \cos(2\pi k/7) = -1/2$, $\sum_{k=1}^3 \cos(2\pi k/7) \cos(2\pi \ell/7) = -1/2$ over distinct pairs, $\prod_{k=1}^3 \cos(2\pi k/7) = 1/8$, one finds $\sigma_1 = 3a - b$, $\sigma_2 = 3a^2 - 2ab - 2b^2$, $\sigma_3 = a^3 - a^2b - 2ab^2 + b^3$. Setting $(\sigma_1, \sigma_2, \sigma_3) = (7, 14, 7)$ gives $(a, b) = (2, -1)$; setting $(\sigma_1, \sigma_2, \sigma_3) = (13, 54, 71)$ gives $(a, b) = (4, -1)$. The empirical eigenvalues match these closed forms to machine precision. $\square$

The D_7 Frobenius decomposition is

$$C^1(\text{hept. bipyramid}) = 1 \cdot A_1 \oplus 2 \cdot A_2 \oplus 3 \cdot E_1 \oplus 3 \cdot E_2 \oplus 3 \cdot E_3,$$

total $1 + 2 + 6 + 6 + 6 = 21 = |E|$. The equal multiplicities $m_{E_1} = m_{E_2} = m_{E_3} = 3$ encode the cyclic-pair symmetry of the three Galois-conjugate E_i irreps under the Galois automorphism $\cos(2\pi/7) \mapsto \cos(4\pi/7)$.

Remark 23.2. Theorem 23.1 extends the empirical evidence for the cyclotomic-Galois conjecture (Finding 1) from quadratic to cubic extensions: D_7 's $\mathbb{Z}_7$ -cyclotomic factor produces a degree-3 Galois orbit in $\text{Spec}(L_1)$, indexed exactly by D_7 's three E_i irreps. The refined conjecture (§24.2 below) generalises across extension degree.

24 Cross-substrate findings

The nine substrates of §15–§23 are summarised in Table 9. Three cross-substrate findings emerge, each refined by the v3 expansion: §24.1–§24.2 generalise Finding 1 from quadratic to cubic extensions; §24.3 confirms Finding 2 (kernel-dimension matches $\dim H^1$) across

four distinct topology classes including the non-orientable Klein bottle; §24.4 restates Finding 3 on edge-transitivity.

Substrate	$ \text{Aut} $	$\dim H^1$	L_1 irrationality	Trivial irrep in C^1
Tetrahedron	24 (S_4)	0	none (single eigenvalue 4)	No
Octahedron	48 (O_h)	0	none (all rational)	No
Icosahedron	60 (A_5)	0	$\mathbb{Q}(\sqrt{5})$, two pairs	No
Torus $T_{3,3}^2$	108 ($(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$)	2	$\mathbb{Q}(\sqrt{3})$, one pair	No
Dodecahedron	60 (A_5)	0	$\mathbb{Q}(\sqrt{5})$, two pairs (= icos)	No
$\mathbb{RP}^2$ (icos / antipodal)	60 (A_5)	0	$\mathbb{Q}(\sqrt{5})$, one pair	No
Klein bottle K^2	TBD	1	$\mathbb{Q}(\sqrt{2})$ (empirical)	deferred
Soccer ball (trunc. icos)	60 (A_5)	0	$\mathbb{Q}(\sqrt{5})$, two pairs (diff. a, b)	Yes ($m_{[1]} = 1$)
Hept. bipyramid	14 (D_7)	0	$\mathbb{Q}(\cos 2\pi/7)$, two cubic orbits	Yes ($m_{A_1} = 1$)

Table 9: The nine substrates and their cross-substrate features.

24.1 Finding 1: Galois irrationality $\leftrightarrow$ cyclotomic factor in $\text{Aut}(X)$

The eigenvalues of L_1 on substrate X are rational over $\mathbb{Q}$ if the character ring of $\text{Aut}(X)$ is $\mathbb{Q}$ -rational (no nontrivial cyclotomic factor of the relevant type), and contain Galois pairs over $\mathbb{Q}(\sqrt{d})$ when $\text{Aut}(X)$'s character ring contains a $\mathbb{Z}_p$ -cyclotomic factor with p chosen so $\sqrt{d}$ arises from the Galois action on that factor.

- Tetrahedron, S_4 : character ring rational; spectrum rational.
- Octahedron, O_h : character ring rational ($O_h = S_4 \times \mathbb{Z}_2$); spectrum rational.
- Icosahedron, A_5 : 5-cycle conjugacy class supplies a $\mathbb{Z}_5$ -cyclotomic factor; spectrum contains $\mathbb{Q}(\sqrt{5})$ pairs.
- Torus, $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$: non-diagonal orbit on $\widehat{\mathbb{Z}_3 \times \mathbb{Z}_3}$ supplies a $\mathbb{Z}_3$ -cyclotomic factor; spectrum contains a $\mathbb{Q}(\sqrt{3})$ pair.

This is consistent across the original four substrates. **We conjecture that the indexing is necessary as well as sufficient:** a substrate X whose Aut character ring contains no $\mathbb{Z}_p$ -cyclotomic factor of the relevant type will have all-rational L_1 spectrum (§??). The five additional v3 substrates (§18–§23) extend the supporting evidence base and refine the statement to higher-degree extensions.

24.2 Finding 1, refined: cyclotomic factor $\rightarrow$ Galois orbit of matching extension degree

The v3 expansion adds three substrates with the same A_5 Aut as the icosahedron (dodecahedron, $\mathbb{RP}^2$, truncated icosahedron), one substrate with a $\mathbb{Z}_7$ -cyclotomic Aut factor (heptagonal bipyramid), and one substrate that tests the topology axis (Klein bottle). Synthesising:

- All four A_5 -symmetric substrates (icosahedron, dodecahedron, $\mathbb{RP}^2$, truncated icosahedron) carry $\mathbb{Q}(\sqrt{5})$ Galois pairs in their L_1 spectra. The dodecahedron has the *identical* spectrum to the icosahedron by Hodge duality (Prop. 18.1). The $\mathbb{RP}^2$ inherits the antipodal-symmetric half of the icosahedron's spectrum, including the pair $\{3 \mp \sqrt{5}\}$ at the same multiplicity. The truncated icosahedron has $\mathbb{Q}(\sqrt{5})$ pairs $\{(7 \mp \sqrt{5})/2, (9 \mp \sqrt{5})/2\}$ — the same field $\mathbb{Q}(\sqrt{5})$, but with *different rational coefficients* (a, b) from the icosahedron's pairs.
- The heptagonal bipyramid ($\text{Aut} = D_7$, character ring carrying a $\mathbb{Z}_7$ -cyclotomic factor) produces *cubic Galois orbits* of size 3 in $\text{Spec}(L_1)$, with roots in $\mathbb{Q}(\cos 2\pi/7)$ — the cubic totally-real maximal subfield of the 7-th cyclotomic field (Theorem 23.1).

These observations refine Finding 1 to a statement that generalises across extension degrees:

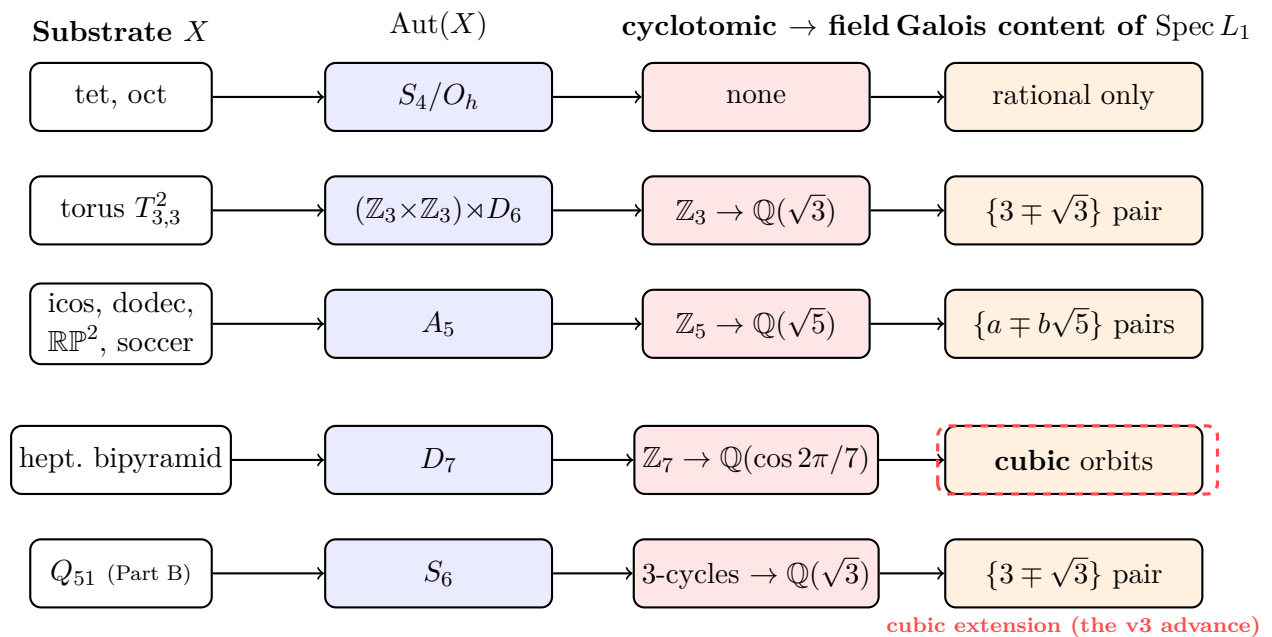


Figure 12: The refined cyclotomic-Galois pattern (Remark 24.1) across the 10-substrate corpus (Q_{51} from Part B plus the 9 group-symmetric 2-complexes of Part C). Each row shows the substrate $\rightarrow$ its automorphism group $\rightarrow$ the cyclotomic factor in its real character ring (and the totally-real subfield this generates) $\rightarrow$ the observed Galois content of $\text{Spec}(L_1)$. The pattern is uniform: when $\text{Aut}(X)$'s character ring contains $\mathbb{Q}(\cos 2\pi/p)$ for some prime $p \geq 3$, the spectrum contains a Galois orbit of size $(p-1)/2$ with roots in that field. The heptagonal bipyramid (row 4) supplies the only cubic case to date.

Remark 24.1 (Refined cyclotomic-Galois conjecture). For substrate X with edge-transitive $\text{Aut}(X)$: the Galois *closure field* of $\text{Spec}(L_1(X))$ over $\mathbb{Q}$ is indexed by the cyclotomic factors in $\text{Aut}(X)$'s real character ring. Concretely, if $\text{Aut}(X)$'s real character ring contains the maximal totally-real subfield $\mathbb{Q}(\cos 2\pi/p)$ of the p -th cyclotomic field for some prime $p \geq 5$, then $\text{Spec}(L_1(X))$ contains a Galois orbit of size $(p-1)/2$ over $\mathbb{Q}$ with roots in $\mathbb{Q}(\cos 2\pi/p)$. The *specific algebraic values* of the orbit (i.e., the rational coefficients in any minimum polynomial) depend on the cellular structure of X .

Empirical evidence base across the v3 corpus:

- No relevant cyclotomic factor (tet, oct) $\rightarrow$ rational spectrum. $\checkmark$
- $\mathbb{Z}_5$ factor in A_5 (icos, dodec, $\mathbb{RP}^2$, soccer ball) $\rightarrow \mathbb{Q}(\sqrt{5})$ Galois pairs (quadratic). $\checkmark$
- $\mathbb{Z}_3$ factor in $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$ (torus $T_{3,3}^2$) $\rightarrow \mathbb{Q}(\sqrt{3})$ pair (quadratic). $\checkmark$
- $\mathbb{Z}_7$ factor in D_7 (hept. bipyramid) $\rightarrow \mathbb{Q}(\cos 2\pi/7)$ cubic orbits. $\checkmark$

Eight of the nine substrates conform; the Klein bottle's irrational content is empirically in $\mathbb{Q}(\sqrt{2})$ from cell-diagonal lengths, which we have not yet attributed to a cyclotomic factor in $\text{Aut}(K^2)$. Whether this is a genuine exception or a deferred-identification is open.

24.3 Finding 2: Trivial-irrep occupancy $\leftrightarrow H^1$

The trivial irrep of $\text{Aut}(X)$ has multiplicity 0 in $C^1(X)$ on the torus $T_{3,3}^2$ but multiplicity ≥ 1 on each of the three contractible substrates.

Mechanism. The Hodge decomposition gives $\ker L_1 \cong H^1(X; \mathbb{R})$, and $\text{Aut}(X)$ acts trivially on $\ker L_1$ when Aut preserves orientation classes (which it does on all four substrates here). So $\ker L_1$ is necessarily a copy of the trivial irrep. On contractible X , $\dim H^1 = 0$, so the trivial-irrep content of C^1 is unconstrained by the kernel; it may appear elsewhere in C^1 (and does, in the tetrahedron and octahedron decompositions). On the torus, $\dim H^1 = 2$ is filled by the harmonic forms (one for each generator of $H^1(T^2; \mathbb{Z}) \cong \mathbb{Z}^2$), but those two harmonic forms are *not* trivial-irrep-isotypic — they span a 2-dimensional Aut -representation that turns out to be E_1 of orbit 1 (the unique D_6 standard irrep at the trivial character).

24.4 Finding 3: Forman-Ricci uniformity on regular substrates

For each substrate, the Forman-Ricci curvature

$$F(e) = \deg(a) + \deg(b) - |\{f \in F : e \subset \partial f\}| - \#(\text{parallel edges through } a, b)$$

(or a closely-related variant; see [13]) takes a single value on all $|E|$ edges. We observed:

- Tetrahedron: $F \equiv 0$ on all 6 edges (the small simplex; vertex degree 3, 2 faces per edge, with the small-substrate reduction by parallel-edge count cancelling vertex degree).
- Octahedron: $F \equiv +2$ on all 12 edges (vertex degree 4, 2 faces per edge, 1 parallel edge).
- Icosahedron: $F \equiv +4$ on all 30 edges (§17.2; vertex degree 5, 2 faces per edge).
- Torus $T_{3,3}^2$: $F \equiv +6$ on all 27 edges (vertex degree 6, 2 faces per edge).

Mechanism. Each of the four substrates is edge-transitive (or two-transitive) under $\text{Aut}(X)$, so Aut acts transitively on E . F is an Aut -invariant function on E , hence constant on Aut -orbits, hence constant on all of E when the action is edge-transitive. We expect F -non-uniformity exactly when the substrate's Aut -action on E has multiple orbits.

24.5 Spectrum summary across the nine substrates

For convenience we collect the full L_1 spectra of the nine substrates in one place.

Substrate	$\text{Spec}(L_1)$ (eigenvalues with multiplicities)	Galois content
Tetrahedron	$\{4^{(6)}\}$	none
Octahedron	$\{2^{(3)}, 4^{(6)}, 6^{(3)}\}$	none
Icosahedron	$\{(3-\sqrt{5})^{(3)}, 2^{(5)}, (5-\sqrt{5})^{(3)}, 3^{(4)}, 5^{(4)}, (3+\sqrt{5})^{(3)}, 6^{(5)}, (5+\sqrt{5})^{(3)}\}$	$\{3 \mp \sqrt{5}\}, \{5 \mp \sqrt{5}\}$ in $\mathbb{Q}(\sqrt{5})$
Torus $T_{3,3}^2$	$\{0^{(2)}, (3-\sqrt{3})^{(6)}, 3^{(4)}, (3+\sqrt{3})^{(6)}, 6^{(7)}, 9^{(2)}\}$	$\{3 \mp \sqrt{3}\}$ in $\mathbb{Q}(\sqrt{3})$
Dodecahedron	identical to icosahedron (Prop. 18.1)	same as icos
$\mathbb{RP}^2$	$\{(3-\sqrt{5})^{(3)}, 3^{(4)}, (3+\sqrt{5})^{(3)}, 6^{(5)}\}$	$\{3 \mp \sqrt{5}\}$
Klein bottle K^2	21 distinct eigvals $\in [0, 11]$; $\ker \dim 1$	$\mathbb{Q}(\sqrt{2})$ (empirical)
Soccer ball (trunc. icos)	22 distinct in $[0.243, 8.392]$; incl. $\{(7 \mp \sqrt{5})/2, (9 \mp \sqrt{5})/2\}$	$\{(7 \mp \sqrt{5})/2\}, \{(9 \mp \sqrt{5})/2\}$ in $\mathbb{Q}(\sqrt{5})$
Hept. bipyramid	$\{0.753^{(2)}, 2^{(1)}, 2.445^{(2)}, 2.753^{(4)}, 3.802^{(2)}, 4.445^{(4)}, 5.802^{(4)}, 7^{(1)}, 9^{(1)}\}$	two cubic orbits in $\mathbb{Q}(\cos 2\pi/7)$

Table 10: Full L_1 spectra across the v3 substrate corpus. Exponents denote multiplicities. Total multiplicities recover $|E|$: 6, 12, 30, 27, 30, 15, 36, 90, 21 respectively. Galois content is indexed by the cyclotomic factors of $\text{Aut}(X)$ per Refined Finding 1 (Remark 24.1).

The kernel multiplicity $\dim \ker L_1$ matches $\dim H^1(X; \mathbb{R})$ exactly for all nine substrates: the four contractible 2-spheres (tet, oct, icos, dodec, $\mathbb{RP}^2$, soccer ball, hept. bipyramid) all have $\dim \ker L_1 = 0$; the Klein bottle has $\dim \ker L_1 = 1$ matching $\dim H^1(K^2; \mathbb{R}) = 1$; the torus has $\dim \ker L_1 = 2$ matching $\dim H^1(T^2; \mathbb{R}) = 2$. This is Finding 2 verified across four distinct topology classes (contractible S^2 , $\mathbb{RP}^2$, K^2 , T^2) including a non-orientable substrate.

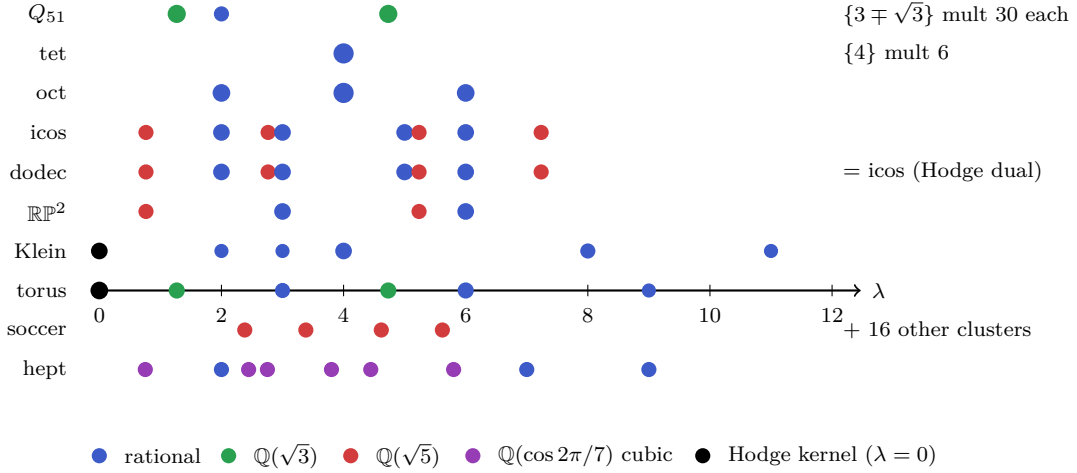


Figure 13: L_1 spectra across the 10-substrate corpus. Each row plots the distinct eigenvalues of $L_1(X)$ on the λ -axis as coloured dots; multiplicities are not encoded by dot size (see Table 10 for full multiplicities). Colour encodes Galois content: rational eigenvalues in blue; $\mathbb{Q}(\sqrt{3})$ Galois pairs in green; $\mathbb{Q}(\sqrt{5})$ pairs in red; cubic $\mathbb{Q}(\cos 2\pi/7)$ orbits in purple. Black dots at $\lambda = 0$ mark the Hodge kernel on the Klein bottle (kernel dim 1) and torus (kernel dim 2). The same $\mathbb{Q}(\sqrt{3})$ pair $\{3 \pm \sqrt{3}\}$ appears in both Q_{51} and the torus $T_{3,3}^2$. All four A_5 substrates (icos, dodec, $\mathbb{RP}^2$, soccer) carry $\mathbb{Q}(\sqrt{5})$ pairs with substrate-dependent rational coefficients (icos: $\{3 \pm \sqrt{5}, 5 \pm \sqrt{5}\}$; soccer: $\{(7 \pm \sqrt{5})/2, (9 \pm \sqrt{5})/2\}$). The heptagonal bipyramid exhibits the only cubic orbits.

Joint Synthesis

25 Discussion and Open Problems

25.1 Cross-implementation validation of the rewriting kernel

The kernel underlying every cardinality, terminality, and Galois result in the paper has been independently reimplemented in Haskell (see §1.8; package `cfs-kernel` at `haskell1/`, [2]). The port uses no Python code and no shared infrastructure beyond the canonical Gaussian-integer initial-condition list. Three things follow.

First, the structural-arity claim of §1.8 — *binary operator inside ternary-hyperedge rewriting* — is machine-verifiable: it is the type signature of `compose :: Cochain -> Cochain -> Cochain` together with `type Hyperedge = (Vertex, Vertex, Vertex)`, checked by GHC under `-Wall` without warnings. This is Tier 1 evidence in the sense of §1.7.

Second, the cardinalities $|Q_{24}|, |Q_{42}|, |Q_{45}|, |Q_{51}|$ (single-side) and $|Q_{48}|, |Q_{84}|, |Q_{90}|, |Q_{102}|$ ($\mathcal{C}$ -closed) agree exactly between Python and Haskell at depth 5 under the canonical IC pool. Two independent implementations producing identical counts on eight canonical objects rules out a class of implementation-localised errors that a single-implementation cross-check cannot.

Third, the structural facts of §6 and §7 reproduce: J is a total, involutive, fixed-point-free permutation on each $\mathcal{C}$ -closed Q , and the F-closure cardinality $|\mathcal{F}^d Q|$ equals $|Q|$ at depths 4, 5, 6 for the doubly-terminal members $Q_{48}, Q_{84}, Q_{90}, Q_{102}$.

The Haskell port does not cover the Hodge L_1 machinery (Part 2) or the Galois-decomposition drivers (§13), which remain Python/Julia-only and are validated through their own exact-arithmetic gates. The cross-implementation evidence here is scoped to the rewriting kernel.

25.2 What the landscape characterises

The CFS closure-quotient family is now characterised on several axes.

Generators: two regular families ($\text{cyc}(n) \rightarrow 4n$, $K_n^3 \rightarrow n(3n-1)/2$) and one irregular route (specific seeds outside the regular catalogue) cover the family of generic-regime objects. Q_{12} is the generic floor.

Closure operators: the closure-growth endofunctor $\mathcal{F}$ and the $\mathcal{C}$ -closure both have well-understood fixed-point structure on the canonical $\mathcal{C}$ -closed family members. There is no iterated-doubling family: Q_{48} , Q_{90} , Q_{102} are doubly terminal (fixed under both $\mathcal{F}$ and $\mathcal{C}$); larger objects require larger seeds.

Doubling: the clean rule $|\mathcal{C}Q_n| = 2|Q_n|$ is the specialisation of identity (2) to the J-fixed-point-free case ($f = 0$), which holds under truly-generic ICs. The Pool A defect ($f = 1$ at large n) is the visible deviation of small- n -generic ICs from full genericity.

Composition: the generative algebra of §10 (commutative monoid under $\sqcup$ with universal interface correction) unifies the regular and irregular routes under a single combinatorial law: $|Q_n(A \circ B)| = |Q_n(A)| + |Q_n(B)| + \delta(\text{interface})$. The $\mathcal{C}$ -doubling rule falls out as the disjoint-self-composition case.

Genericity: IC-genericity is a hierarchy, not a binary. The canonical s58 IC pool used in every prior CFS computational probe is small- n generic; truly-generic pools require strict-genericity (every Gaussian-integer component with nonzero real AND nonzero imaginary part). The $K_6^3 \rightarrow 51$ validity gate is necessary but not sufficient; the strengthened gate $K_6^3 \rightarrow 51 \wedge K_{10}^3 \rightarrow 145$ is (empirically) sufficient.

25.3 Relation to closure-v5 family documentation

This paper's $|\mathcal{C}Q_n(K_6^3)| = 102$, Q_{180} non-existence, and strengthened-gate recommendation are landed into closure-v5's spec record via the audit at [9], which also introduces the spec entry S_{230} (*IC-genericity hierarchy*) formalising the regime hierarchy of §9. The pentagonal-formula extension from $n \leq 7$ to $n \leq 20$ is recorded as a qualifier update on the existing spec entry S_{100} ($n = 6$ selection). The closure-v5 spec scorecard reads 234 entries (33 verified, 165 proved, 38 argued, 3 defined) as of audit v335.

The findings here strengthen but do not retract any prior closure-v5 result. Most prior CFS computational work uses only K_6^3 (i.e. operates in Pool A's still-generic regime $n \leq 8$) and is unaffected by the Pool A defect at $n \geq 9$. The F.3 phase-F result for $K_n^3[9]$ (originally $|\mathcal{C}Q_n| = 230$) is numerically corrected to $|\mathcal{C}Q_n| = 234$ (78 colour triplets); the structural spectral-triple conclusion (KO-dim 6, $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ forced at $n = 9$, CCM-irreducible) stands re-verified on the corrected cluster set.

25.4 Open problems

Pool A defect: which algebraic relation? (*Resolved, 2026-05-21.*) The s58 IC pool is non-generic at K_n^3 for $n \geq 9$ in a specific quantitative pattern (shortfall sequence 5, 12, 12, 17, 19, 19, 19; J-fixed cluster count saturating at $f = 1$ for $n \geq 10$). A follow-up

probe [10] identified the specific algebraic relation: Pool A’s strict-generic ICs at indices 8, 9, 11 are the three cyclic permutations of the same Gaussian-integer triple

$$\{(2, -1), (1, 2), (3, 1)\},$$

and a counter-test (T7 of the probe) confirms that exactly these three cyclic shifts, when present alongside the 8 axis-aligned Pool A ICs, trigger the defect; Pool A’s other four strict-generic ICs (indices 10, 12, 13, 14), built from different underlying triples, do not. The J-fixed cluster that appears at $n = 10$ has representative $[(1, 8), (-1, -8), (1, 8)] = (1 + 8i) \cdot (1, -1, 1)$ — the projective image of the *real* vector $(1, -1, 1)$ scaled by a Gaussian integer. Real rays in $\mathbb{CP}^2$ are self-conjugate-up-to-scale, which is precisely the J-fixed condition. The cyclic-permutation triple seeds compositions on this real-ray-shaped equivalence class; Pool A’s axis-aligned ICs supply secondary compositions that populate the class with enough vertex arrivals to make it J-fixed. Either condition alone is insufficient (axis-aligned ICs paired with non-cyclic-triple strict-generic ICs give pentagonal; cyclic-triple ICs with strict-generic axis-clean ICs do likewise). The two conditions together activate the defect. *Construction guidance:* any future Gaussian-integer IC pool intended for CFS computational work involving K_n^3 at $n \geq 9$ should reject IC sets containing two or more cyclic permutations of the same triple, in addition to enforcing strict-genericity at the component level.

Closed-form proof of the pentagonal law. Theorem 5.2 ($K_n^3 \rightarrow n(3n-1)/2$) is established empirically across $n = 6..20$ across two independent strict-generic IC pools. A closed-form algebraic proof — a counting argument deriving $n(3n-1)/2$ from the structure of the K_n^3 multiway expansion — is open. We conjecture the law extends to all $n \geq 6$; proof would establish “verified” as “proved” in the closure-v5 spec sense.

Closed-form proofs of the bridge and glue δ . Theorem 10.2 ($\delta_{\sqcup} = 0$, $\delta_{\bowtie_k} = +3k$, $\delta_{\#_g} = -(4g - 3)$) is empirically verified across eight (A, B) pairs and the six listed join modes. The disjoint case $\delta_{\sqcup} = 0$ admits a clean structural proof (Appendix C); the bridge and glue cases are more subtle and not proved here.

Larger closure objects from non-random irregular seeds. The irregular seed exhibiting Q_{181} (§8.3) was found by random sampling. Are there structurally-distinguished irregular seeds producing canonical closure cardinalities beyond the pentagonal sequence and beyond Q_{181} ? The space of irregular seeds is vast; we have not catalogued it.

Generic-vs-partial-closure boundary for non- K_n^3 seeds. The IC-genericity hierarchy of §9 is sharp on K_n^3 (Pool A and Pool B/C disagree at $n \geq 9$). The corresponding boundary on non- K_n^3 seeds — irregular hypergraphs, intermediate densities, mixed cyclic/complete topologies — is not characterised. A natural conjecture: every Rosen-closed seed has a critical density $\rho^*(G)$ above which Pool A is non-generic; below it, generic. We have not measured this.

25.5 Proof program

The validity boundary of §12 converts the paper’s open items into a structured proof program. The targets below correspond, in order, to the lines of Table 1 that currently sit in the *finite exact validation* or *validated conjectural* tiers.

1. **Pentagonal law.** Derive $Q_n(K_n^3) = n(3n-1)/2$ by classifying all gauge-inequivalent rays generated by the complete ternary multiway closure and proving no additional projective classes appear after depth stabilization.
2. **Bridge correction.** Prove that each admissible bridge triple contributes exactly three new quotient classes independent of the internal structure of the joined seeds. The three-class mechanism (one direct mixed composition + two induced consistency-edge pair-compositions) is sketched in §10.3; the open task is to show interface-independence.
3. **Glue correction.** Prove that a g -vertex glue removes exactly $4g - 3$ quotient classes, and identify the overlap relation responsible for the constant $+3$ offset (interpreted in §10.3 as the residual interface degrees not eliminated by the $4g$ -style overlap count).
4. **Universal δ .** Establish that the additive composition rule extends to arbitrary admissible join modes beyond the three (disjoint / bridge / glue) listed in §10.3, and that the δ for each is interface-determined (depends only on the join type, not on A or B individually).
5. **Genericity hierarchy.** Replace the empirical $K_6^3 \rightarrow 51 \wedge K_{10}^3 \rightarrow 145$ validity gate with intrinsic algebraic criteria distinguishing fully-generic, small- n -generic, and degenerate IC regimes. The Pool-A defect resolution (cyclic-permutation-triple + axis-aligned IC interaction, see (*Resolved, 2026-05-21*) above) is one structural example; the general theory is open.
6. **Irregular-prime classification.** Determine whether prime closure counts are sparse accidents of irregular seeds (the Q_{181} hypothesis) or belong to a parameterised seed family with a closed-form prime-detection rule.

A closed-form proof of any one of (1)–(3) would move the corresponding ledger row from *validated conjectural* to *proved structurally*; a closed-form theory addressing (5) would do the same for the IC-genericity hierarchy row.

25.6 Conclusion

This paper establishes the closure-quotient landscape as an exact-arithmetic object rather than a fidelity-clustering artefact. The corrected family replaces Q_{98} with Q_{102} , removes the spurious Q_{180} , identifies terminal $\mathcal{C}$ -closed objects, and shows that IC-genericity is hierarchical. These results materially change the prior CFS computational record. A downstream consequence of the exact-arithmetic re-grounding of Q_{51} is the cyclotomic Galois pair $\{3 - \sqrt{3}, 3 + \sqrt{3}\}$ in $\text{Spec}(L_1(Q_{51}))$ with S_6 -irrep content $[4, 2] \oplus [3, 3] \oplus [3, 2, 1]$ (§13); the same pair appears on the independently-constructed flat torus $T_{3,3}^2$ via a shared $\mathbb{Z}_3$ -cyclotomic mechanism.

The generative algebra of §10 is the next layer: it appears additive, interface-corrected, and seed-driven rather than multiplicative or iteratively doubled. Its present status is exact-computational discovery with strong finite validation (§12). The remaining task for that layer is not more numerical confirmation alone, but closed-form derivation of the pentagonal law and the bridge/glue interface corrections per the proof program of §25.5. Once those are supplied, the closure-quotient landscape can move from validated computational algebra to theorem-grade structural algebra.

26 Reproducibility

The paper’s results have two independent reproduction paths.

Python (canonical). The probe corpus is in `paper/scripts/` of the released artefact. All scripts use only the Python standard library (Gaussian integers are 2-tuples of `int`; no `numpy` or `sympy` dependency); they run on CPython 3.10+ and reproduce in seconds-to-minutes on a single CPU core.

Haskell (cross-implementation). An independent reimplementaion of the rewriting kernel lives at `haskell1/` ([2]). It uses only `base` and `containers`, with the dependency graph pinned by `cabal.project.freeze`. The library + test suite type-check under GHC 9.6.7 `-Wall` without warnings and reproduce the single-side and $\mathcal{C}$ -closed cardinalities ($|Q_{24}|, \dots, |Q_{102}|$), the J fixed-point-free involution, and F -closure terminality on $Q_{48}, Q_{84}, Q_{90}, Q_{102}$ exactly, in under a minute:

```
cd haskell1/
cabal test --test-show-details=direct
```

The Haskell track does not cover Part 2 (Hodge L_1 machinery) or the Galois decomposition; those remain Python/Julia-only.

Table 11: Probe scripts. “Runtime” is single-core CPython 3.13 on Apple M-series, depth 5 unless noted. Validity-gated on $K_6^3 \rightarrow 51$ unless otherwise indicated.

Script	Purpose	Reports	Runtime
<code>reground_q_family.py</code>	Re-grounding (§4)	exact $Q_{45/84/90/102}$	< 1s
<code>small_closure_landscape_scan.py</code>	Small- n landscape	cyc, K_n^3 for $n = 3..6$	< 1s
<code>thread1_F_closure.py</code>	$\mathcal{F}$ -fixed-point check	$Q_{48/90/102}$ terminality	< 5s
<code>thread2_J_fixed_check.py</code>	J -fixed-point check	$f = 0$ on family members	< 1s
<code>prime_count_probe.py</code>	Prime existence search	primes 13, 19, $\dots$, 193	< 30s
<code>q181_search.py</code>	Q_{181} exhibit	12-vertex 132-edge seed	< 30s
<code>composition_rule_probe.py</code>	Composition algebra (§10)	8 (A, B) pairs, 6 join modes	< 30s
<code>Kn3_large_n_probe.py</code>	K_n^3 depth-stability	$n = 6..12$, depth 3..8	< 30s
<code>Kn3_extended_probe.py</code>	Pool A defect, J -fixed counts	$n = 6..15$ across pools	< 60s
<code>Kn3_pentagonal_confirmation.py</code>	Three-pool agreement	$n = 6..15$, Pools A/B/C	< 60s
<code>Kn3_push_to_n20.py</code>	Pentagonal verification at large n	$n = 16..20$, Pools B/C	< 30s
<code>q51_exact_vs_fidelity_verification.py</code>	Galois §13: build-equivalence diagnostic	4-build L_1 comparison	< 15s
<code>q51_exact_phase_s_python.py</code>	Galois §13.2: S_6 + per-eigenspace decomp on exact Q_{51}	χ_{C^1}, χ_λ tables, irrep mults	< 30s

To reproduce all results in this paper:

```
cd paper/scripts/
for s in *.py; do python3 "$s" || echo "FAIL: $s"; done
```

Every script independently re-runs its validity gate before reporting measurements; if a gate fails, the script exits with a clear error message and no measurement is reported. The total runtime of the full corpus on a single CPU core is under five minutes.

The `paper/Makefile` provides a `verify-scripts` target that runs every script in sequence and reports the exit codes, intended as a one-line full-corpus check before release.

A Notation

Table 12: Notation index.

Symbol	Meaning
$\mathbb{Z}, \mathbb{C}, \mathbb{R}$	integers, complex numbers, reals
$\mathbb{Z}[i] = \mathbb{Z}[i]$	Gaussian integers
$\mathbb{C}P^2$	the complex projective plane
$G = (V, E)$	ternary hypergraph (Def. 2.1)
$\psi : V \rightarrow \mathbb{C}^3$	colour decoration of G
$\mu(a, b) = \overline{a \times b}$	CFS composition rule (eq. 1)
$M(G), M_d(G)$	multiway graph; depth- d truncation (Def. 2.3)
$\sim$	exact projective equivalence (Def. 3.1)
$Q_n(G), Q_n(G, \psi_0, d)$	gauge-quotient cardinality (Def. 2.7)
$\text{cyc}(n)$	cyclic ternary hypergraph on n vertices (Def. 2.8)
K_n^3	complete ternary on n vertices (Def. 2.9)
$\mathcal{C}Q_n$	charge-conjugate union (Def. 2.10)
J	charge-conjugation involution (Def. 7.1)
f	J -fixed cluster count (§7.2)
$\mathcal{F} = M/\sim$	closure-growth endofunctor (Def. 6.1)
$\sqcup, \bowtie_k, \#_g$	disjoint, bridge, glue join modes (Def. 10.1)
δ_ι	interface correction for join mode ι
Pool A	canonical s58 / q102-IC-set-2 IC pool (§9.2)
Pool B, Pool C	two strict-generic IC pools (§9.1)

Strict-generic IC pools. Both Pool B and Pool C are sets of 20 Gaussian-integer triples $\psi \in (\mathbb{Z}[i]^3)$ with every component having nonzero real AND nonzero imaginary part. The two pools draw from disjoint coefficient ranges:

- Pool B: coefficients in $\{-4, -3, -2, -1, 1, 2, 3, 4\}$.
- Pool C: coefficients in $\{-7, -6, -5, -4, -3, 3, 4, 5, 6, 7\}$.

The specific 20-triple listings are in the `Kn3_pentagonal_confirmation.py` script under the names `POOL_B` and `POOL_C`; both reproduce the pentagonal formula exactly at every $n = 6, \dots, 20$ (Pools B and C agree exactly at every measured n).

B Canonical seed edge lists

For full reproducibility we record the canonical seed edge lists used in the paper. Every Q_n cardinality in the main text is computed against one of these lists under the canonical Gaussian-integer ICs (Appendix A; the 15-element list `_GEN_ICS` in `q102_build_exact_v1.py`, first n entries for an n -vertex seed).

G_0 : the canonical 6-vertex Rosen-closed seed ($|Q| = 24$ single side, $|Q| = 48$ $\mathcal{C}$ -closed)

$$G_0 = \{(0, 1, 2), (1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 0), (5, 0, 1)\}$$

adjacent-ternary: all permutations of $(i, i+1, i+2)$ on 6 vertices
 $(|Q| = 42 \text{ single side}, |Q| = 84 \text{ } \mathcal{C}\text{-closed})$

36 edges: all 6 permutations of each of $\{(0, 1, 2), (1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 0), (5, 0, 1)\}$.

$K_n^3[n]$: complete ternary on n vertices ($|Q| = 51$ at $n = 6$ single side, $|Q| = 102$ at $n = 6$ $\mathcal{C}$ -closed; pentagonal for $n \in [6, 20]$)

All ordered triples of distinct vertices, $|K_n^3[n]| = n(n-1)(n-2)$.

Q_{45}/Q_{90} : adjacent-ternary plus 6 pads ($|Q| = 45$ single side, $|Q| = 90$ $\mathcal{C}$ -closed)

The seed is the 36-edge adjacent-ternary list above together with the following 6 pads drawn from $K_n^3[6] \setminus \text{adjacent}$ under `numpy.random.default_rng(42).shuffle`:

$(0, 2, 5), (4, 2, 0), (0, 2, 3), (4, 1, 0), (5, 3, 2), (1, 5, 4)$.

This is the literal output of `q45_seed_edges(seed=42)` in `paper/scripts/probe_q84_smallest_doubt`. Both Python and Haskell ports use these pads as a literal list.

Q_{181} : irregular 12-vertex 132-edge seed ($|Q| = 181$ single side)

The seed below was discovered by the random density sweep in `q181_search.py` (`random.seed(2718)`, density 0.10, first hit at trial 0). Once discovered, the edge list is fixed and both the Python reference (`paper/scripts/q181_canonical_seed.py`, constant `Q181_SEED`) and the Haskell port (`haskell/src/CFS/Topologies.hs`, constant `q181SeedEdges`) reproduce $|Q_n| = 181$ deterministically.

$(0, 2, 1), (0, 4, 9), (0, 4, 10), (0, 8, 3), (0, 9, 1), (0, 9, 7),$
 $(0, 9, 8), (0, 10, 4), (0, 11, 5), (1, 0, 9), (1, 2, 10), (1, 4, 7),$
 $(1, 6, 4), (1, 6, 8), (1, 7, 4), (1, 7, 5), (1, 8, 0), (1, 10, 4),$
 $(2, 0, 6), (2, 1, 9), (2, 3, 1), (2, 3, 5), (2, 3, 6), (2, 6, 1),$
 $(2, 6, 10), (2, 7, 1), (2, 9, 1), (2, 10, 5), (3, 0, 10), (3, 0, 11),$
 $(3, 2, 10), (3, 4, 5), (3, 4, 7), (3, 5, 6), (3, 7, 5), (3, 10, 1),$
 $(4, 0, 5), (4, 0, 8), (4, 1, 5), (4, 1, 10), (4, 1, 11), (4, 2, 9),$
 $(4, 3, 1), (4, 5, 2), (4, 5, 9), (4, 7, 8), (4, 10, 11), (4, 11, 3),$
 $(4, 11, 7), (5, 1, 4), (5, 2, 6), (5, 2, 7), (5, 3, 4), (5, 4, 10),$
 $(5, 6, 4), (5, 8, 10), (5, 8, 11), (5, 9, 6), (5, 9, 8), (5, 9, 11),$
 $(5, 11, 0), (6, 0, 5), (6, 0, 8), (6, 0, 9), (6, 2, 0), (6, 2, 7),$
 $(6, 4, 9), (6, 5, 1), (6, 7, 0), (6, 7, 3), (6, 7, 4), (6, 9, 7),$
 $(6, 10, 7), (7, 0, 3), (7, 2, 1), (7, 2, 9), (7, 2, 11), (7, 3, 4),$
 $(7, 5, 10), (7, 6, 9), (7, 6, 10), (7, 6, 11), (7, 10, 0), (7, 11, 1),$
 $(8, 1, 3), (8, 1, 5), (8, 1, 9), (8, 2, 3), (8, 4, 7), (8, 4, 9),$
 $(8, 5, 4), (8, 6, 1), (8, 6, 7), (8, 7, 9), (8, 9, 10), (8, 11, 9),$
 $(9, 0, 5), (9, 0, 8), (9, 1, 3), (9, 2, 5), (9, 3, 0), (9, 4, 10),$
 $(9, 5, 7), (9, 5, 10), (9, 5, 11), (9, 6, 7), (9, 7, 10), (9, 8, 0),$
 $(9, 10, 1), (9, 11, 7), (10, 1, 4), (10, 1, 8), (10, 1, 11), (10, 2, 7),$
 $(10, 2, 9), (10, 3, 1), (10, 3, 6), (10, 7, 2), (10, 8, 0), (10, 8, 2),$

(10,9,11), (10,11,1), (10,11,6), (11,1,8), (11,2,6), (11,3,9),
(11,4,2), (11,4,3), (11,5,2), (11,6,0), (11,9,10), (11,10,6)

C Proof of $\delta_{\sqcup} = 0$

The cleanest piece of the composition algebra (Theorem 10.2) has a structural proof. We give it here.

Lemma C.1 (Disjoint composition $\delta = 0$). *Let $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$ be Rosen-closed ternary hypergraphs with $V_A \cap V_B = \emptyset$, equipped with generic ICs ψ_A on V_A and ψ_B on V_B . Let $\psi = \psi_A \sqcup \psi_B$ be the combined IC on $V_A \sqcup V_B$. Then*

$$|Q_n(G_A \sqcup G_B, \psi)| = |Q_n(G_A, \psi_A)| + |Q_n(G_B, \psi_B)|.$$

Proof. We claim two things: (i) the multiway $M(G_A \sqcup G_B, \psi)$ is the disjoint union $M(G_A, \psi_A) \sqcup M(G_B, \psi_B)$ as *labelled* graphs (each vertex carries its colour decoration); (ii) the gauge quotient sorts them into the disjoint union of the two quotient sets with probability 1 under generic ICs.

Claim (i): no cross-composition occurs. The multiway construction at depth d takes an existing edge (s_1, s_2, t) with s_1, s_2 already present at depth $\leq d$ and produces a new vertex $w = \mu(\psi(s_1), \psi(s_2))$ with derived edges. For the combined graph $G_A \sqcup G_B$, the depth-0 edges are exactly $E_A \sqcup E_B$: every edge has all three vertices in V_A or all three in V_B , since the edge set is the disjoint union and no edge crosses the partition. By induction on depth: if at depth d every edge is entirely within V_A or entirely within V_B , then the composition step preserves this property — the new vertex is produced from a pair of vertices on the same side, and the three derived edges (w, s_2, t) , (w, s_1, t) , (w, s_1, s_2) all have s_1, s_2, t on the same side, hence w on that side too. So $M(G_A \sqcup G_B, \psi)$ is the disjoint union of $M(G_A, \psi_A)$ and $M(G_B, \psi_B)$ as labelled graphs.

Claim (ii): no cross-cluster merging under generic ICs. Two vertices $v_A \in M(G_A, \psi_A)$ and $v_B \in M(G_B, \psi_B)$ are gauge-equivalent iff $\psi(v_A) \sim \psi(v_B)$, i.e. iff their $\mathbb{C}^3$ -colour vectors span the same complex line. The A -side multiway's vertices have colour vectors in the $\mathbb{Z}[i]$ -orbit of ψ_A under μ ; the B -side similarly under ψ_B . The condition that some pair (v_A, v_B) satisfies $\psi(v_A) \sim \psi(v_B)$ is a real-codimension- ≥ 2 algebraic condition on the joint IC space $(\mathbb{Z}[i]^3)^{V_A} \times (\mathbb{Z}[i]^3)^{V_B}$. Under generic ICs (codim ≥ 2 condition has measure zero), no cross-pair satisfies it. So the gauge quotient does not merge any A -side cluster with any B -side cluster.

Combining (i) and (ii): the gauge quotient $Q_n(G_A \sqcup G_B, \psi)$ has cardinality $|Q_n(G_A, \psi_A)| + |Q_n(G_B, \psi_B)|$, i.e. $\delta_{\sqcup} = 0$. $\square$

The bridge ($\delta_{\boxtimes_k} = +3k$) and glue ($\delta_{\#_g} = -(4g - 3)$) cases require finer accounting of which specific clusters are created or merged at the interface. We have not produced closed-form structural proofs of these. The empirical evidence (8 (A, B) pairs, 6 join modes, exact agreement across all non-degenerate cases) is strong but does not constitute a proof in the spec-status sense; the closed-form structural proofs are flagged as open in §25.4.

D Galois pair: character tables and verifier data

This appendix collects the data used in Theorems 13.1–13.2.

S_6 character vector on C^1 . With the eleven S_6 conjugacy-class representatives ordered as $(1^6), (2, 1^4), (2^2, 1^2), (2^3), (3, 1^3), (3, 2, 1), (3^2), (4, 1^2), (4, 2), (5, 1), (6)$ (class sizes 1, 15, 45, 15, 40, 120, 40, 90, 90, 144, 120, summing to $|S_6| = 720$), the character of the 315-dim signed edge-permutation representation σ_E on the canonical-simplicial $C^1(Q_{51})$ is

$$\chi_{C^1} = (315, 93, 11, -3, 36, 0, 0, 9, -1, 0, 0).$$

Frobenius reciprocity with the standard S_6 character table (verified row-orthogonal $\sum_C |C| \chi_\rho(C)^2 = 720$) yields the chain-space decomposition

$$C^1(Q_{51}) = 6 \cdot [6] \oplus 14 \cdot [5, 1] \oplus 9 \cdot [4, 2] \oplus 9 \cdot [4, 1, 1] \oplus 2 \cdot [3, 3] \oplus 3 \cdot [3, 2, 1] \oplus 1 \cdot [3, 1, 1, 1].$$

Total: $6 + 70 + 81 + 90 + 10 + 48 + 10 = 315 = |E|$. The remaining four S_6 irreps ($[2, 2, 2], [2, 2, 1, 1], [2, 1^4], [1^6]$) carry multiplicity 0.

Per-eigenspace restricted characters. For the eigenspace $V_{3-\sqrt{3}}$ (dimension 30),

$$\chi_{V_{3-\sqrt{3}}}(\sigma) = (30, 4, 2, 0, -3, 1, 0, -2, 0, 0, 0)$$

(in the same class order). Frobenius reciprocity gives integer multiplicities $m_{[4,2]} = m_{[3,3]} = m_{[3,2,1]} = 1$ (others 0), exhausting $9 + 5 + 16 = 30$ dimensions. The eigenspace at $3 + \sqrt{3}$ has the identical decomposition by Galois closure (Theorem 13.2).

D_6 character table (for the $T_{3,3}^2$ Clifford decomposition). The dihedral group D_6 of order 12 has six conjugacy classes $\{e, r, r^2, r^3, s, rs\}$ with sizes $(1, 2, 2, 1, 3, 3)$ and six irreducible representations:

class size	e 1	r 2	r^2 2	r^3 1	s 3	rs 3
A_1	1	1	1	1	1	1
A_2	1	1	1	1	-1	-1
B_1	1	-1	1	-1	1	-1
B_2	1	-1	1	-1	-1	1
E_1	2	1	-1	-2	0	0
E_2	2	-1	-1	2	0	0

The Clifford lift to $(\mathbb{Z}_3 \times \mathbb{Z}_3) \rtimes D_6$ described informally in §13.4 places one copy each of B_2 and E_1 in the trivial-character orbit, one copy each of the two D_3 irreps in the $\{(1, 1), (2, 2)\}$ orbit, and one $\hat{1}$ plus two $\widehat{\text{sgn}}$ in the six-element non-diagonal orbit. These three contributions sum in dimension to $1 + 2 + 2 + 4 + 6 + 12 = 27 = \dim C^1(T_{3,3}^2)$. The $\{3 \mp \sqrt{3}\}$ pair lives entirely in the non-diagonal orbit's $\widehat{\text{sgn}}$ -isotypic.

E Cross-substrate contrast cases

We support Theorem 13.3's framing of the cyclotomic pair as a *sufficient algebraic ingredient* with two negative controls.

Octahedron under O_h . The octahedron (6 vertices, 12 edges, 8 faces, $\chi = 2$) carries the full octahedral symmetry group O_h of order 48, which has no order-3 central element generating a $\mathbb{Z}_3$ -cyclotomic factor in its character ring. A direct computation of L_1 on the octahedron gives $\text{Spec}(L_1) = \{0, 2, 4, 6\}$ with multiplicities $(2, 3, 3, 4)$ totalling $12 = |E|$. All eigenvalues are rational; $\{3 \mp \sqrt{3}\}$ is absent. The C^1 decomposition under O_h is $[2, 2]_+ \oplus [2, 1, 1]_+ \oplus [1^4]_+ \oplus [3, 1]_- \oplus [2, 1, 1]_-$ ($\pm$ indices labelling parity under the orientation-reversing sublattice), each multiplicity 1.

Icosahedron under A_5 . The icosahedron (12 vertices, 30 edges, 20 faces, $\chi = 2$, 5-regular) carries A_5 of order 60. Its L_1 spectrum contains two Galois pairs $\{3 \mp \sqrt{5}, 5 \mp \sqrt{5}\}$ (over $\mathbb{Q}(\sqrt{5})$) plus four rational eigenvalues $\{2, 3, 5, 6\}$. The irrationalities are $\sqrt{5}$ -type, not $\sqrt{3}$ -type: A_5 's character ring carries a $\mathbb{Z}_5$ -cyclotomic substructure (via the 5-cycle conjugacy class) but no $\mathbb{Z}_3$ -cyclotomic factor of the relevant type. The chain-space decomposition is $C^1 = 2 \cdot [3a] \oplus 2 \cdot [3b] \oplus 2 \cdot [4] \oplus 2 \cdot [5]$, uniquely fixed by the constraint $H^1(\text{icos.}) = 0$. Again $\{3 \mp \sqrt{3}\}$ is absent.

Conclusion of contrast. The two contrast cases support reading Theorem 13.3's $\mathbb{Z}_3$ -cyclotomic ingredient as a *sufficient* algebraic condition for the $\{3 \mp \sqrt{3}\}$ pair; necessity is open. Specifically, a substrate whose automorphism group's character ring lacks any $\mathbb{Z}_3$ -cyclotomic factor of the type identified in §13 does not host the pair — but we have not ruled out the possibility of other algebraic ingredients producing the same conjugate pair via a different mechanism.

F Character table of A_5

A_5 has five conjugacy classes (id, (12)(34), (123), $5a$, $5b$) with sizes (1, 15, 20, 12, 12), summing to $|A_5| = 60$. Its five irreducible representations have dimensions (1, 3, 3, 4, 5).

class	id	(12)(34)	(123)	$5a$	$5b$
size	1	15	20	12	12
[1]	1	1	1	1	1
[3a]	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$
[3b]	3	-1	0	$(1 - \sqrt{5})/2$	$(1 + \sqrt{5})/2$
[4]	4	0	1	-1	-1
[5]	5	1	-1	0	0

[3a] and [3b] are Galois conjugates over $\mathbb{Q}(\sqrt{5})$. Row orthogonality $\sum_C |C| \chi_\rho(C)^2 = 60$ verified for each row; column orthogonality likewise.

G Character table of S_4

S_4 has five conjugacy classes ((1^4) , $(2, 1, 1)$, $(2, 2)$, $(3, 1)$, (4)) with sizes (1, 6, 3, 8, 6), summing to $|S_4| = 24$. Irrep dimensions (1, 3, 2, 3, 1).

class	(1 ⁴)	(2, 1, 1)	(2, 2)	(3, 1)	(4)
size	1	6	3	8	6
[4]	1	1	1	1	1
[3, 1]	3	1	−1	0	−1
[2, 2]	2	0	2	−1	0
[2, 1, 1]	3	−1	−1	0	1
[1 ⁴]	1	−1	1	1	−1

H Character table of $O_h = S_4 \times \mathbb{Z}_2$

O_h has 10 conjugacy classes (the 5 S_4 classes each split by $\mathbb{Z}_2$ -parity) and 10 irreducible representations of dimensions (1, 3, 2, 3, 1, 1, 3, 2, 3, 1). Construction: tensor each S_4 -irrep with each $\mathbb{Z}_2$ -irrep (+ or −); the table is the direct product of Appendix G and the $\mathbb{Z}_2$ table $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. We omit the full 10×10 table; see [14] or any standard reference.

I Character table of D_6

The dihedral group D_6 of order 12 has six conjugacy classes $\{e, r, r^2, r^3, s, rs\}$ with sizes (1, 2, 2, 1, 3, 3) and six irreducible representations of dimensions (1, 1, 1, 1, 2, 2).

class	e	r	r^2	r^3	s	rs
size	1	2	2	1	3	3
A_1	1	1	1	1	1	1
A_2	1	1	1	1	−1	−1
B_1	1	−1	1	−1	1	−1
B_2	1	−1	1	−1	−1	1
E_1	2	1	−1	−2	0	0
E_2	2	−1	−1	2	0	0

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The audit-and-reclassification framing of this paper (§1.5–§1.6, §12, §25.5) and the explicit topological treatment of bridge / glue / $\mathcal{C}$ -conjugacy mechanisms in §10 and §7 emerged from a sustained collaborative review process between the authors during 2026-05-21 to 2026-05-25. The Galois-pair observation of §13 surfaced during that review when the fidelity-clustered Q_{51} build then in use was caught and replaced by the exact-arithmetic $\mathbb{Z}[i]$ ray-equivalence build of §3.2; the resulting cross-repo substrate-provenance discipline [12] is the audit-on-audit infrastructure that prevents recurrence of the original $Q_{98} \rightarrow Q_{102}$ class of error in future cross-project substrate reuse.

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